

What Populist Voters Want: A Nonparametric Revealed Preference Approach

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Abstract

This paper proposes a nonparametric revealed-preference approach for recovering voter preferences from aggregate election returns in a multidimensional spatial voting model. The approach exploits the convex partitions of the policy space induced by candidate positions and requires only that a voter's utility decrease in the distance between the candidate's position and the voter's ideal point. It is portable across settings and countries—for instance, to simultaneous U.S. elections. I apply it to municipal-level results from Germany's four Bundestag elections between 2013 and 2025, modelling political competition along left-right and anti-elite dimensions. The results reveal a transformation in the basis of support for the radical-right populist party Alternative for Germany (AfD): what began primarily as anti-elite voting became increasingly aligned with the party's right-wing positions. By 2025, the left-right dimension accounts for as much as 70% of AfD support. Counterfactuals point to an apparent but illusory opportunity for the centre-right Christian Democratic Union (CDU) party. If the AfD's position is held fixed, a modest shift to the right by the CDU—for example, on immigration—attracts more than one-fifth of AfD voters. When the AfD can respond strategically, however, it moves further right and reverses the CDU's gains; in equilibrium, the CDU does not move right. These results may explain why establishment parties fail to win back populist voters despite a seeming ideological opening.

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1 Introduction

Politics is multidimensional. Politicians build their electoral campaigns around cultural and social issues, confrontation with the elites, demands for institutional reforms and climate policy, among other topics. Voters, too, tend to base their electoral choices at least partly on issues unrelated to the economic policy dimension (Gethin, Martínez-Toledano and Piketty, 2022; Danieli et al., 2023; Gennaioli and Tabellini, 2025; Longuet-Marx, 2025; Noy and Rao, 2025; Acemoglu, Egorov and Sonin, 2026). Understanding voters’ preferences in such multidimensional environments is essential for a better understanding of voters’ policy demands, and for crafting appropriate responses.

Credibly learning voters’ preferences requires reliable data on voting choices and a plausible model of voting behaviour. The first points to aggregate electoral data rather than surveys; the second to the spatial voting model, the canonical framework of voting choice (Downs, 1957). Bringing the two together in a multidimensional environment is non-trivial: existing identification results rely on full support of candidate positions on the policy space, and estimation requires restricting the class of preference distributions (Merlo and De Paula, 2017). This paper introduces a new identification and estimation approach for a multidimensional spatial voting model that relaxes both requirements, adapting the nonparametric approach to demand estimation of Tebaldi, Torgovitsky and Yang (2023); Lieber, Tebaldi and Torgovitsky (2024) to a political economy setting.

This approach could be applied to different settings in which voters face different ballots in the same elections, such as simultaneous elections for several offices (the U.S., Italy), ranked-choice voting (Australia, Ireland) or district-based representation.

I exploit the latter setting in the context of Germany in order to study the rise of populism. While the literature has established many causal drivers of populist voting, it remains divided on the underlying dimension of this rise: is populist voting grounded in a rejection of the elites and of how representative democracy currently functions, or in the ideological Left–Right positions of populist candidates (Guriev and Papaioannou, 2022)?

I recover the preferences of voters of the radical-right populist party Alternative for Germany (AfD) over 2013–2025, in a voting model with Left–Right and Anti-Elite dimensions. The results show that the party’s earlier support was grounded in anti-elite sentiment, whereas its more recent support is rooted in the party’s radical-right stance. Counterfactual simulations suggest that in the most recent election, establishment parties could have won about one third of the AfD electorate by moving moderately to the right on immigration policy alone; this opportunity, however, is illusory once the AfD’s response is allowed for.

In the first part of the paper, I develop an approach to recovering voters’ preferences. I adopt

a spatial voting model in which voter i 's valuation of candidate j depends only on the weighted Euclidean distance between the voter's ideal point (bliss point) X_i and the candidate's position Y_j in the policy space. Weights, represented by a diagonal weight matrix W , capture the importance voters attach to each dimension. Voters are grouped into types defined by observable characteristics, and the distribution of ideal points is allowed to differ across types. I formalize a nonparametric approach to recovering voters' preferences (the common dimension weights W and the type-specific distributions of ideal points X_i) from aggregate election returns. The approach relies solely on the geometry of choice implied by the multidimensional spatial voting model, imposes no distributional assumptions, and exploits variation in candidate positions across choice environments. In my application, this variation comes from differences in candidate positions across electoral districts within a state.

Two observations are key. First, the weight matrix W determines which joint voting patterns a given type can display across different electoral districts: since a single type-specific distribution of ideal points must generate that type's vote shares in every district, a given W can rationalize only some joint patterns and not others. Weights are therefore set-identified: any W that cannot rationalize the joint voting patterns observed in the data is ruled out, and the identified set consists of those that can.

Second, there are generally infinitely many ideal-point distributions consistent with the data. This fundamental underidentification can be addressed by adding structure—either by strengthening assumptions on the choice model or by imposing structure on unobserved heterogeneity.¹ An alternative approach is to answer the questions of interest without recovering a single best-fitting distribution of ideal points, at the cost of obtaining bounds rather than point estimates. This paper adopts the latter approach, building on [Tebaldi, Torgovitsky and Yang \(2023\)](#). Specifically, I focus on bounding *target parameters*: interpretable functionals of the underlying distribution of preferences (W, X_i) . A target parameter could be, for example, the share of voters whose ideal points lie in a given area of the policy space, or the share of voters who would abandon a party if it shifted its position by a given amount.

In the second part of the paper, I apply the framework to Germany, where the electoral landscape has undergone a pronounced transformation over the past two decades, with the traditional establishment parties losing ground to the populists. The central empirical object in this paper is the preferences (W, X_i) of the electoral base of the very successful populist party Alternative for Germany (AfD), which by mid-2026 leads national polls with a 6 p.p. margin across the country.² This party combines radical-right positions with an explicitly anti-elite appeal, which motivates the choice of the

¹This is the approach taken, among others, by [Merlo and De Paula \(2017\)](#); [Iaryczower, Montero and Kim \(2024\)](#).

²Politico, National Parliament Voting Intentions – [Link](#). Accessed: 6th July 2026.

two dimensions used in the paper – the Left–Right and Anti-Elite dimensions.³

The analysis leverages rich within-state variation in the political supply of local candidates across electoral districts. I estimate candidate positions on two dimensions (Left–Right and Anti-Elite), using the German Longitudinal Election Study (GLES) Candidate Study and a complementary candidate survey distributed by the nonpartisan platform *Wen Wählen* (*Whom to Vote For?*). Because the same survey items recur across election years, the resulting positions are directly comparable over time. On the voters’ side, the unit of analysis is a demographic group, defined as an age–gender–education bin within a state and year (e.g. female, aged 18–24, higher education, Saxony, election year 2021). I combine municipality-level party-list vote shares for the six main parties (AfD, CDU, FDP, Greens, Left, SPD) with municipality-level joint demographic data for 24 such groups. Together, these data allow the model to relate observed vote shares to latent group-specific preferences.

The identifying variation comes from differences in candidate positions across electoral districts within a state and year. The identifying assumption is that the within-state variation in candidate positions is unrelated to unobserved voter heterogeneity across districts, conditional on the observed age–gender–education composition of municipalities. This is consistent with institutional details: single-member district candidates are approved by the state party leadership, which is typically more concerned with advancing preferred candidates to the Bundestag than with responding to specific local voter demands. I test the robustness of my results to this assumption in four ways: first, by focusing on very small municipalities, which are least likely to be targeted by candidates; second, by restricting the analysis to districts with non-contender AfD candidates; third, by allowing for preference heterogeneity between electoral districts in which AfD candidates are more or less conservative (an instrument similar in spirit to the one used in [Tebaldi, Torgovitsky and Yang \(2023\)](#)); fourth, by focusing on municipalities near district borders within states. The results remain quantitatively similar across all specifications.

I start by estimating the weight of the Anti-Elite dimension relative to the Left–Right dimension among AfD voters. The results suggest that in Western Germany the Left–Right dimension carries roughly three to five times the weight of the Anti-Elite dimension throughout the period. The same does not hold in Eastern Germany, where the Anti-Elite dimension carries substantial weight in every year and is estimated to outweigh the Left–Right dimension in 2013 and 2025.

I then recover the *motivations* behind the AfD vote, in an exercise that combines the estimated weights with the underlying preference distribution. Suppose the AfD split into two parties: one

³These dimensions also connect to how the political science literature conceptualizes populism. A body of work distinguishes the role of *thin* and *thick* ideologies in populist support, that is, anti-elite appeals and substantive ideological stances. See [Mudde and Kaltwasser \(2017\)](#); [Vries and Hobolt \(2020\)](#) for a discussion, and [Guriev and Papaioannou \(2022\)](#) for an overview of the political economy literature.

keeping its radical-right economic position but moving towards the establishment, the other keeping its anti-establishment appeal but moderating its Left–Right position. What share of voters would each attract? I find that AfD support is increasingly grounded in the party’s conservative Left–Right positions rather than in its anti-elite appeal, with up to 70% of voters supporting the AfD in 2021–2025 due to its radical-right ideology.

I next run counterfactual repositioning exercises to understand the intensity of competition between the parties of the establishment and the AfD. In the first, I explore the sensitivity of the AfD electorate to shifts in the AfD’s own position towards more pro-elite or more moderate Left–Right stances. The results suggest that until 2025 voters were relatively insensitive to small shifts in party positions. This indicates that competition between the AfD and other parties used to be relatively weak. The picture changes in 2025, when even a small moderation in the AfD’s Left–Right position would lead to substantial electoral losses.

In the second exercise, I ask how establishment parties could compete with populist challengers. To do so, I focus on the largest establishment party to the right of center (CDU) and estimate how much of the AfD electorate it could capture by shifting towards the AfD—that is, towards more conservative and more anti-elite positions. The results suggest that a small repositioning (0.2 estimated units) on the Left–Right dimension would capture no more than 10% of the AfD electorate before 2025. In 2025, however, the same shift would allow the CDU to capture about one third of the AfD electorate. Crucially, hardening the party’s anti-elite positions is not nearly as effective in any year.

Finally, I allow for strategic interaction between the parties: both the CDU and the AfD can reposition, while the positions of the remaining parties are held fixed. I consider a Stackelberg game in which the CDU leads and the AfD follows. If the AfD stays put, it is optimal for the CDU to move rightward, because the gains on the right flank (from the AfD) exceed the losses on the left. The AfD’s optimal response to a rightward move by the CDU, however, is to move further right itself, recapturing the most right-wing (and relatively pro-elite) voters. The Stackelberg equilibrium therefore does not feature a rightward move by the CDU, but rather staying put or moving left. This exercise offers one answer to the question of why mainstream parties do not adjust their platforms as much as a static calculation would suggest: the anticipation of the populists’ response.

I conclude by discussing the choices underlying the framework and what they imply for the results. I show that (i) the estimated model can be read as a probabilistic voting model in which the error distribution is induced by the distribution of bliss-point shocks rather than assumed directly; (ii) measurement error in candidate positions shifts the bounds towards the uninformative composition without widening them; (iii) the consequences of omitting a dimension depend on its relevance and on

its correlation with the included ones; and (iv) the main results are not driven by party repositioning, but reflect demand-side evolution. I also re-estimate the model with Economics and Culture as the two dimensions, and find a rise in cultural concerns relative to economic ones.

Related Literature. This paper is related to several strands of literature. First, to a growing body of research that estimates structural models of voting in elections (Coate and Conlin, 2004; Ujhelyi, Chatterjee and Szabó, 2021; Iaryczower, Montero and Kim, 2024; Kawai and Sunada, 2025; Berry, Cox and Haile, 2025; Longuet-Marx, 2025). This paper differs from these in that its underlying framework is a multidimensional spatial voting model, in which individuals vote for the candidate closest to them in the weighted policy space.⁴

The paper is closely related to Degan and Merlo (2009), who first formalise the partitioning of the policy space under the multidimensional spatial voting model using Voronoi cells, and to Merlo and De Paula (2017), who provide nonparametric identification and estimation of voter preferences under such a partition. Conceptually, this paper is closest to the latter. Merlo and De Paula (2017) obtain identification under a large-support assumption on candidates' positions and estimate the model with a sieve minimum-distance estimator. This paper instead relaxes the full-support assumption and asks what the observed configurations alone identify; the weights are identified from which cells the data require to be non-empty, without restricting the bliss-point distribution, rather than jointly with a sieve density. The cost is set rather than point identification of both.

The approach builds on Tebaldi, Torgovitsky and Yang (2023), who characterise sharp identified sets in a discrete choice model through the minimal relevant partition and linear programming and whose estimator I adopt, and on Lieber, Tebaldi and Torgovitsky (2026), who demonstrate that the composition of the minimal relevant partition (which cells it contains) can be used to learn about the weights on product characteristics in an additive random utility model, since these weights shift the partition. Relative to both, the spatial model is a pure-characteristics model without an outside option, so the latent object is a multi-dimensional bliss point and only relative dimension weights are identified; I show that after rescaling the policy space by the weights, their identification reduces to the enumeration problem similar to that of Lieber, Tebaldi and Torgovitsky (2026). The data are also ecological rather than aggregate: municipal shares are known mixtures over demographic groups whose choices are unobserved, which makes compositional variation across municipalities a source of identification.

Second, this paper is related to a literature that employs a range of methodologies to recover

⁴Of these papers, only Iaryczower, Montero and Kim (2024) estimates a spatial voting model, and only in one dimension. As noted in Merlo and De Paula (2017), estimation of a single-dimensional model is considerably easier, since the problem of separately identifying *weights* from bliss points does not arise.

voters' preferences, including the use of survey responses on political attitudes and self-placement, which are then mapped into ideological positions (Fiorina and Abrams, 2008; Draca and Schwarz, 2024; Algan et al., 2017; Gethin, Martínez-Toledano and Piketty, 2022; Hangartner et al., 2019; Desmet, Ortuno-Ortín and Wacziarg, 2024). Such methods face two common shortcomings: the views voters hold do not necessarily translate into voting choices, and some views may be stigmatized or socially desirable, and thus not reported truthfully. Survey-based stated preferences can therefore differ substantially from the revealed preferences inferred from electoral outcomes, and the extent of this divergence can only be assessed by estimating preferences from actual election data.

Third, this paper contributes to the large and growing literature on the causes of the rise and success of populist parties and leaders in the Western world. An extensive body of work examines the various drivers of populist voting, including economic factors such as trade and the 2008–2009 Global Financial Crisis (Autor et al., 2020; Colantone and Stanig, 2018*b,a*; Algan et al., 2017; Guiso et al., 2021; Dippel et al., 2022), immigration (Dustmann, Vasiljeva and Piil Damm, 2019; Alesina and Tabellini, 2024), and cultural factors (Margalit, 2019; Algan et al., 2018; Gennaioli and Tabellini, 2025), among others. While these drivers can be mapped onto distinct ideological dimensions (see, e.g., Rodrik, 2021, for a clear formalization of this logic and a meta-analysis of mechanisms related to globalization shocks), they do not reveal the relative importance of each, or their interplay. This paper addresses this gap by estimating at large voters' ideal points and dimension weights in a multidimensional spatial model, and tracing the evolution of preferences across demographic groups.

In this respect, the paper is closely related to Danieli et al. (2023), who recover the preferences of voters of radical-right populist parties in Europe. Unlike the present paper, however, that study relies on survey-based self-reported vote choices and issue preferences; moreover, while its conceptual framework can be cast as a bliss-point model, it cannot separate a strengthening association between individual characteristics and ideal points on a given dimension from a change in the weight on that dimension.

Lastly, this paper contributes to the debate on the relative importance of ideological and anti-elite dimensions in populist voting. A large part of the literature emphasizes the role of host ideologies in shaping populist support (Acemoglu, Egorov and Sonin, 2013; Mudde, 2013; Norris and Inglehart, 2019; Guenther and Nunnari, 2025), whereas another strand adopts an *ideational* approach and highlights the role of populist attitudes distinct from Left–Right ideology (Mudde, 2004; Akkerman, Mudde and Zaslove, 2014; Van Kessel, 2015). Several studies have sought to compare the importance of these two factors directly using survey and experimental data (Silva, Fuks and Tamaki, 2022; Neuner and Wrátil, 2022; Neuner et al., 2025). This paper shows that populist voting can take different forms, with different factors prevailing across regions and over time.

The rest of the paper is structured as follows. Section 2 presents the nonparametric model of choice, and Section 3 describes the scope of the approach. Section 4 describes the institutional context and the data. Section 5 covers identification and estimation. Section 6 presents the main results. Section 11 concludes.

2 Nonparametric Model of Party Choice

This section develops a nonparametric model of choice in a voting environment and outlines its identification. It draws on Merlo and De Paula (2017), which formalizes the logic of Voronoi partitions in the estimation of voter preferences, and on Tebaldi, Torgovitsky and Yang (2023), which develops a nonparametric approach for discrete choice models. Relative to the former, the framework differs in two key respects: identification does not rest on the assumption of a large support for candidate positions, and it does not aim to simultaneously recover a single distribution of voter preferences together with weights, but does so separately. Relative to the latter, it shifts attention from a product-valuation model in a price space to a pure characteristics model in a policy space, in which the main object of interest is individuals' preferences⁵, and extends that model by estimating the relative importance of the different dimensions.

The section proceeds in three steps. Subsection 2.1 introduces the primitives, the identifying assumptions, and the target parameters (functionals of bliss-point distributions). Subsection 2.2 shows that for known weights the estimation problem for target parameters is equivalent to a finite-dimensional linear problem. Subsection 2.3 establishes the set identification of weights, and combines the two steps into a single algorithm.

2.1 Setup

Individual i chooses a candidate⁶ j from a set $\mathcal{J} = \{1, \dots, J\}$. Each choice environment is characterized by a vector of potentially endogenous candidate positions $Y_i = (Y_{i1}, \dots, Y_{iJ})$, which I model as discretely distributed. An individual may face one or more such environments at the same time. In an electoral setting, the ballot provides a natural representation of this set. Individual i has a bliss point X_i known to the individual but latent to the researcher.

I adopt a spatial voting model (Downs, 1957; Hinich and Munger, 1996) and assume that valuations are a function of the distance between the individual's own bliss point and the positions of the candidates:

$$V_{ij} = V(X_i, Y_{ij}) = U(d(X_i, Y_{ij})),$$

⁵See Berry and Pakes (2007) for an extensive discussion of the differences between these models.

⁶*Candidate* is used here to denote any choice option appearing on the ballot, including parties, individual candidates, or referendum choice options, among others.

where $d(X_i, Y_{ij})$ denotes the Euclidean distance between the two points in space, and $U(\cdot)$ is a strictly decreasing function. Note that this functional form does not presuppose any specific cardinal loss function, and is invariant to strictly monotone transformations of the (weighted) Euclidean distance.⁷

For simplicity of exposition, in this and the next subsection I consider the unweighted Euclidean distance ($d(X_i, Y_{ij}) = \sqrt{(X_i - Y_{ij})'(X_i - Y_{ij})}$). Subsection 2.3 extends the analysis to the weighted Euclidean distance and discusses separate identification of the weights.

The policy space $\mathcal{R}$ is assumed to be nonempty and compact. For the purposes of this paper I focus on the two-dimensional policy space, $\mathcal{R} \subset \mathbb{R}^2$, and, hence, $X_i, Y_{ij} \in \mathcal{R}$. An individual i 's choice is:

$$\text{Choice}_i = \text{Choice}(X_i, Y_i) = \arg \max_{j \in \mathcal{J}} V_{ij} \quad (1)$$

The primitive object of interest is the conditional distribution of bliss points X_i , given candidate positions Y_i , market (region) M_i , and demographic covariates D_i . I assume throughout that this conditional distribution is absolutely continuous, so that, for each realization $Y_i = y$, $M_i = m$, and $D_i = d$, it admits a density $f(x \mid y, m, d)$; since the set of bliss points indifferent between any two candidates has Lebesgue measure zero, ties between choices then occur with probability zero. Let $\mathcal{F}$ denote the set of all such conditional density functions.

Conditioning on Y_i allows for the possibility that the choice environment faced by an individual is *endogenous*: candidate positions may reflect voter characteristics beyond the observed demographics d and market m , in which case $f(x \mid y, m, d)$ varies with y . I assume there exists an observed variable Z_i (an instrument) such that, for all (z, m, d) and all y in the support of Y_i given $(Z_i, M_i, D_i) = (z, m, d)$, the conditional density of bliss points satisfies

$$f(x \mid y, z, m, d) = f(x \mid z, m, d) \quad \text{for all } x. \quad (2)$$

The leading case is when candidate positions are unrelated to voters' unobserved characteristics conditional on (m, d) alone; the assumption then is $f(x \mid y, m, d) = f(x \mid m, d)$ for all y . To save on notation, I proceed under this special case throughout; when a non-degenerate instrument is required, every argument below applies with z added to the conditioning set (m, d) , and objects conditional on (m, d) alone are recovered by averaging over the observed distribution of the instrument, $f(x \mid m, d) =$

⁷See [Kendall, Nannicini and Trebbi \(2015\)](#); [Cruz et al. \(2024\)](#) for the estimation of the functional form of the loss function. The analysis is not invariant, however, to arbitrary alternative notions of distance. This is because the identification argument relies on the geometry of the regions of the ideological space consisting of points closest to each candidate, i.e. Voronoi cells. Alternative metrics may generate different choice regions. For example, in a multidimensional space, using the sum of absolute coordinate-wise differences between a bliss point and a candidate position corresponds to imposing Manhattan, rather than Euclidean, geometry.

$\mathbb{E}[f(x \mid Z_i, m, d) \mid M_i = m, D_i = d]$. The assumption has identifying content only if the instrument is *relevant*, i.e., if candidate positions vary with the instrument. Section 3 discusses institutional settings, and identification strategies from the literature, under which Assumption 2 is plausible.

Common quantities of interest can be written as functionals of f . For instance, one could be interested in estimating the share of individuals with given demographics whose bliss points are left-leaning on a given dimension, $\underline{\mathcal{X}} = \{x \in \mathcal{R} : x_1 \leq \underline{x}_1\}$ for a known threshold $\underline{x}_1$:

$$\theta(m, d; f) = \int_{\underline{\mathcal{X}}} f(x \mid m, d) dx$$

I refer to quantities of interest such as the one above as *target parameters*. Other quantities of interest could characterize support for political alternatives or the consequences of institutional rules – for instance, the share of candidate j 's supporters with bliss points in a given region, or counterfactual vote shares under alternative positions or rules. The goal is to infer the values of $\theta(m, d; f)$ that are consistent with both the observed data and the assumptions imposed.

The first assumption is that the support of f is contained in a known subset of $\mathcal{R}$. In the case of the voting application, this naturally translates into assuming a bounded policy space, preventing unreasonably extreme ideal points:

$$\int_{\bar{\mathcal{X}}} f(x \mid m, d) dx = 1 \quad \text{for all } (m, d), \text{ where } \bar{\mathcal{X}} \text{ is a known subset of } \mathcal{R} \quad (3)$$

In voting contexts, only vote shares aggregated over specific administrative units (e.g. municipalities, precincts) are observed; I refer to these units as *submarkets* of a market and index them by s , with $\mathcal{S}(m)$ denoting the submarkets of market m . For each submarket a researcher observes its demographic composition; let $q_{s,d}$ denote the share of the group with $D_i = d$ in submarket s , with $\sum_d q_{s,d} = 1$. The observed data for the submarket s in market m with candidate positions y are then:

$$s_j(y, s) = \sum_d q_{s,d} \mathbb{P}[\text{Choice}_i = j \mid Y_i = y, M_i = m, D_i = d]$$

Each density of bliss points $f \in \mathcal{F}$ also implies a set of vote shares:

$$s_j(y, s; f) = \sum_d q_{s,d} \int_{\mathcal{X}_j(y)} f(x \mid m, d) dx$$

where $\mathcal{X}_j(y) = \{x \in \bar{\mathcal{X}} : V(x, y_j) > V(x, y_k) \ \forall k \neq j\}$ is the attraction region of candidate j

under positions y , with y_j the position of candidate j in configuration y . A density f is consistent with the observed choice shares if

$$\underbrace{s_j(y, s)}_{\text{observed}} = \underbrace{s_j(y, s; f)}_{\text{predicted}} \quad \text{for all } j \text{ and all pairs } (y, s) \text{ observed in the data} \quad (4)$$

The sharp identified set of bliss-point densities is defined as the subset $\mathcal{F}^*$ of $f \in \mathcal{F}$ that satisfy (2), (3), (4). The main object of interest in the paper throughout is the sharp identified set for the target parameter, constructed as:

$$\Theta^* = \{\theta(f) : f \in \mathcal{F}^*\}$$

2.2 Equivalent finite-dimensional problem

How could one compute Θ^* ? Note that $\mathcal{F}$ is an infinite-dimensional object, which does not allow solving the problem via grid search. [Tebaldi, Torgovitsky and Yang \(2023\)](#) develop an approach for checking whether there exists some $f \in \mathcal{F}^*$ that would generate a given conjectured value $t = \theta(f)$ for the target parameter. Their paper proves that the existence of such an f is equivalent to the existence of a solution to a finite-dimensional system of linear equations.

The finite-dimensional system is constructed by first partitioning the policy space into the smallest number of sets needed to describe choice behavior under all relevant candidate positions. Let $\mathcal{Y}$ denote a finite set of candidate position vectors that contains the observed candidate positions, as well as any additional positions relevant for evaluating the target parameter. The Minimal Relevant Partition (MRP) is the coarsest partition of the policy space $\mathcal{R}$ with the following property: any two bliss points in the same subset produce the same choice behavior for all $y \in \mathcal{Y}$, while any two bliss points in different subsets produce different choice behavior for at least one $y \in \mathcal{Y}$. Formally, the MRP is the coarsest partition $\mathbb{X}$ of $\mathcal{R}$ such that

$$x, x' \in \mathcal{X} \text{ for some } \mathcal{X} \in \mathbb{X} \iff \text{Choice}(x, y) = \text{Choice}(x', y) \text{ for all } y \in \mathcal{Y}.$$

In other words, whenever two different bliss points x, x' belong to the same set from the MRP, this implies that x, x' yield the same choice behavior across all choice environments in $\mathcal{Y}$. Note two distinctive features of this setup. First, it does not require researchers to normalize any of the options. Intuitively, this is because individuals' valuations of the options are well defined relative to one another (see discussion in [Merlo and De Paula \(2017\)](#) on identification of the spatial voting model). Second, because only relative comparisons between candidates are exploited, the framework does not naturally

accommodate the turnout decision, which would require an absolute utility threshold for abstention.⁸

Figures 1(a) and 1(b) illustrate the partitions of the policy space generated by two hypothetical position vectors, y_1 and y_2 , in a setting where individuals face different candidate choice sets across elections. For consistency with the application that follows, the illustration uses a two-dimensional policy space. Figure 1(c) shows the two partitions overlaid.

This mirrors the structure of the application ahead, where voters face different choice environments across geographic areas: individuals in the same market m confront different candidate sets (y_1 and y_2 in Figure 1) depending on the electoral district to which they belong. The identifying variation comes from the differences in choice environments of individuals with plausibly similar preferences (same market m , demographics d).

The MRP is obtained by partitioning the continuum of bliss points into sets that generate identical choice behavior for all positions in $\mathcal{Y}$. When the object of interest is behavior under these candidate positions, attention can therefore shift from densities to *mass functions defined over the sets of the MRP*.

To illustrate this, consider the target parameter defined above, the share of voters with left-leaning bliss points. Figure 1(d) graphically represents this quantity as the share of voters whose bliss points lie in area T , where $T = \{x \in \bar{\mathcal{X}} : x \leq (0, 0)\}$. In the MRP representation, evaluating such a target parameter requires refining the partition by the boundary of T , so that T can be represented as a union of MRP cells. In Figure 1(d), T consists of three subsets, $\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3$. Hence, the target parameter can be written as:

$$\theta(m, d; f) = \int_{\mathcal{X}_1} f(x | m, d) dx + \int_{\mathcal{X}_2} f(x | m, d) dx + \int_{\mathcal{X}_3} f(x | m, d) dx,$$

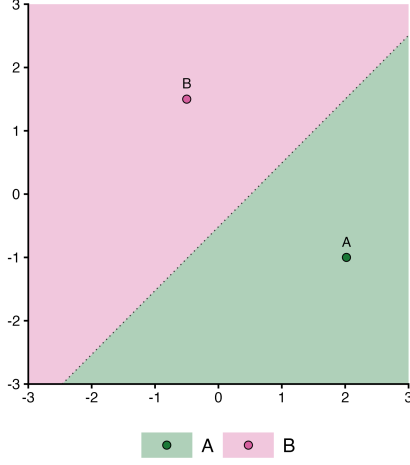
which depends not on f per se, but only on the mass that f places on $\mathcal{X}_1, \mathcal{X}_2, \mathcal{X}_3$. Focusing on these masses transforms the intractable infinite-dimensional problem of searching over all $f \in \mathcal{F}$ into a tractable finite-dimensional problem of searching over a finite set of non-negative numbers that sum to unity.

Denote the set of conditional mass functions on $\mathbb{X}$:

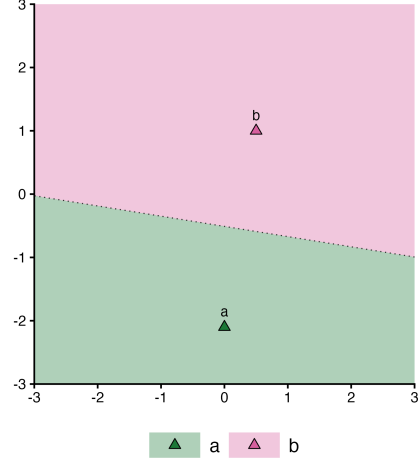
$$\Phi = \{\phi \in \mathbb{R}_+^{d_\phi} : \sum_{\mathcal{X} \in \mathbb{X}} \phi(\mathcal{X} | m, d) = 1 \ \forall (m, d) \in \text{supp}(M_i, D_i)\},$$

where d_ϕ is the cardinality of $\mathbb{X} \times \text{supp}(M_i, D_i)$. Denote by $\mathbb{X}_j(y)$ the sets in the MRP associated

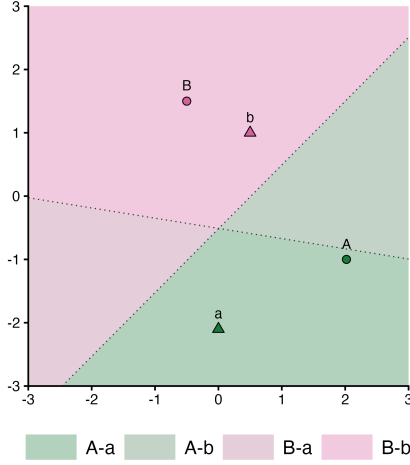
⁸In the application ahead, I account for the differential turnout rates across demographic groups across markets.



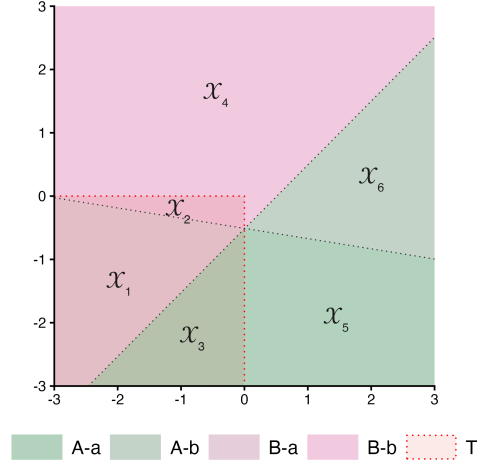
(a) Election 1



(b) Election 2



(c) Overlay



(d) MRP: Overlay with Target Parameter

Notes: The figure illustrates a hypothetical setting with elections involving two candidates in each contest, (A, B) and (a, b) . In the two upper panels, the shaded pink region denotes the attraction area of candidates B and b , the green region – of candidates A and a in Elections 1 and 2, respectively. Panel (c) shows the overlay of the partitions in panels (a) and (b). Panel (d) presents the same overlay together with the region corresponding to a target parameter. $\mathcal{X}_n$ denotes the unique cells that characterize observed behavior in Elections 1 and 2 or a target parameter.

Figure 1. Example of partitions of the policy space in two elections with two parties ((a), (b)) and their overlay (c) with a target parameter (d).

with choosing j under y . Then $\phi \in \Phi$ is consistent with some f that satisfies (4) only if it satisfies:

$$\underbrace{s_j(y, s)}_{\text{observed}} = s_j(y, s; \phi) = \underbrace{\sum_{\mathcal{X} \in \mathbb{X}_j(y)} \sum_d q_{s,d} \phi(\mathcal{X} \mid m, d)}_{\text{predicted}} \quad \forall j \text{ and observed } (y, s). \quad (5)$$

Similarly, $\phi \in \Phi$ is consistent with an f that satisfies (3) only if it satisfies:

$$\sum_{\mathcal{X} \in \bar{\mathbf{X}}} \phi(\mathcal{X} \mid m, d) = 1 \text{ for all } (m, d) \quad (6)$$

where $\bar{\mathbf{X}}$ is the subset of cells of $\mathbb{X}$ that are contained in $\bar{\mathcal{X}}$ (the partition can always be refined so that every cell lies either inside or outside $\bar{\mathcal{X}}$).

For any $f \in \mathcal{F}$, define $\bar{\phi}(f) \in \Phi$ as

$$\bar{\phi}(f)(\mathcal{X} \mid m, d) = \int_{\mathcal{X}} f(x \mid m, d) dx \quad \text{for all } \mathcal{X} \in \mathbb{X}, (m, d) \in \text{supp}(M_i, D_i).$$

Next, assume that the target parameter is representable in terms of the cell masses: there exists a known function $\bar{\theta} : \Phi \rightarrow \mathbb{R}^{d_\theta}$ such that

$$\theta(f) = \bar{\theta}(\bar{\phi}(f)) \quad \text{for all } f \in \mathcal{F}. \quad (7)$$

This assumption can always be satisfied by choosing $\mathcal{Y}$ (and hence the partition) to include all positions and regions relevant for evaluating the target parameter; in the leading example above, $\bar{\theta}(\phi) = \phi(\mathcal{X}_1 \mid m, d) + \phi(\mathcal{X}_2 \mid m, d) + \phi(\mathcal{X}_3 \mid m, d)$.

The main propositions in [Tebaldi, Torgovitsky and Yang \(2023\)](#) can then be adapted to the voting context as follows:

Proposition 1. *Suppose that (7) is satisfied. Let $\Phi^* \equiv \{\phi \in \Phi : \phi \text{ satisfies (5), (6)}\}$, and for any $t \in \mathbb{R}^{d_\theta}$, let $\Phi^*(t) \equiv \Phi^* \cap \{\phi \in \Phi : \bar{\theta}(\phi) = t\}$. Then $t \in \Theta^*$ iff $\Phi^*(t) \neq \emptyset$.*

The nontrivial direction of Proposition 1 is sufficiency: any $\phi \in \Phi^*(t)$ can be extended to a density $f \in \mathcal{F}^*$ with $\theta(f) = t$ by distributing the mass assigned to each cell arbitrarily (for instance, uniformly) within that cell; see the Appendix of [Tebaldi, Torgovitsky and Yang \(2023\)](#) for the formal argument.

Proposition 2. *If $\bar{\theta}$ is continuous on Φ and $d_\theta = 1$, then $\Theta^* = [t_{lb}^*, t_{ub}^*]$, where*

$$t_{lb}^* = \min_{\phi \in \Phi^*} \bar{\theta}(\phi), \quad t_{ub}^* = \max_{\phi \in \Phi^*} \bar{\theta}(\phi).$$

Existence of the extrema and the interval form follow because Φ^* is a compact convex polytope and $\bar{\theta}$ is continuous; when $\bar{\theta}$ is linear in ϕ , as in the leading example, both bounds are computed by linear programming.

2.3 Weighted spatial voting model

Assume individual valuations of choice options can be expressed as:

$$V_{ij} = V(X_i, Y_{ij}; W_i) = U(d_{W_i}(X_i, Y_{ij})),$$

where W_i is a weighting matrix whose diagonal elements govern the relative importance of the dimensions, and whose off-diagonal elements determine the interaction between the two dimensions. For computational and identification purposes, I set the off-diagonal elements to zero:

$$W_i = \begin{pmatrix} w_{i1} & 0 \\ 0 & w_{i2} \end{pmatrix}, \quad w_{i1}, w_{i2} > 0.$$

In the two-dimensional setup the distance is $d_{W_i}(X_i, Y_{ij}) = \sqrt{(X_i - Y_{ij})^\top W_i (X_i - Y_{ij})}$. The case considered before is a special case with $w_{i1} = w_{i2}$. Since choices depend only on the relative weights assigned to the dimensions, not on their absolute levels, I normalize $w_{i1} = 1$ without loss of generality and treat $w_{i2} \in \mathbb{R}_+$ as the relative weight of the second dimension.

Throughout this section I impose that the weighting matrix is common across individuals in a given market and denote: $W_i = W = \text{diag}(1, w_2)$ for all i . In the application, w_2 is allowed to differ across region-years but is common within them.

This subsection demonstrates that one can separately identify the weight w_2 and the probability mass function of bliss points ϕ and derive the sharp bounds for the target parameters using those.⁹

⁹The closest argument in the literature is [Merlo and De Paula \(2017\)](#), who prove nonparametric point identification of weights and distributions of bliss points in a spatial voting model. Their analysis relies on a large support assumption for candidate positions. In particular, to establish joint identification of bliss points and weights, they rely on the results of [Ichimura and Thompson \(1998\)](#), who prove nonparametric identification in a binary discrete choice model using the Cramér–Wold device: there exists a two-candidate configuration such that the affine hyperplane separating the two candidates delivers different election outcomes under two distinct preference distributions. Building on these results, they extend them to 1) include multiple candidate positions, and 2) introduce weights. Both 1) and 2) are based on the large support assumption, i.e. that one could sample arbitrary candidate configurations from $\mathbb{R}^2$. This assumption is infeasible in political-economy environments, where the set of candidate positions is typically finite and confined to a bounded subset of $\mathbb{R}^2$. Hence, the setup of this paper accommodates the fact that there are just finitely many observed

The two observational-equivalence conditions (4) and (5) for f and ϕ , respectively, can now be recast as follows. The predicted vote share of candidate j is:

$$s_j(y, s; f, w) = \sum_d q_{s,d} \int_{\mathcal{X}_{j;w}(y)} f(x \mid m, d) dx,$$

where $\mathcal{X}_{j;w}(y) = \{x \in \bar{\mathcal{X}} : V(x, y_j; w) > V(x, y_k; w) \ \forall k \neq j\}$. It matches the data if

$$s_j(y, s; f, w) = s_j(y, s) \quad \forall j \text{ and observed } (y, s). \quad (8)$$

Likewise, for the equivalent finite-dimensional problem, denote by $\mathbb{X}_w$ the MRP generated by $\mathcal{Y}$ under weight w , and by $\mathbb{X}_{j;w}(y)$ its cells on which j is chosen under y ; then

$$\sum_{\mathcal{X} \in \mathbb{X}_{j;w}(y)} \sum_d q_{s,d} \phi(\mathcal{X} \mid m, d) = s_j(y, s) \quad \forall j \text{ and observed } (y, s) \quad (9)$$

The algorithm for obtaining bounds for the target parameter in the presence of an unknown weight is then the following. First, obtain the identified set of weights over a candidate space $\mathcal{W} \subseteq \mathbb{R}_+$,

$$W^* = \{w \in \mathcal{W} : \exists \phi \in \Phi \text{ such that (9) and (6) are satisfied}\}.$$

Second, for each $w \in W^*$ obtain $\Theta^*(w)$ by applying Propositions 1–2 to the partition $\mathbb{X}_w$. The identified set for the target parameter is then $\Theta^* = \bigcup_{w \in W^*} \Theta^*(w)$. Since each $\Theta^*(w)$ is an interval, Θ^* is contained in

$$\left[\min_{w \in W^*} t_{lb}^*(w) ; \max_{w \in W^*} t_{ub}^*(w) \right],$$

and these are the bounds I report. Appendix F describes this estimation procedure in the application.

Note two features of this problem. First, the equations in the finite-dimensional problem (9) are defined by the available partition cells in a minimal relevant partition $\mathbb{X}_{j;w}(y)$, and these cells vary with the weight. Second, given the cells, the distribution of bliss points is matched by fitting $\phi(\mathcal{X} \mid m, d)$ over the available cells. Intuitively, the former is the source of identification for the weights, and the latter the source of identification for the bliss-point distribution: the weight is identified by *which* cells the data require, the mass function by *how much* mass those cells must carry. I now provide intuition for the argument for weight identification.

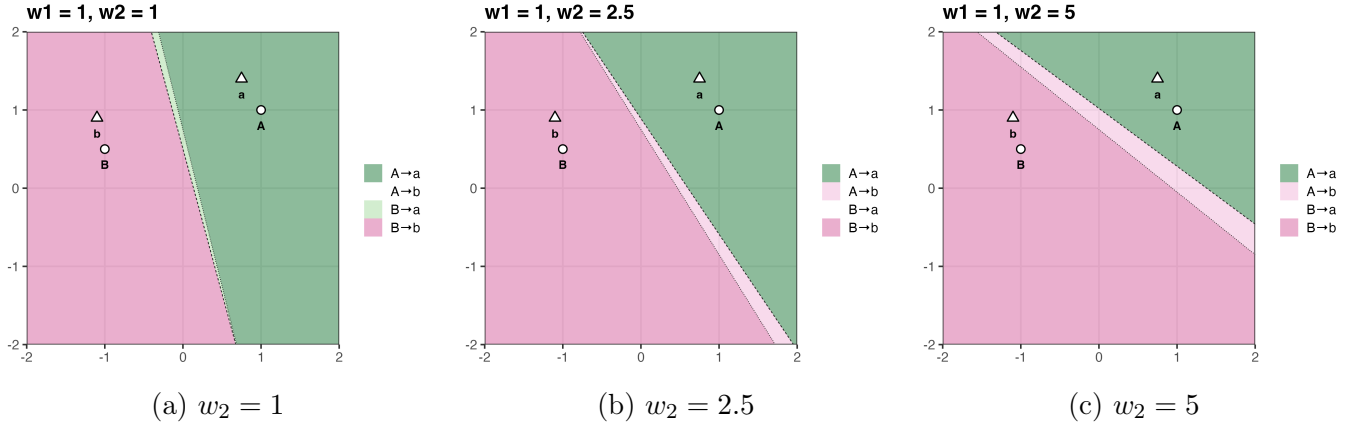
Imagine a researcher observes the results of two elections with two candidates each (candidates A, B and a, b , respectively). The goal is to use these two elections to obtain information about f and W . The observed marginals (vote shares) – s_A, s_B, s_a, s_b – are informative about the joint values configurations in a bounded policy space.

$(s_{Aa}, s_{Ab}, s_{Ba}, s_{Bb})$ of interest. Consider the hypothetical scenario from Table 1, in which candidate A and candidate b obtain the majority of votes in their respective elections.

	a	b	Total
A	$[0, 0.1]$	$[0.6, 0.7]$	0.7
B	$[0, 0.1]$	$[0.2, 0.3]$	0.3
Total	0.1	0.9	1

Table 1. Marginals (observed) and Fréchet bounds on the joint cell shares

This scenario implies that the cell A – b hosts at least 60% of all voters. Hence, the minimal relevant partition of those elections must contain a cell that would rationalize such voter behavior. The claim is that such a cell might not appear under every configuration of weights, and that the weights yielding the partition that contains this cell are best suited to explain the behavior of voters. Figure 2 illustrates this point and demonstrates that, in this hypothetical example, the weight configuration $w_2 = 1$ is not well suited to explain the data. Moreover, Figure 2 illustrates the nature of the set-identification of weights in this setup: $w_2 = 2.5$ and $w_2 = 5$ effectively yield the same partition of the policy space and cannot be told apart.



Notes: The figure illustrates a hypothetical setting with elections involving two candidates in each contest, (A, B) and (a, b) . Each panel overlays the two elections in the policy space $\mathbb{R}^2$: triangles mark the positions of candidates A and a , circles mark candidates B and b . The shaded regions denote the attraction areas of the joint outcomes: $A \rightarrow a$ (dark green), $A \rightarrow b$ (light green), $B \rightarrow a$ (light pink), and $B \rightarrow b$ (dark pink), where $X \rightarrow y$ denotes voters who prefer X in Election 1 and y in Election 2. The cells present in a given panel constitute the minimal relevant partition for that weight configuration. Across panels, the dimension weight w_2 increases ($w_1 = 1$ throughout).

Figure 2. Illustration of minimal relevant partitions induced by the same candidate configurations (A, B) , (a, b) under different values of w_2 , $w_1 = 1$.

The example shows that weights are identified by the available cells in the minimal relevant partition. Appendix A proves that each cell of the minimal relevant partition is non-empty only for weights in an interval, so the weights consistent with the cells the data require form an intersection of

such intervals—itself an interval containing the true weight. Thus the identified set for w_2 is contained in an interval on $\mathbb{R}_+$. The informativeness (length) of this interval is then an empirical question. [Lieber, Tebaldi and Torgovitsky \(2024\)](#) provide a similar argument about the set-identification of relative weights for different characteristics in an additive RUM.

In practice there need not be a unique partition that best explains the data: because the number of feasible cells (over which voter mass is estimated) weakly exceeds the number of observed vote-share marginals, the problem is an under-identified linear system, and several partitions may rationalize the data equally well. Two observations are worth highlighting. First, as [Lemma 1](#) in [Appendix A](#) shows, each cell is generated only by weights lying in an interval. Identifying power then comes from Fréchet-type necessity constraints¹⁰. Second, additional elections with distinct vote-share configurations weakly increase the number of such constraints. The admissible weights must lie in the intersection of the intervals associated with every cell required to carry positive mass; since the data were generated at the true weight, every cell the data require is feasible there, so this intersection is a non-empty interval containing the truth, and it weakly shrinks as elections accumulate. The size and shape of the identified set are thus an empirical object.

[Appendix B](#) reports simulations comparing the approach of this paper—which exploits the fact that different weights generate different cells—with that of [Merlo and De Paula \(2017\)](#), who recover the weights jointly with the distribution of bliss points using a sieve minimum-distance estimator. The simulations suggest that the true weight value belongs to the recovered set under the approach of this paper, whereas the weights recovered jointly with the bliss-point distribution fall outside the identified set. The intuition is that identification of the weights from the cell structure requires no assumption on the distribution of bliss points, so it is not contaminated when that distribution is misspecified; the sieve estimator, by contrast, must approximate the bliss-point density, and any residual misspecification is absorbed into the weight estimate. [Appendix B](#) also demonstrates that the identified set for the weights shrinks with the number of elections. Introducing noise in voters’ perceptions of candidate positions leaves this conclusion unchanged: the identified region tightens around the true weight, while the sieve estimates remain outside it.

The identification argument above does not rely on the weight being common across demographic groups within a market, and a similar argument goes through when W varies across demographic groups too. In practice, however, group-specific weights are considerably more demanding on the data. Informative group-level intervals would require either group-level choice data or far richer variation in candidate configurations than is available in the application.

¹⁰For example, if candidate (or coalition) i receives a share of 0.52 in election e and candidate j receives 0.51 in election e' , at least 3% ($0.52 + 0.51 - 1$) of voters must choose the pair (i, j) , so the corresponding cell must be non-empty.

3 Scope of the Approach

This section discusses the settings in which the identifying assumption of the approach introduced in the previous section is plausible, highlights its broad applicability, and considers possible extensions and generalizations of the model, including the incorporation of strategic voting.

The approach offers a method for estimating voters' preferences within a spatial voting framework using real-world aggregate voting data. Identification rests on comparing the behavior of voters who share the same preference distribution across different choice environments: either directly, $f(x \mid y, m, d) = f(x \mid m, d)$ for all y , or conditional on an instrument z for candidate positions, $f(x \mid y, z, m, d) = f(x \mid z, m, d)$. While such settings may at first seem uncommon, they are in fact pervasive, and closely related identifying assumptions are already used in the political economy literature and beyond. I organize the discussion around three types of settings.

First, consider simultaneous voting, where the same voters face different choice environments because several elections take place at the same time. The main advantage of this setting is that the identifying condition $f(x \mid y, m, d) = f(x \mid y', m, d)$ holds by construction. Its principal limitation, however, is that the number of simultaneous elections is typically small, resulting in a coarse partition of the policy space. Consequently, the inferences that can be drawn using this approach may be less informative.

Among other cases, simultaneous voting is used in the United States, where elections for multiple offices often take place on the same day; in Australia, where voters cast ballots for both chambers of parliament simultaneously; in Italy, where general elections fill both the Chamber of Deputies and the Senate on the same day; or in Switzerland, where voters may face multiple electoral ballots as well as referendum questions on the same election day.

Second, *ranked-choice voting (RCV)* provides another setting in which the same voters face varying choice environments. A ranking can be read as a sequence of choices from shrinking choice sets: the ballot contains the same candidates throughout, but the relevant choice environment changes across ranking stages, since a candidate who has already been ranked is no longer available in the voter's subsequent choice set. As a result, lower-ranked choices can be especially informative for identifying voters' positions in the policy space. As with simultaneous elections, the identifying condition holds by construction.

RCV is used in national elections in Australia, Ireland, Malta, and Sri Lanka, among others, and has been adopted for state and federal elections in Maine and Alaska as well as for local races in several US cities.

Third, consider the case when *different* voters face *different choice environments across geo-*

graphic areas. Such variation is generated by design in most electoral systems, since geographically based representation is a standard feature: examples of first-past-the-post systems across districts include U.S. Senate and House of Representatives elections and House of Commons elections in Canada and the United Kingdom; examples of mixed systems, combining elements of first-past-the-post and proportional representation, include legislative elections in Germany, Hungary, Mexico, Japan, and South Korea, among many others.

In this setting, however, the identifying condition no longer holds by construction. The political economy literature posits that politicians’ and voters’ preferences are closely related, whether because of politicians’ opportunistic incentives, intrinsic preference congruence, or voters’ responsiveness to politicians’ cues (Persson and Tabellini, 2002; Osborne and Slivinski, 1996; Callander and Carbajal, 2022). A researcher should therefore justify the comparison of different choice environments. In practice, this can be addressed in several complementary ways: (1) by identifying contexts or subcontexts in which the condition holds directly; (2) by instrumenting candidate positions; or (3) by relaxing the equality condition and assessing the sensitivity of the results to such relaxation. I now discuss each approach with reference to papers that adopt it.

(1) Contexts where the condition holds directly. Longuet-Marx (2025) exploits congressional district borders within states, following the approach of Spenkuch and Toniatti (2018), and compares only contiguous precinct pairs, which lie in different electoral districts but are very similar otherwise. In the terminology of this paper, he posits that $f(x \mid y, m, d) = f(x \mid y', m, d)$ holds within each precinct pair.¹¹ This implies that the overlay of choice situations should be done only within the pairs of precincts, which limits the available identifying variation. However, by considering many such pairs, a researcher may still be able to recover more about the underlying primitives of interest than from single elections alone. I illustrate this with a simple example in Appendix C.

Merlo and De Paula (2017) and Danieli et al. (2023) treat national-level party positions across European countries as exogenous to voters’ preferences and estimate voter preferences by exploiting cross-country variation in party positions. While the former is closer to the main setup considered here and relies on aggregate voting data, Danieli et al. (2023) combine survey data with party positions. Nevertheless, the same logic of preference estimation applies in this setting as well.

(2) Instrumenting candidate positions. Iaryczower, Montero and Kim (2024) exploit variation in candidates’ positions across Brazilian municipalities to estimate voters’ preferences. Their empirical strategy relies on using the policy positions of mayoral and state-level candidates in the most recent local electoral cycle as an instrument z for current local party positions – the non-degenerate case of

¹¹In terms of Assumption 2, this amounts to letting the instrument z index the precinct pair: conditional on the pair, the two precincts share the same preference distribution while facing different candidate sets.

Assumption 2.

(3) Relaxing the equality condition. [Tebaldi, Torgovitsky and Yang \(2023\)](#) suggest a way to relax the equality assumption on preference distributions by allowing for pointwise differences in the density. In the voting context, one adaptation of their idea would be to impose $f(x \mid y, m, d) \in [(1 - \kappa)f(x \mid y', m, d), (1 + \kappa)f(x \mid y', m, d)]$ for all x and all y, y' , where κ is a researcher-chosen parameter governing the degree of difference across preference distributions in different regions. Setting $\kappa = 0$ recovers the baseline condition, and the identified set is weakly increasing in κ , so reporting bounds across several values of κ amounts to a sensitivity analysis with respect to the identifying assumption.

The model can also be extended to incorporate other important features of voting behavior, such as strategic voting. Under strategic voting, some individuals may abandon their most preferred candidate if a vote for that candidate is perceived as *wasted*. The key implication is that such behavior generates different effective candidate choice sets across voters. Those who vote for underdog candidates can plausibly be treated as sincere and therefore as facing the full choice set. By contrast, for voters who support viable contenders, it is unclear whether they are choosing their most preferred candidate overall or only their most preferred candidate among viable alternatives. For these voters, the relevant choice set should therefore be restricted to viable candidates. Although this coarsens the partition and may therefore widen the resulting bounds, it makes them more robust to strategic voting.

I now turn to describing an application of the model to one of the most widespread electoral regimes: the mixed electoral system.

4 Application: The Rise of Populism in Germany

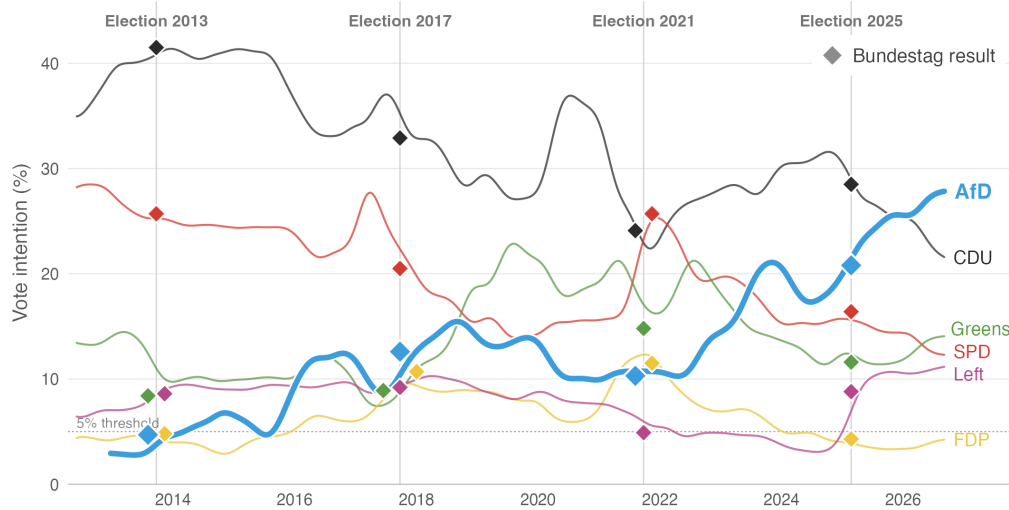
One of the main puzzles in the political economy literature is the rise of populism, and in particular radical-right populism, in Western democracies. The empirical literature has largely focused on identifying the causal mechanisms underlying the rise of populist politics, examining their effects on voters' preferences, and analyzing the contextual and individual-level factors that shape these preference responses (see [Guriev and Papaioannou \(2022\)](#) for an overview).

While understanding specific mechanisms is essential, it is equally important to develop a broader understanding of the preferences of populist voters, so as to be able to draw policy implications and inform competition strategies for the mainstream parties. One recent attempt to recover populist voters' preferences is [Danieli et al. \(2023\)](#), who study the rise of populist radical-right parties in Europe using survey evidence over time. To my knowledge, this paper is the first

to provide revealed-preference evidence on the preferences of populist voters at large. Focusing on revealed-preference data rather than surveys is valuable because it sidesteps concerns about survey misreporting, in particular the possible underreporting of populist vote choice, and makes it possible to recover the relative importance of different dimensions of political competition, which can be difficult to identify from survey data alone.¹²

4.1 Context

Over recent decades, the German electoral landscape has undergone important changes, most notably the rise of the Alternative for Germany (AfD), a populist radical-right party. For the first time since the 1950s, a party to the right of the Christian Democrats (CDU/CSU) surpassed the 5% threshold and entered the Bundestag. In the most recent federal election, in 2025, the AfD finished second, ahead of several parties of the traditional political establishment, including the Social Democratic Party (SPD), one of Germany’s two major parties. Figure 3 demonstrates the rise of the AfD over 2012–2026 and plots the responses to the polling question “Which party would you vote for if the Bundestag election were held next Sunday?”, alongside the results of the four Bundestag elections used in the paper (2013, 2017, 2021 and 2025).



Notes. Lines show the share of respondents intending to vote for each party in the question “Which party would you vote for if the Bundestag election were held next Sunday?”, smoothed across 3,435 national polls by eight polling institutes (Allensbach, Forsa, Forschungsgruppe Wahlen, GMS, Infratest dimap, INSA, Kantar/Emnid, YouGov), compiled from *wahlrecht.de*. The smoothed value on a given day is a Gaussian-weighted mean of all polls with a standard deviation of 60 days. Diamonds mark the official second-vote shares in the Bundestag elections of 2013, 2017, 2021 and 2025. The dotted line marks the five per cent threshold for parliamentary representation. CDU denotes CDU/CSU; Greens and Left denote Bündnis 90/Die Grünen and Die Linke.

Figure 3. Federal voting intentions, June 2012 – September 2026.

¹²Voters tend to adjust their reported positions towards those of their party, particularly when the issue is not especially important to them. When relative weights are estimated using survey data, such proximity in reported positions may then be read as evidence that the dimension carries greater weight for voters than it in fact does.

Alternative for Germany was established in 2012 to oppose the policies pursued by the German government to fight the euro crisis. From the beginning, the party was distinctive in its radical-right and anti-establishment stance. In the run-up to the 2013 Bundestag elections, its main concerns were the euro crisis and the Greek bailout, as well as adherence to the rule of law and stronger participatory democracy (AfD, 2013). In 2015, the party split; many founding members resigned, and the conservative, anti-immigrant wing took over the leadership, which it has held since.

Throughout the study period (2013–2025), the AfD was consistently ranked as the most right-wing and anti-elite among Germany’s main parties (Jolly et al., 2022), and it stood out as the most conservative and anti-elite party with a realistic chance of winning seats in the Bundestag. This motivates the choice of the two dimensions of policy space in my application – Left–Right and Pro–Elite–Anti-Elite (I refer to the latter simply as *Anti-Elite* throughout).¹³ I discuss what I mean by those dimensions below in the data section.

My analysis draws on data from German federal parliamentary (Bundestag) elections. Germany is divided into 299 electoral districts across 16 states (e.g. Bavaria, Saxony). Under Germany’s mixed electoral system, each voter casts two votes: a first vote for a direct candidate in a single-member district, and a second vote for a state-level party list. First votes are likely to be heavily shaped by the strategic incentives of first-past-the-post competition, whereas second votes among viable parties can more plausibly be treated as sincere (more on this below). Although the two votes are distinct, the candidates appearing on district ballots and on party lists largely overlap: around two-thirds of candidates on state lists also run in single-member districts, and the district candidates are on average higher-ranked on the lists (Appendix K.1 provides more detail). In what follows, I rely exclusively on second-vote party shares.

My analysis is based on the assumption that *candidates’ positions in single-member districts shape perceived party positions*. In other words, I use estimated candidate positions (see below) as proxies for parties’ positions in each district. This allows me to exploit cross-district variation in candidate positions while preserving the assumption of sincere voting for the party list. To support this assumption, I show that, within federal states (which I later treat as separate *markets*) and within parties, the estimated Left–Right position of a party’s district candidate predicts respondents’ Left–Right placement of that party (Table II). The relationship holds for the party a respondent votes for with the second vote and is strongest among AfD voters. It is absent for parties other than the chosen one, and the candidate’s position has no bearing on the respondent’s own Left–Right self-placement, which speaks against the correlation reflecting local sorting of voters rather than

¹³This aligns with the literature in political science that highlights that populist parties build their agenda on two main dimensions: ideological and anti-system (see Mudde (2004); Guriev and Papaioannou (2022), among others), and the companion paper Nikiforova (2026), which identifies Left–Right and Anti-Elite as the two central dimensions of voting in Germany in recent years.

candidate-driven perceptions. The results are robust to controlling for gender, age, and education. Additionally, survey data collected around the 2021 Bundestag election show that voters interact with and know their local candidates well (Figure I2).¹⁴ Both exercises use the 2021 Bundestag election, for which the richest survey data are available.

Two additional institutional features are important for the analysis. First, state-level party elites play a substantial role in candidate selection across electoral districts. Although German election law formally assigns nominations in single-member districts to local (district) party chapters, state party organizations act as *de facto* gatekeepers within the state. They can influence nominations both (i) by blocking district candidates they consider unsuitable and (ii) by determining the state party list for Bundestag elections. The latter matters because candidates may enter the Bundestag either through district races or through state lists; in practice, around two-thirds of candidates on state lists also run in single-member districts. Candidate selection is moreover often uncontested, as elites tend to preselect candidates in advance—particularly when incumbents are involved (Detterbeck, 2016; Schüttemeyer et al., 2024). These considerations both motivate the choice of states as separate markets and inform the identification argument. Appendix K.1 gives further descriptive evidence on the parties’ nomination practices, documenting persistence in nominations and the interplay between single-member districts and state lists.

Second, electoral districts are drawn primarily on the basis of population size. Unlike in the United States, redistricting in Germany is nonpartisan and district boundaries are largely stable. A permanent Constituency Commission reviews population changes each electoral cycle and recommends adjustments to the Bundestag. Constituencies should remain within 15% of the average population, must be redrawn if deviations exceed 25%, and must lie entirely within a single state. In practice, only a small share of constituencies changes in any given electoral period.¹⁵

4.2 Data

In the application that follows, a state ($N=16$) constitutes a market; markets are subdivided into electoral districts ($N=299$); and each electoral district contains multiple municipalities (submarkets, $N \approx 11,000$). Voting and demographic data are observed at the municipality level. Each electoral district features a distinct set of candidates. Figure I1 illustrates this division. I now describe the data used for estimation.

¹⁴In the political science literature, the influence of single-member district candidates on parties’ list-tier outcomes in mixed electoral systems is known as the *contamination effect* (Ferrara, Herron and Nishikawa, 2005). It is documented across mixed systems (Herron and Nishikawa, 2001; Cox and Schoppa, 2002) and, for Germany, by Klingemann and Wessels (2001), Hainmueller and Kern (2008) and Gschwend and Zittel (2015); this literature mostly examines the effects of incumbency, candidate evaluations and campaign effort. My assumption differs in that I focus on policy positions and assume that candidates shape voters’ perception of the party’s position.

¹⁵See, for example, Die Bundeswahlleiterin (n.d.) for changes made prior to the 2025 Bundestag elections.

The demographic group taken as the unit of analysis consists of an age–gender–education bracket in a given state, and the distribution of preferences is estimated separately for each group within each state in a given year. An example of a group is *female, aged 18–24, with higher education, in Saxony in 2021*. To proceed with the estimation I need data on age–gender–education composition and voting at the same level. I now turn to describing such data.

Municipal-level data. I collect data on turnout and second vote shares (party-list vote shares) for the six main parties at the level of municipalities¹⁶¹⁷: *AfD, CDU/CSU, FDP, Greens, Left, SPD*. I abstract from coalition considerations and treat those parties as *viable* in the years studied, which implies that if an individual truly prefers one of the considered parties, then there are no strategic considerations not to vote for this party.¹⁸ In turn, this implies that my estimates elicit true voting preferences among the voting population.

I also collect demographic data at a municipal level for each electoral year. I construct municipality-level information on the population composition of eligible voters in 24 age–gender–education brackets, comprising 6 age groups (18–24, 25–34, 35–44, 45–59, 60–69, 70+), two genders, and a binary distinction between individuals with and without higher education.¹⁹ These demographics are measured precisely for 2021, drawing on the 2022 Census. To construct demographics for 2013, 2017 and 2025, I use changes in the size of each age–gender group at the district level ($N \approx 400$) and project these changes onto municipalities within each district, using the 2021 municipal composition as the baseline.

I restrict the benchmark sample to municipalities with fewer than 10,000 inhabitants.²⁰ Cross-market variation in the demographic distribution is what identifies preferences separately by demographic group (Berry and Haile, 2021); smaller units retain more distinctive compositions, widening that variation. I test the sensitivity of my results to this threshold value below. Table I3 reports how voting patterns differ across municipalities below and above 10k. Small municipalities lean more strongly towards the AfD (+2.1pp) and the CDU/CSU (+4.4pp) at the expense of the other parties, but this partly reflects state and district composition effects: once those are accounted for, the gaps fall to +0.7 and +3.6pp.

¹⁶The cumulative party vote shares for the studied viable parties are 94%, 95%, 91% in years 2013, 2017 and 2021 respectively. This implies that the analysis excludes only a small group of voters who supported non-viable parties.

¹⁷The municipality is the smallest unit of analysis in Germany that can be geo-coded.

¹⁸It also assumes that voters expected these parties to clear the 5% threshold; the exceptions across years and parties are FDP/AfD (2013), Left (2021) at 4.7–4.9%, and FDP (2025) with 4.3%.

¹⁹To do so, I use data on the age–gender composition of the population at the municipality level and complement them with data on the age–gender–education composition, which are available only for larger municipalities (with more than 10,000 inhabitants), municipal associations (*Gemeindeverbände*), or higher-level administrative units (*Kreise*).

²⁰This restriction effectively captures almost exclusively rural municipalities, making a separate urban–rural distinction unnecessary in this context. It also naturally excludes the city-states (Berlin, Bremen, and Hamburg) from the analysis.

Candidate positions. A key feature of the considered setting is that voters face distinct party-affiliated candidates across single-member districts. To estimate candidates’ ideological positions, I combine two candidate surveys:

- 1) the GLES Candidate Study for the Bundestag elections 2013–2025, administered by the German Society for Electoral Research and the Leibniz Institute for the Social Sciences (GLES, 2023);
- 2) Wen Wählen (*Whom to Vote For?*), a nonpartisan platform that fields a standardized questionnaire to election candidates.

Both surveys elicit candidates’ own positions on a battery of policy issues, and both target candidates from parties with a high likelihood of clearing the 5% threshold. Consequently, candidates from all parties considered in my analysis were contacted for participation in each election year.

First, using the GLES Candidate Study, I construct a left–right score and an anti-elite score for each candidate from multiple survey items corresponding to each dimension (see Table K2 for the exact wording).²¹ I treat the left–right dimension as a bundle of economic, cultural and social policies, and construct it from questions spanning economic issues (e.g., whether the government should reduce income differences, or stay out of the economy), cultural issues (e.g., whether same-sex marriage should be banned, or whether women should be free to decide about abortion), and immigration (e.g., whether immigrants should adapt to German culture, or whether immigration benefits the German economy). In line with the literature, I treat the anti-elite dimension as the axis capturing antagonism toward elites and the claim that the people (rather than professional politicians or technocrats) should hold authority, as well as general distrust in democratic institutions (Mudde, 2004; Guriev and Papaioannou, 2022).²² To quantify a candidate’s position on this dimension, I use items such as whether democracy offers citizens enough opportunity to participate, whether there should be nationwide referendums, and whether citizens have lost trust in politicians. See Table K2 for the list of questions used from the GLES Candidate Study.

The same questions are used in each election year.²³ I standardize each item using z-scores and compute, for each candidate, the average across all available items within a given dimension to obtain a single index.²⁴

Participation in the candidate survey is voluntary, and the response rate among single-member

²¹I use the terms *anti-establishment* and *anti-elite* interchangeably.

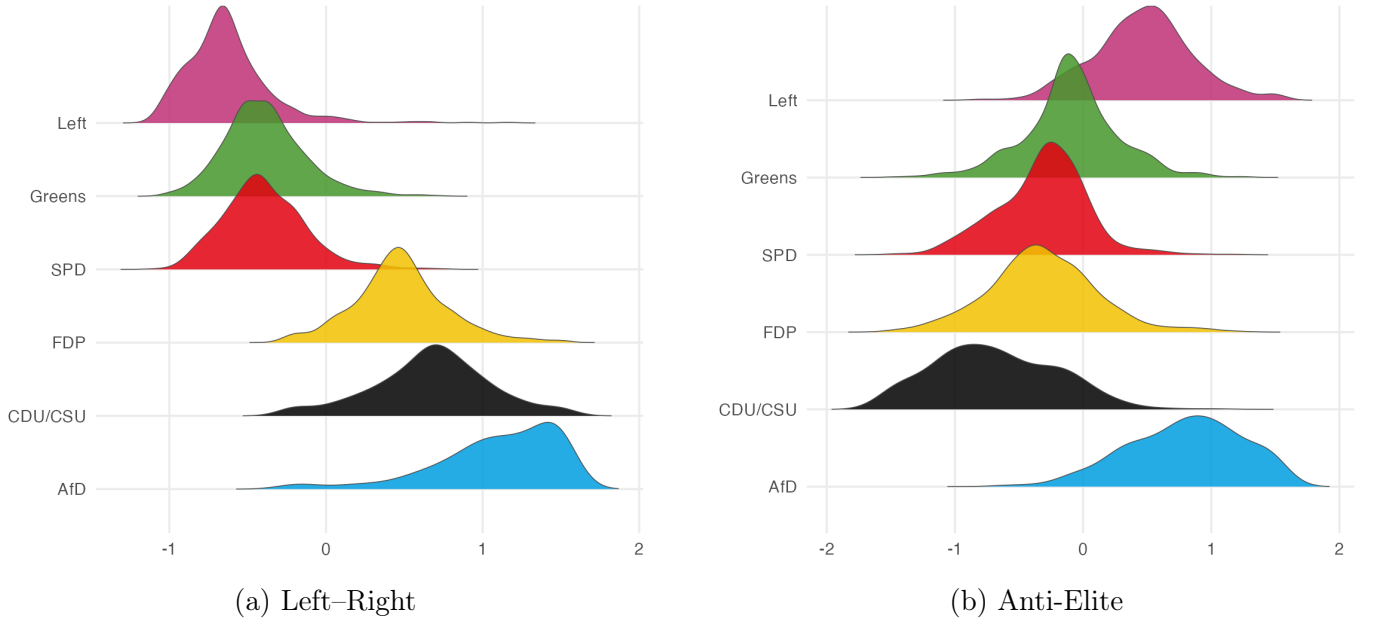
²²An important part of the AfD’s rhetoric is nationalism. While it could be attributed to either dimension, I treat it as belonging to the Left–Right dimension. As is clear from the imputation coefficients reported in Appendix K.2, the item most closely related to economic nationalism (“Reintroduce the Deutsche Mark!”) is a strong predictor of a candidate’s Left–Right placement in 2013.

²³Several questions are not available in some years.

²⁴I prefer this method for its transparency. The correlation with indices based on Item Response Theory exceeds 0.85 across all years and dimensions.

district candidates remains moderate – about 35% across the years. To improve data coverage, I supplement the GLES Candidate Study with a second source, *Wen Wählen* (Whom to Vote For?), a nonpartisan platform that collects standardised policy questionnaires from election candidates across all major parties. Because the two surveys use distinct question sets, I bridge them using an elastic net regression (Appendix K.2 provides details on this).

Figure 4 plots the distribution of candidate positions by party on the Left–Right and Anti-Elite dimensions, pooled across all four elections (2013–2025). Within each party candidate positions span a broad range on both dimensions, while the overall ordering of parties is preserved: the Left is furthest left and the AfD furthest right, with the Greens, SPD, FDP and CDU/CSU in between. On the Anti-Elite dimension, the Left and the AfD stand out as clearly more anti-elite, while the remaining parties sit on the pro-elite side. Figure K1 disaggregates these distributions by election year; Figure K2 plots the median values of candidate positions by party, dimension, and year.²⁵



Notes. Kernel density estimates of the positions of single-member district candidates, by party, pooling the 2013, 2017, 2021 and 2025 elections and all federal states. Positions are estimated from the GLES Candidate Survey and the non-partisan platform *Wen Wählen*. Each density uses a Gaussian kernel with the bandwidth selected by Scott’s rule, computed separately for each party and dimension.

Figure 4. Distribution of candidate positions by party, pooled across elections

Combining the two surveys yields comparable positions for about 60% of candidates across

²⁵External validation of such measures is difficult; one available benchmark is the within-party evolution of positions over time in the Chapel Hill Expert Survey (CHES) (Bakker et al., 2015). The within-party dynamics implied by my estimates are broadly consistent with CHES. In particular, the rightward and pro-elite shift of the CDU between 2021 and 2025, and the moderation of the AfD’s anti-elite positioning over the same period, are both mirrored in the change between the 2019 and 2024 CHES waves. Similarly, the hardening of AfD positions between 2017 and 2021 is consistent with the change between the 2014 and 2019 waves. Note that the CHES waves (2014, 2019, 2024) do not coincide with the German federal elections (2013, 2017, 2021, 2025), so the correspondence is approximate.

the years. Response rates differ by party: the CDU is lowest and the Greens highest (37% and 77% respectively), with about 50% of AfD candidates covered. Importantly, respondents and non-respondents obtain comparable first-vote shares and have similar winning probabilities within each party. Missing positions are imputed with the median position of the same party’s candidates in other districts of the same state; the national median is used only when no state-level information is available, which occurs for fewer than 2% of the candidates retained. For the estimation that follows I keep only districts with at most three imputed candidates, which leaves about 70% of districts across years, in which roughly 30% of candidate positions are imputed.

An important concern is that these two dimensions may be too few. While the framework in principle allows for estimation with more dimensions, in practice this is infeasible with the data available. Moreover, many candidate dimensions—nationalism, globalism, immigration—are highly correlated with the two dimensions considered, which makes them harder to use for preference estimation. I document the correlations between expert-estimated party positions across relevant policy dimensions in Europe (Figure K3), and show that in my data the Left–Right and Anti-Elite dimensions are essentially uncorrelated, whereas both correlate considerably with a measure of European Union attitudes (Figure K4). I discuss these issues, and how they could affect my results, in more detail in Section 10.

I also construct second-best party choices using survey data across federal states and across elections. Standard discrete choice models are prone to predicting consumers’ substitution behaviour (second choices) poorly, and second-best choices used as explicit target moments are known to be very helpful in fitting substitution patterns (Conlon and Gortmaker, 2023). While stated voting behaviour is biased, I believe that second-best party choices are less so. See the companion paper Nikiforova (2026) for a discussion. Figure I4 plots the evolution over time of the second-best party choices of those who indicate the AfD as their first choice, at the national level.

5 Estimation & Identification

5.1 Estimation

Appendix F describes how the empirical restrictions implied by the spatial voting model are translated into a finite-dimensional optimization problem, and how (i) the weights and (ii) the bounds on the target parameters are computed.

Importantly, when computing the target parameters I construct a single linear program for each part of the country (East/West) in a given year, introducing coupling of the preference distribution across states. This allows me not to confine the analysis to individual states but to explore richer

heterogeneity both within and across them, under a weak assumption on preference similarity between demographic groups across states. See Appendix F for more details.

Appendix F.3 discusses the fit of the model. The correlation between predicted and observed AfD vote shares exceeds 0.9 for 2017–2025 and is 0.59 in 2013 (Figures F1 and F3). Moreover, the correlations of predicted and observed municipal AfD shares with municipalities’ demographic characteristics are almost identical (Figure F2). This suggests that the model fits the data well and can be used for the preference estimation that follows.

5.2 Identification

The approach in this paper exploits variation in candidate positions across electoral districts within a state in a given year to estimate preferences of AfD voters. The identification assumption then posits that *in a given year variation in candidates’ positions across electoral districts within a state (market) m is unrelated to districts’ unobserved voter heterogeneity*, $f(x \mid y, m, d) = f(x \mid y', m, d)$. I first discuss the reasons why this assumption may be viewed as plausible, and then discuss how I test the validity of this assumption in later sections.

First, survey evidence suggests that age–gender–education brackets explain a large share of the variation in individuals’ self-identification among the voters of one party. To demonstrate this I focus on a more restrictive version of the model using the self-perceived Left–Right position from survey data, and estimate an OLS *mean-prediction* regression predicting the Left–Right position separately for each demographic group across states within parties. The R^2 of such regressions across federal states takes values between 0.3 and 0.85, with a mean of 0.519.

Second, as discussed in the data section, candidate nominations are largely controlled by regional party elites and influenced by state-list considerations, which limits local discretion in candidate selection. More generally, the political science literature highlights the relatively small pool of potential candidates and the constraints this places on candidate choice (see, e.g., Höhne (2023)). I provide more information on candidate nomination practices in Appendix K.1.

I further restrict the sample to municipalities with fewer than 10,000 inhabitants. Because the marginal electoral return to targeting a locality is low when its electorate is small, candidates have little incentive to tailor their platforms to these voters. Observed preferences in small municipalities are therefore less likely to reflect strategic candidate positioning than those in larger, more contested urban centers.

I also present two additional pieces of observational evidence that strengthen this assumption. First, across the municipalities used in the estimation, the estimated candidate positions of a given

party are largely uncorrelated with local socio-economic characteristics within a state.²⁶ Note, however, that at the district level positions do correlate with a district’s socio-economic characteristics within party and state, and with the expected signs. For example, in districts with higher unemployment or larger shares of benefit recipients, candidates are more left-leaning. Figure I5 reports the balance of socio-economic and demographic characteristics at the municipal level, and Figure I6 at the district level. Second, Table I2 demonstrates that candidates’ positions are uncorrelated with individuals’ self-reported views in surveys.

I assess the robustness of the identification assumption in four ways: (i) by restricting the analysis to small municipalities (fewer than 5,000 inhabitants), (ii) by excluding AfD candidates with stronger incentives to cater to local voters, (iii) by relaxing the exogeneity assumption and separating the districts into conservative and moderate (an approach similar to Tebaldi, Torgovitsky and Yang (2023)), (iv) by restricting the analysis to the municipalities that lie on the border of the electoral districts within the same state, and assuming that only the municipalities on the border share the same preference distribution. I discuss these approaches in detail when presenting the results.

6 Results: Voters’ Preferences

The main goal of this section is to provide new insights into the rise of populism using the revealed-preference evidence from the approach developed above. To do so, I first estimate voters’ preferences: the weight on the Anti-Elite dimension relative to the Left–Right dimension, w_{AE} , and the distribution of ideal points in a policy space defined by Left–Right and Anti-Elite dimensions separately for demographic groups d in markets m , $f(x \mid m, d)$. I next use those estimates for the counterfactual analysis.

Throughout the section I often present the results separately for eastern and western Germany. There are many reasons for this distinction, such as different histories and present-day institutional environments, as well as different levels of AfD support.²⁷ Crucially, the results from both parts of the country, analyzed separately, could inform our understanding of the rise of populism across countries with different institutional environments, such as those in eastern and western Europe.

²⁶Correlations remain significant for three variables after controlling for state-by-year and party-by-year fixed effects. Two of these are demographic characteristics, which I therefore account for in the estimation.

²⁷At the moment of writing this draft, the AfD tops the polls nationwide with almost 30% support and a 6 p.p. winning margin. The within-country heterogeneity is striking. In eastern Saxony-Anhalt, which holds elections in September 2026, the AfD is expected to win an outright majority, whereas in western Rhineland-Palatinate and Baden-Württemberg the party obtained less than 20% in March 2026.

6.1 Relative importance of dimensions

I begin by describing how the relative importance of the Left–Right and Anti-Elite dimensions has evolved over time. The weights come first because they determine the shape of the partition of the policy space used to evaluate the target parameters below.

Only relative weights can be recovered, and I normalize the weight on the Left–Right dimension to $w_{LR} = 1$ and search for the best-fitting value of w_{AE} . Precise estimation of weights requires especially rich data, as demonstrated in the simulations in Appendix B. To increase the observational richness of the analysis, I estimate the weights separately for East and West Germany, and I allow ideal-point distributions to vary across states – that is, each demographic group d in each market m (state) has its own bliss-point distribution $f(x \mid m, d)$, while I assume the same weight w_{AE} for all groups in a given year and region (East or West).

The constructed grid to search for the best weight includes relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios.²⁸ The best relative weight is then defined as the weight that yields the smallest model-fit error.

Figure 2 presents examples of the partition of the policy space for different weight configurations. It illustrates the intuition behind the weighting: a lower (higher) weight on the anti-elite dimension means that individuals tolerate a larger (smaller) distance along that dimension for a given distance along the left–right dimension between their bliss point and a candidate’s position.

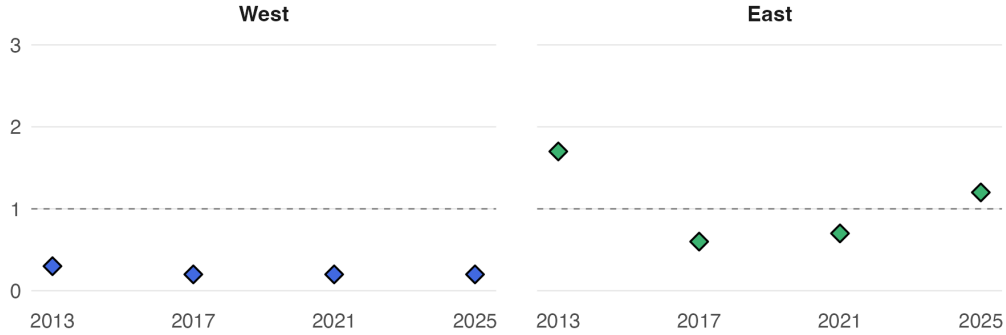
The model’s ability to yield the best fit in the context of changing partitions (i.e. changing relative weights) depends on the number of cells available under each setup, which, in turn, depends on the patterns of co-location of candidate positions on the anti-elite and left–right dimensions.²⁹ To alleviate this concern, I fix the number of available cells at 500, restricting attention to the AfD attraction region. Specifically, I retain the 500 largest cells and merge all remaining smaller cells into the larger cells with which they share the longest boundary. Additionally, Figure I7 shows that the correlation between left–right and anti-elite positions differs little between Eastern and Western Germany, both within parties (Figure I7a) and across parties (Figure I7b); and Table I5 shows that the standard deviations also do not differ much across years and parts of the country. This suggests that the relative positioning of parties on these two dimensions is unlikely to account for the regional

²⁸While it is possible to construct wider bounds, I do not do so for two reasons. First, wider bounds would amplify the measurement error in the estimated candidate positions. Second, the weights are identified only insofar as they meaningfully change the partition of the space, which for each pair of candidates depends on the weight and the ratio of their position differences, $\frac{|x_j^{LR} - x_k^{LR}|}{|x_j^{AE} - x_k^{AE}|}$. More than 80% of the values of this ratio across pairwise comparisons involving AfD candidates lie within $[0.2, 5]$.

²⁹Note that this argument is analogous to a standard result on variance decomposition in linear regression: for a fixed regressor X_1 , the improvement in R^2 from adding a second regressor X_2 is smaller the larger the correlation between the two regressors.

differences reported below and that the results are instead driven by demand-side differences in the relative importance attached to each dimension.

Figure 5 presents the results for the optimal weights. Points denote the point estimates corresponding to the best-fitting weight. Because the data are noisy, the optimization problem always yields a single best-fitting estimate. To reflect the fact that the weights are set-identified, I also report intervals that allow for discrepancies of up to 1% in the objective-function error (Figure J1).



Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (state) has its own bliss-point distribution $f(x | m, d)$, while imposing the same weight w_{AE} for all groups within a given year and region. To run this exercise, the grid of possible weights is constructed. The grid is symmetric around 1 and implies relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. See Figure J2 for full results over the grid. Points denote the point estimates corresponding to the best-fitting weight.

Figure 5. Weights for the Anti-Elite dimension w_{AE} relative to Left–Right $w_{LR} = 1$ across years and parts of the country

The results suggest that the Anti-Elite dimension carried more weight in Eastern Germany than in Western Germany throughout the period. In Western Germany, the Left–Right dimension remains consistently the more important of the two. In Eastern Germany, the Anti-Elite dimension carried a larger weight than the Left–Right dimension in 2013 (although this estimate is imprecise once a 1% difference in the objective-function error is allowed for, see Figure J1) and in 2025. In 2017 and 2021, the Left–Right dimension is relatively more important, though the weight on the Anti-Elite dimension remains sizable.³⁰

These results are in line with the differential role played by both dimensions across the two parts of the country. A broad literature documents the saliency with which Eastern Germans perceive themselves as neglected by federal elites, a perception grounded in measurable underrepresentation: as of 2024, Eastern Germans held only 12.1% of positional elite roles across twelve societal sectors, far below their population share of roughly 20% ([Bundesministerium für Wirtschaft und Energie](#),

³⁰Note that here and in what follows the evolution of preferences of AfD voters is a combination of at least two factors – (i) genuine within-person evolution, and (ii) compositional change.

2025). The Anti-Elite dimension is much more muted in the West, where this region-wide sense of elite underrepresentation is far less pronounced.

Figure J2 plots the objective-function value for different values of the weight on the second dimension.³¹ Figures J4 and J5 demonstrate the consistency of the estimates across subsamples by plotting the objective-function value on different subsamples of electoral districts across years and parts of the country. I interpret these results as evidence supporting the validity of the exercise.

In the following analysis I use the values for the estimated weights to define the partition of the policy space across the two regions unless stated otherwise.

6.2 Distribution of ideal points

The next preference object is the distribution of voters' ideal points $f(x|m, d)$. To provide evidence on the evolution of this object, I focus in this subsection on the location of AfD voters in the weighted policy space.

6.2.1 Voters' ideal points in fixed areas of policy space

The natural question to ask about voters' ideal points is *Where are voters' ideal points located in the policy space?* In particular, I ask this question in relation to AfD voters in 2013–2025. To study the bliss-point distribution of voters I then partition the policy space into equally spaced intervals with a step of 0.5 separately for the Anti-Elite and Left–Right dimensions, and ask what share of individuals lie in each of these intervals.

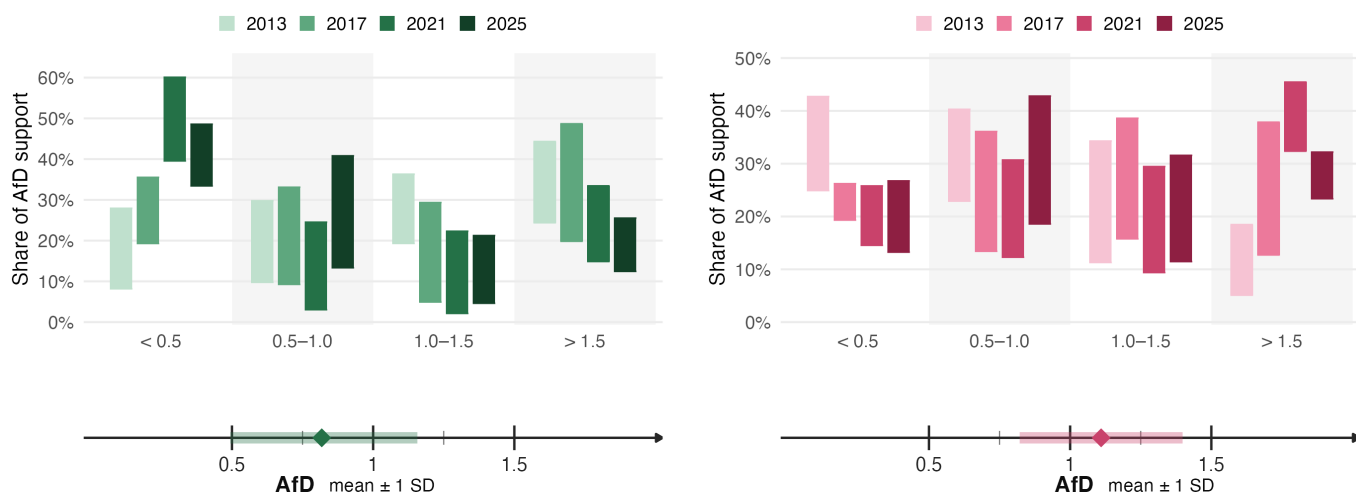
Let $f(x | m, d)$ be the density of ideal points among eligible voters in state m and demographic group d , and $F_{M,D}$ their joint (observed) distribution. The target parameter is the share of AfD voters in each interval. For example, the share of individuals with bliss points in $[0.5, 1]$ on the Anti-Elite dimension is:

$$\theta_{\text{AE}, [0.5 - 1]} = \iint_{X_{\theta_{\text{AE}, [0.5 - 1]}}} f(x | m, d) dx dF_{M,D}(m, d)$$

Because $f(x | m, d)$ is only partially identified from the municipal vote shares and demographic composition, I report bounds on $\theta_{\text{AE}, [0.5 - 1]}$ and other target parameters; note that the bounds reported here and below are sharp (see Tebaldi, Torgovitsky and Yang (2023) for details).³² Figure 6

³¹In the West in 2017–2025 the solution 0.2 is the corner value of the grid. Figure J3 shows that when the grid is extended below 0.2 (adding 0.1), a weight of 0.2 remains the best fitting weight in 2021 and 2025, but not in 2017. For the reasons discussed above, I nonetheless retain 0.2 as the best-fitting weight for all of these years.

³²Also note that the fixed masks are contingent, so the lower bounds are the most informative: they give the smallest possible share of voters located in the region (since the estimated objects are the *sharp* bounds on the target parameters). The upper bounds, by contrast, may reflect both the degree to which the problem is underidentified and the mass in the cells shared by contiguous fixed masks.



(a) Anti-Elite dimension

(b) Left-Right dimension

Notes: Each panel plots sharp bounds on the share of AfD voters whose bliss points lie in each 0.5-wide interval of one policy dimension (the other dimension is left unrestricted): panel (a) the Anti-Elite dimension, panel (b) the Left-Right dimension. Within each interval, bars run from 2013 (lightest) to 2025 (darkest). The line below each panel places the intervals on the dimension; the diamond marks the mean position of AfD district candidates pooled over 2013–2025 and the band one standard deviation around it. Bounds are estimated without the second-choice moments, separately for East and West Germany, and aggregated; adjacent intervals share mesh cells, so within each region and year the bounds are rescaled so that the interval midpoints sum to one.

Figure 6. Distribution of AfD supporters' bliss points along each dimension, 2013–2025

plots the results separately for the Anti-Elite and Left-Right dimensions.

The results suggest a recomposition of AfD support between 2013 and 2025. In 2013, AfD support came predominantly from voters with strongly anti-elite views (up to 42% of AfD supporters had a bliss point above 1.5 on the anti-elite dimension) rather than from the radical economic right (at least one fourth of AfD supporters had an economic bliss point below 0.5, and at most 19% lay beyond 1.5). Over the following decade this pattern reversed. The share of strongly anti-elite supporters fell to about one fifth by 2025, while the share with radical-right economic positions rose sharply. Notably, the party's own position moved far less than its electorate: the national AfD median remained between 0.96 and 1.30 on the economic dimension and between 0.67 and 0.97 on the anti-elite dimension throughout—placing the party well below its anti-elite fringe in 2013, and to the left of much of its economic-right support from 2021 onward. The AfD's electorate, in short, transformed around a comparatively stable platform.

These patterns constitute a first piece of evidence on the distribution of AfD voters' bliss points and its evolution, pointing to a changing composition of the party's electorate. While instructive and intuitive, this evidence does not speak directly to voters' motivations for choosing the AfD, nor to how establishment parties might win them back. I now turn to the exercises that resolve this.

6.2.2 Voters' ideal points and their motivations for AfD support

The AfD is distinctive in combining an ideologically radical-right stance with an anti-elite appeal, so a ballot cast for it is uninformative about why the voter chose it. Recovering that motivation matters, because it maps to the policy demands the vote expresses, and could help policymakers craft more appropriate responses. The analysis in this section resolves the ambiguity with a counterfactual: suppose the AfD split into two parties, one retaining its radical-right position on the Left–Right dimension but moving toward the elite, the other retaining its anti-elite appeal but moderating its right radicalism—what share of AfD voters would each attract? Equivalently, it measures the share of AfD voters whose ideal point in a weighted policy space leans anti-elite versus right, relative to where the AfD itself stands.

The estimated composition therefore has a behavioral interpretation: *a voter in the anti-elite region is one for whom protest against the elites, rather than Left–Right ideology, is the pivotal reason to vote AfD, and conversely for the right region.*

Let $\bar{Y} = (\bar{Y}_{LR}, \bar{Y}_{AE})$ be the nation-wide median AfD position. Displacing the AfD by δ along each dimension defines two hypothetical successor parties and partitions the AfD win region $X_{AfD}(\bar{Y})$ into an anti-elite region X_{A-E} (voters nearer the party that retains the anti-elite appeal; green) and a Right region X_{Right} (voters nearer the party that retains the same Left–Right position; pink), with $X_{AfD} = X_{A-E} \cup X_{Right}$.³³ Let $f(x | m, d)$ be the density of ideal points among eligible voters in state m and demographic group d , and $F_{M,D}$ their joint (observed) distribution. The target parameter is the share of AfD voters in each region:

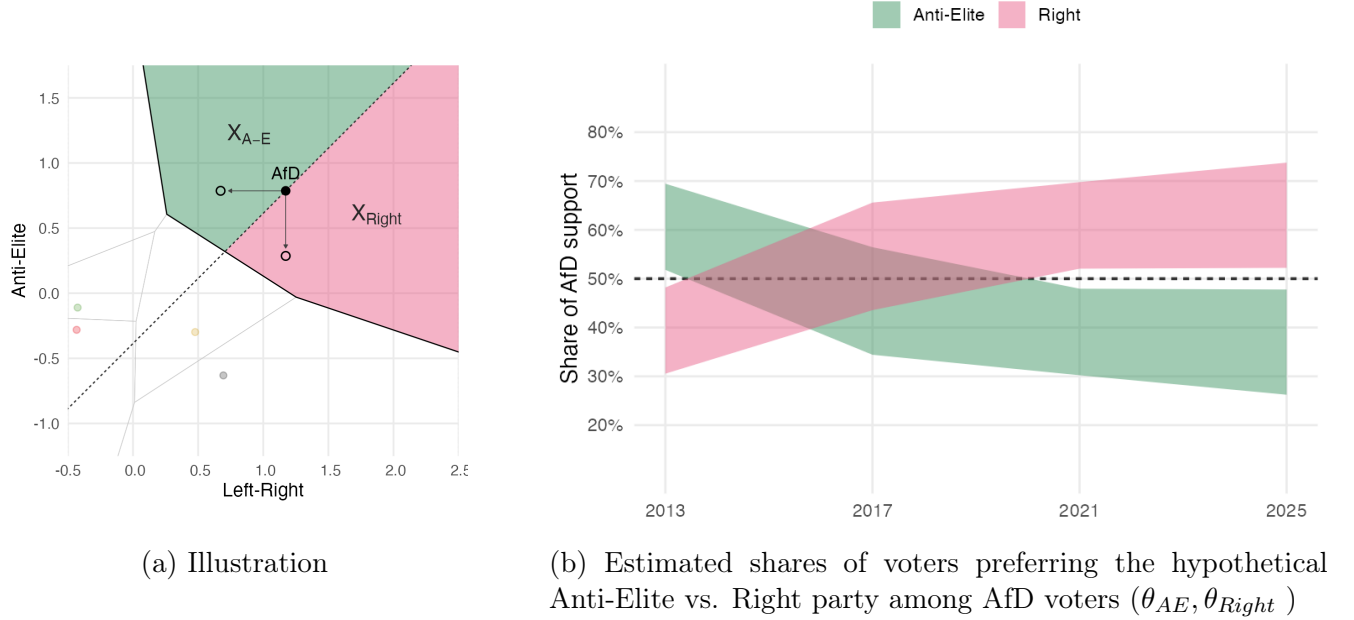
$$\theta_{AE} = \frac{\iint_{X_{A-E}} f(x | m, d) dx dF_{M,D}(m, d)}{\iint_{X_{AfD}} f(x | m, d) dx dF_{M,D}(m, d)}$$

I now report bounds on $\theta_{AE}, \theta_{Right}$. Note that by construction, $\theta_{Right} = 1 - \theta_{AE}$.

Figure 7b provides the bounds for the target parameters over years. The results indicate that AfD support has changed over time. In earlier years, AfD support was driven more by anti-elite motivations; this pattern flipped over time, and in 2021 and 2025 at least 52% of AfD support comes from individuals with stronger ideological Right motivations.

These aggregate results mask substantial regional and subgroup heterogeneity. Figure J9 reports the estimates separately for Eastern and Western Germany. The pattern for Western Germany

³³The shift size used in the estimations that follow is $\delta = 1$. Figure J8 presents the results for the other shifts for θ_{Right} . Results are quantitatively similar for different step values.



Notes: Panel (a) illustrates party positions at their national median, and hollow circles illustrate the counterfactual *AfD shifts*. The areas of AfD voters motivated more by anti-elite (green, X_{A-E}) or right (pink, X_{Right}) ideological stance of AfD are plotted. The illustrative weight used is $w_{AE} = 1$; the weights used in the estimation are those estimated in Section 6.1. Panel (b) plots bounds on the shares of voters whose bliss points lie in the green and pink regions across years. The results are aggregated across the two parts of the country. Panel (b) reports the estimates without the second-choice moments; Figure J10 reports the estimates with them.

Figure 7. Illustration of a target parameter (a) and results (b)

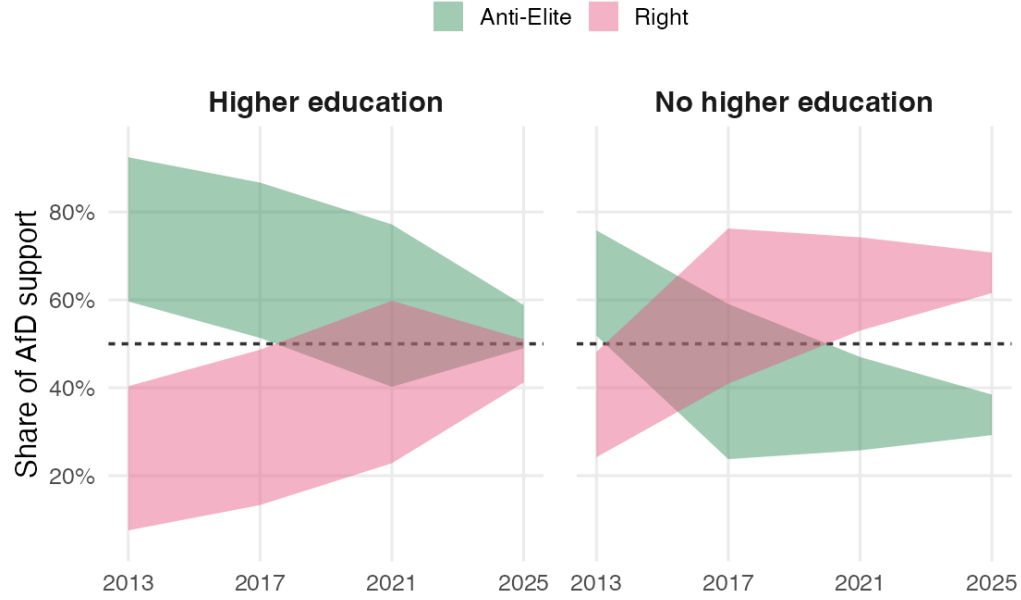
resembles the national-level pattern presented above; in turn, voters in Eastern Germany are more strongly driven by anti-elite attitudes than voters in Western Germany, especially in 2017.³⁴ The patterns are clearer and more precisely identified for Western Germany, reflecting the fact that it is larger in population and offers more electoral districts across which to exploit variation.

I now highlight two further subgroup-heterogeneity results among Western voters over time, as Western Germany allows rich heterogeneity patterns to be detected in the data. Figure 8 shows the shares of AfD voters with and without higher education belonging to the Anti-Elite and Right areas of Figure 7a (these target parameters are similar to those above, but also take into account the relative sizes of the groups in each year); Figure 9 provides the same analysis for young (under 35) and older (35 and over) individuals.

These figures are intended to highlight that individuals in different demographic groups have different rationales for supporting the same party, and that these rationales may evolve over time. In particular, for individuals without higher education the party's radical-right stance came to outweigh its anti-elite appeal only in later years. This could suggest that voters with lower educational at-

³⁴Except in 2013, when anti-elite sentiment dominated in the West. The party's anti-elite appeal then was arguably of a different nature—rooted in opposition to the Eurozone bailouts rather than in the hostility towards the establishment and democratic institutions that came to define the party after 2015.

tainment hold preferences that are more malleable, or more responsive to party messaging. Another important pattern is the rising share of young AfD voters whose preferences lie in the Right group. The growing support for radical (and in particular radical-right) political movements among young people has been documented in the political science literature, though the motivations behind these choices remain contested (van der Brug, Hobolt and Popa, 2026). The results of this paper suggest that young individuals may increasingly support these parties because of their radical-right ideology rather than their anti-elite stance.

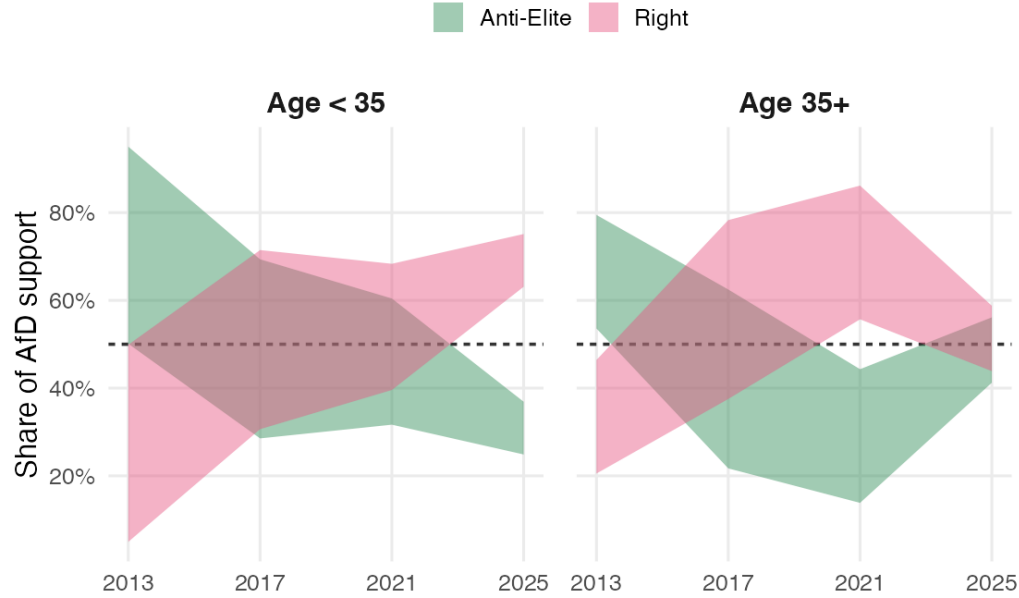


Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions across years are plotted. Results are for Western Germany only.

Figure 8. Estimated shares of AfD voters in both subspaces of policy space by educational attainment

A large literature in political economy relies on stated vote choices and stated issue preferences to estimate voters' preferences and their relationship to electoral behavior (Le Pennec and Pons, 2019; Gethin, Martínez-Toledano and Piketty, 2022), including in the context of support for populist politicians (Guiso et al., 2017; Algan et al., 2017; Danieli et al., 2023; Bartels, 2023). A natural question, then, is how the preferences inferred from survey data differ from those recovered using the revealed-preference approach developed in this paper. Appendix D presents this analysis. The main takeaway is that the shares of voters located in the Anti-Elite and Right regions display patterns similar to those obtained from survey data. Note that this comparison concerns bliss points only: the weights used in the exercise are those recovered from the revealed-preference framework, since they cannot be estimated from survey data (see Appendix D for more details).

The estimates above do not use the second-choice moments, which are not widely available in other settings. Re-estimating the analysis with them (Figure J10) yields slightly tighter bounds and



Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions in Figure 7a across years are plotted. Results are for Western Germany only.

Figure 9. Estimated shares of AfD voters in both subspaces of policy space by age

the same pattern, suggesting that the second-choice moments do not heavily restrict the distribution of voters' bliss points, and that the aggregate data alone provide enough identifying variation for this exercise.

The results presented in this subsection speak to the changing nature of AfD support over time, across regions, and across demographic subgroups. As I regard these as one of the paper's main contributions, I now verify that they are robust to relaxing the selection-on-observables assumption used to identify these patterns.

Robustness. I provide four robustness exercises that relax a selection-on-observables assumption on candidate positions. First, I run the main analysis on municipalities with fewer than 5,000 inhabitants (Figure J11). The idea is that parties are likely to target more populous, more electorally relevant constituencies; small municipalities therefore face a set of candidates pre-selected for larger constituencies. The national results remain qualitatively similar.

The second approach exploits the idea that politicians' incentives to cater to local voters vary with their electoral prospects, as competitive candidates are more likely to tailor their positions to constituency preferences. I restrict the analysis to districts in which AfD candidates were not competitive for victory. Specifically, I exclude every district in which the AfD candidate trailed the winning candidate by no more than 5 (10) percentage points in the Eastern (Western) part of the country. This removes 12 districts in 2017 and 28 in 2021 in the East, and 22 in the West in 2025.

The different thresholds reflect the party’s greater popularity in the East, where a uniform 10-point threshold would remove up to 80% of all districts in 2021. Moreover, for the East the analysis is restricted to 2017 and 2021: in 2013 there were no serious contenders, while in 2025 there were too many even under the 5-point threshold, as AfD candidates won almost every district. Figure J12 presents the results separately for Eastern and Western Germany. The results for the presented years remain quantitatively similar.

Third, I exploit an instrument similar in construction to the one in Tebaldi, Torgovitsky and Yang (2023), which directly relaxes the assumption of preference similarities across districts with different candidates. In particular, I allow that the mass of voters of a given demographic group g within a state in each partition cell between conservative and moderate districts could differ by at most a factor of $1 + \kappa$ for some fixed κ . I set $\kappa = 2$, so that masses may differ by no more than a factor of three in either direction. Figure J13 presents the results. Relaxing the assumption on preference similarity does not quantitatively change the results, but provides somewhat wider bounds.

Finally, I restrict the analysis to municipalities located on the boundaries between electoral districts within a federal state. This relaxes the maintained assumption that preferences are identical within a federal state: instead, I require the preference distribution of a given demographic group to be the same only across municipalities that lie on the same district boundary (the median number of such municipalities is seven). Across these border units within a state, preference distributions are allowed to differ up to a specified tolerance. All other assumptions remain as in the benchmark model. Figure J14 presents results for a within-state tolerance level of 0.2, which means that the preference distribution within the state across border pairs is allowed to differ by up to 20% in both directions from the state mean. The estimates are noticeably noisier, but the main qualitative results presented above remain similar.

The bounds reported above reflect partial identification only and do not account for any statistical uncertainty. Appendix F.4 quantifies sampling uncertainty by re-estimating the bounds on bootstrapped samples of municipalities.

The analysis so far has focused on the evolution of AfD voters’ preferences. The next subsections turn to counterfactual exercises, examining how changes in the positions of the (i) AfD and (ii) its center-right, pro-elite rival (the Christian Democratic Union, CDU) would affect the AfD vote share.

7 Spatial Competition Counterfactuals

7.1 What explains the support of AfD in competition with other parties: conservative positions or anti-elite stance?

The party does not exist in isolation, but competes with other parties for voters. Answering the question of which dimension matters more for voters requires assessing the sensitivity of voters' choices to counterfactual party positions.

To assess to what extent the AfD's success is driven by right-wing ideology or by anti-elite sentiment, I combine the estimated bliss-point distributions of AfD voters, the estimated relative weights, and the median positions of all the other parties, and ask: *How sensitive are AfD voters to the party shifting its positions along the Left-Right and Anti-Elite dimensions?* Put differently, the question is how many of the AfD's baseline supporters the party would lose if it adopted a more pro-elite or a more left-wing position.

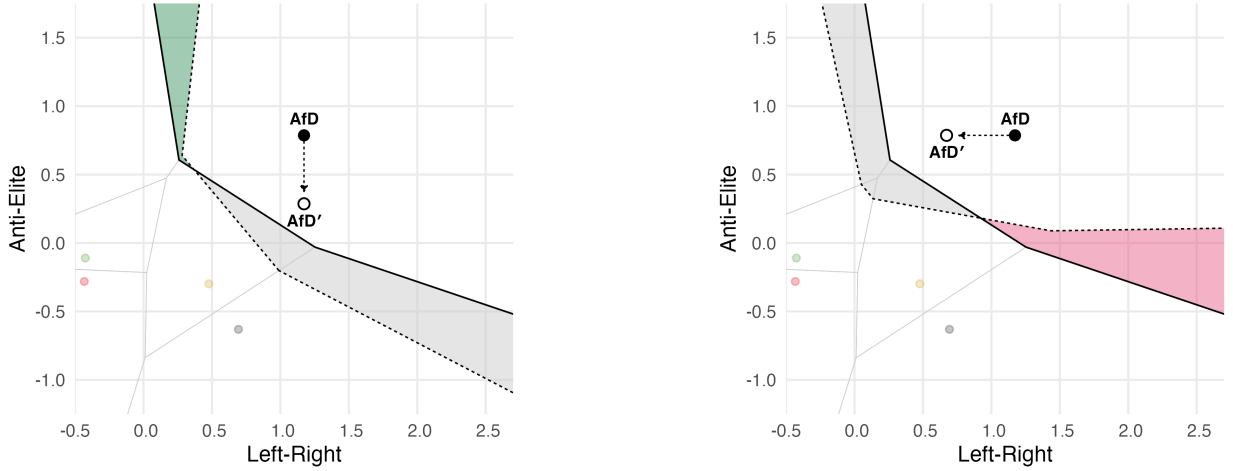
I hold the estimated preferences fixed and shift the AfD's position relative to its national-level median in each year, $\bar{Y}$, by 0.25/0.5/1 units along each dimension.³⁵ Figure 10 illustrates the shifts along both dimensions under median party positions across years and isotropic weights for clarity.

Before presenting the results, I provide a substantive interpretation of what a shift of 0.25, 0.5, or 1 unit actually means for a party. To do so, I trace back the scoring pipeline to determine what such a shift would require in terms of changes in candidates' stated opinions. Note that the units of the dimensions—and the scoring procedure as a whole—are identical across years, so these steps are comparable over time.

A candidate's position on each dimension is the average of her standardized answers to the relevant survey items, so moving a single question by one scale point shifts the dimension score by some σ , unique to each question. On the anti-elite dimension, the most influential single question—*Satisfaction with the way democracy works* (a 1–4 item)—moves the score by about 0.18 per scale point across years, and a typical five-point item such as *Legislation reflects citizens' interests* by about 0.14. A 0.25 step therefore corresponds to revising one such question by roughly 1.5–2 scale points (say, from *rather agree* to *strongly disagree*), or two questions by 1 point each. Moving every anti-elite question by one scale point would imply a strong repositioning of the party, and would change the score by about 0.95, so the 0.25/0.5/1 steps amount to softening (or hardening) a corresponding share of all anti-elite answers.

The Left-Right dimension behaves similarly: potent five-point items such as *Immigration is*

³⁵Positions are standardized by construction, so a given shift is comparable across dimensions.



(a) AfD shift of 0.5 towards a more pro-elite position (b) AfD shift of 0.5 towards a more left-wing position

Notes: The panels present an illustration of the unweighted cropped policy space, with partitions determined by median party positions across years. Shaded dots mark the median positions of parties other than the AfD. **AfD** denotes the AfD’s median position across years, and **AfD’** denotes the shifted AfD position—either towards the pro-elite position (Panel (a)) or towards a more left-wing position (Panel (b)). Solid lines mark the existing partition, dashed lines the counterfactual partition. The illustrated shift is 0.5 units along each dimension. Since positions are standardized by construction, a 0.5-unit shift is comparable across dimensions. The exercise estimates the share of voters the AfD would lose under the shift, i.e. those located in the green or pink areas in each panel.

Figure 10. Illustration of counterfactual moderating shifts by AfD by 0.5 units separately on two dimensions

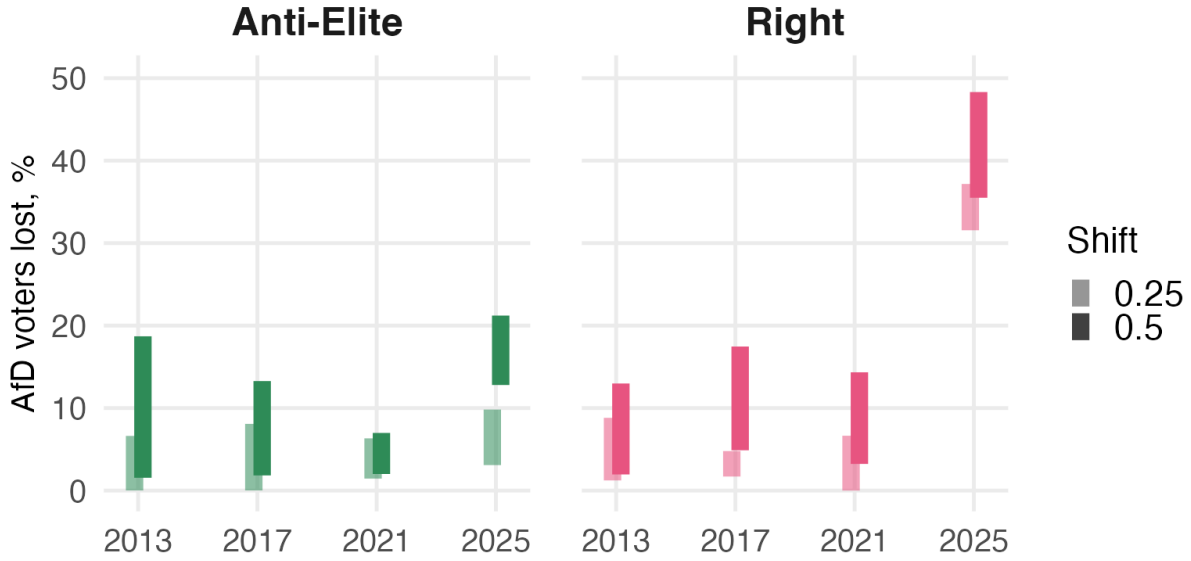
good for the economy or *Marriage only for heterosexual couples* move the score by about 0.10 per point, and moving all economic items by one point shifts the score by about 0.69. In short, a 0.25 step is a modest, one-to-two-question adjustment; a 0.5 step is a substantial repositioning that must touch many questions at once; and a full 1-unit step amounts to broadly shifting one’s stance across an entire dimension. Table I6 provides score sensitivities across questions for 2017.

The target parameter is the share of baseline AfD voters who would abandon the party had it shifted to **AfD’**.³⁶ Figure 11 plots the results separately for different step sizes and directions of the shift.

The results suggest that under small steps of 0.25 (0.5), the share of voters abandoning the AfD is no more than 10% (20%) across years and dimensions before 2025. Shifts of a full step of 1 yield much larger losses, especially along the Right dimension, reaching up to 60%.

The results also suggest that competition between the AfD and other parties became much

³⁶Precisely, $\theta_{\text{Loss, A-E}} = \frac{\iint_{X_{\text{Loss, A-E}}} f(x | m, d) dx dF_{M,D}(m, d)}{\iint_{X_{\text{AfD}}} f(x | m, d) dx dF_{M,D}(m, d)}$, $X_{\text{Loss, A-E}} = X_{\text{AfD}} \setminus X_{\text{AfD'}}$



Notes. Each bar is the identified interval for the share of *baseline* AfD voters who would abandon the party if it moderated by one step along the indicated dimension, holding all other parties at their observed positions. In the left panel (**Anti-Elite**) the AfD moves toward the elite along the anti-elite dimension; in the right panel (**Right**) the AfD moderates along the left–right dimension. Within each year, shading denotes the size of the moderation step (0.25/0.5 in standardized policy-space units; see the discussion just above for its substantive interpretation). National figures aggregate the East and West regional bounds. Partitions are built under the weighted (anisotropic) spatial metric with the estimated relative weight w_{AE} .

Figure 11. Estimated shares of AfD voters lost under counterfactual AfD shifts

tighter by 2025, in the sense that even small AfD shifts (particularly along the Left–Right dimension) would produce large defections of voters to other parties. This was not the case before 2025, indicating that the party had previously existed in relative isolation.

If competition with the establishment parties has indeed intensified, those parties should be able to capture a large mass of AfD voters by shifting their positions only slightly. I now turn to such an analysis.

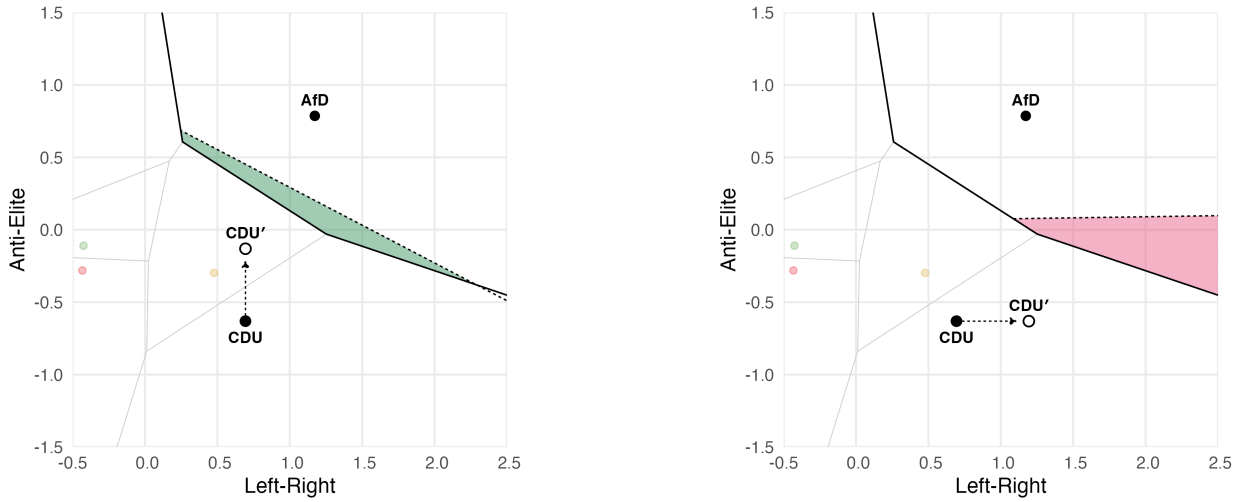
7.2 The response from the parties of the establishment

German establishment parties, which once commanded a large majority of the electorate, are now losing ground to the AfD, which leads the polls nationwide as of mid-2026.³⁷ The development is by no means unique to Germany. Across Western democracies, populist and radical parties have risen sharply over the past two decades, in many cases entering national legislatures and governments. Examples include Italy, France, the Netherlands, Spain and Sweden, among others. Beyond Europe, Donald Trump’s rise is widely regarded as part of the same populist wave.

³⁷As of July 2026, national polling averages place the AfD at roughly 28% and the CDU/CSU at roughly 22%. See Politico’s Poll of Polls, <https://www.politico.eu/europe-poll-of-polls/germany/> (accessed July 2026).

This raises a natural question: how should establishment parties compete with populist challengers? This section examines how effective it would be for them to adopt positions closer to the AfD's in the two-dimensional space defined by the Left–Right and Anti–Elite dimensions. The natural candidate for such analysis is the Christian Democratic Union (CDU), one of Germany's two largest parties and the main establishment party on the right of the Left–Right spectrum, on which I focus throughout this section.

In particular, I conduct an analysis similar to that in the previous subsection, but ask the reverse question: *if the CDU moves a given step towards the AfD along the Left–Right or Anti–Elite dimension, how many AfD voters would it capture?* Figure 12 illustrates these counterfactual scenarios.



(a) CDU shift of 0.5 towards a more anti-elite position

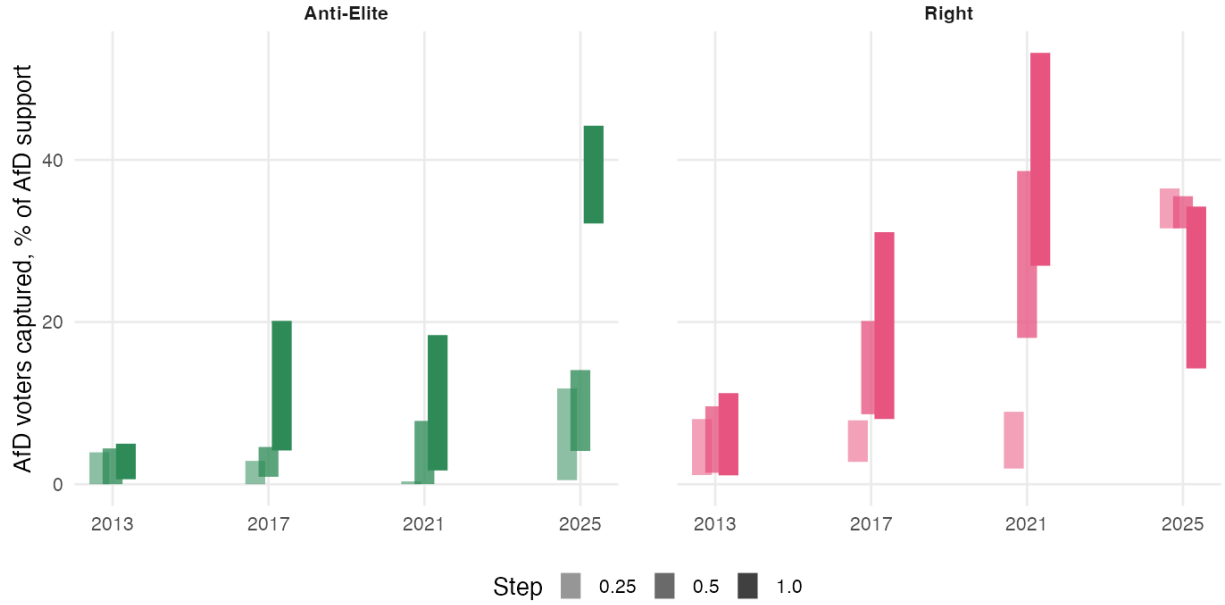
(b) CDU shift of 0.5 towards a more right-wing position

Notes: The panels present an illustration of the unweighted cropped policy space, with partitions determined by median party positions across years. Shaded dots mark the median positions of parties other than the AfD and CDU. **CDU** denotes the median CDU position across years, and **CDU'** denotes the shifted CDU position—either towards the anti-elite position (Panel (a)) or towards a more right-wing position (Panel (b)). Solid lines mark the existing partition, dashed lines the counterfactual partition. The illustrated shift is 0.5 units along each dimension. Since positions are standardized by construction, a 0.5-unit shift is comparable across dimensions. The exercise estimates the share of voters the AfD would lose under the shift, i.e. those located in the green or pink areas in each panel.

Figure 12. Illustration of counterfactual shifts by CDU by 0.5 units separately on two dimensions

Figure 13 plots the results separately for different step sizes and different directions of the shift. The results suggest that even a small shift (0.25) along the Left–Right dimension could allow the CDU to capture up to 10% of AfD support across 2013–2021; shifts of size 0.5 and 1 would yield larger gains. However, as discussed above, such larger steps may be too costly for a party with a relatively stable policy platform over the years, leaving the 0.25 step as the only realistic option.

Note that in 2025 a 0.25 shift in the CDU's position along the Left–Right dimension could allow



Notes. Each bar is the sharp identified interval for the share of *baseline* AfD voters who would abandon the party if the CDU moved by one step along the indicated dimension, holding all other parties at their observed positions. In the left panel (**Anti-Elite**) the CDU moves towards more anti-elite positions; in the right panel (**Right**) the CDU moves towards more right-wing positions. Within each year, shading denotes the size of the step (0.25/0.5/1.0 in standardized policy-space units; see the discussion just above for its substantive interpretation). National figures aggregate the East and West regional bounds. Partitions are built under the weighted spatial metric with the estimated relative weight w_{AE} .

Figure 13. Estimated shares of AfD voters lost under counterfactual CDU shifts

it to capture about 35% of the entire AfD electorate. While this exercise is not designed to solve the CDU’s vote-share maximization problem, it is clear that by 2025 35% of AfD voters constituted a large mass of the electorate, and the CDU would be unlikely to lose a comparable mass of voters on the other (left) side of its attraction area. The next section focuses on this in more detail.

Put into perspective, this result implies that if the CDU had slightly hardened its positions on immigration in the 2025 election, it could have captured about a third of the entire AfD electorate. Concretely, this would mean shifting from “Agree” to “Rather Agree” on “*Immigrants are good for the German economy*”, from “Agree” to “Strongly Agree” on “*Immigrants should adapt to German culture*”, and from 8 to 10 on “*Own position on immigration*”. A 0.25-unit shift to the right on the Left–Right dimension can be expressed in terms of economic policy items in the same way: moving from “Neither agree nor disagree” to “Disagree” on “*Providing social security is a core government task*” and “*The government should reduce income differences*”, from “Neither agree nor disagree” to “Agree” on “*The government should stay out of the economy*”, and from 9 to 10 on “*Own position on taxes vs. social spending*”.³⁸

³⁸All illustrative answers for the immigration and economic policy items are given relative to the median answers of CDU candidates in the 2025 survey. See Table 17 for the contribution of a one-point shift in each survey item.

These results are aligned with the experimental findings of [Guenther and Nunnari \(2025\)](#), who find that perceiving the CDU as one unit more liberal on immigration (on a 0–10 scale) raises the intention to vote for the AfD by an average of 5 percentage points – about 25% of the AfD vote. Putting the results on the same scale, my results suggest that a 2-unit CDU hardening on immigration on the 0–10 scale would capture about 35% of the AfD electorate. The results are also consistent with the documented defection of voters between parties in 2025: about 1.2 million CDU supporters in 2021 voted for the AfD in 2025, making this the largest such flow between any two parties in the country ([Vehrkamp and Posthofen, 2026](#)). The counterfactual is thus likely driven by the fact that the CDU could win these voters back with a slightly hardened Left–Right position.

This section was intended to provide some intuition for what the best strategy might be for establishment parties competing with populists. Notice that the party’s hardening of its anti-elite positions for realistic step values (0.25, 0.5) does not appear to be as effective in any year—in particular, not even in earlier years, when the AfD electorate was less pronouncedly right-wing. The party selected for this exercise was a large center-right establishment party, which could be affecting the results. Nonetheless, the finding that competition with populists could be effective along the Left–Right dimension at least for this party is a reassuring one, since competing on anti-elite rhetoric could, among other things, contribute to the erosion of democratic institutions.

8 Spatial competition between the establishment and the populists: equilibrium analysis

One of the puzzles of the rise of populism in Europe is why establishment parties do not adjust their platforms so as to hinder the entry and growth of populist rivals. Several answers are plausible: the establishment may anticipate that the populists would move in turn, it may fear losses on its other flank, it may be ideologically committed, or repositioning may be costly. This section takes a step towards resolving the puzzle by studying a spatial competition game between the CDU and the AfD in the 2025 Bundestag election, using the estimated distribution of voters’ ideal points as its primitive. The 2025 election is the natural case: as [Figure 13](#) shows, competition between the two parties had intensified, and a small rightward shift by the CDU would have captured about a fifth of the AfD’s electorate had the AfD stood still.

The game. All parties other than the CDU and the AfD are held at their observed 2025 positions, so the set of players is $N = \{\text{CDU}, \text{AfD}\}$. Each player i may shift its position Y_i by at most 0.2 standardised units on each dimension, in steps of 0.1. On Left–Right both parties may move in either direction. On Anti-Elite I only admit moves that are plausible for each party: the CDU may become

less anti-elite but not more, the AfD more anti-elite but not less. The pure-strategy sets are therefore

$$S_{\text{CDU}} = \{-0.2, \dots, 0.2\} \times \{-0.2, -0.1, 0\}, \quad S_{\text{AfD}} = \{-0.2, \dots, 0.2\} \times \{0, 0.1, 0.2\},$$

15 moves each, including the status quo $(0, 0)$. Given that the CDU and the AfD are more separated on the Anti-Elite dimension than they are on Left–Right, even the largest admissible moves keep the parties clearly apart on Anti-Elite, while on Left–Right their positions can overlap after both parties move towards each other. Because the policy space is two-dimensional, the vote consequences of such overlaps are not obvious: they depend on how the voters’ ideal points are distributed in the policy space.

The payoff of a profile $s = (s_{\text{CDU}}, s_{\text{AfD}})$ is the change in i ’s share of the electorate relative to the status quo,

$$u_i(s) = \iint_{\mathcal{X}_i(s)} f(x \mid m, d) dx dF_{M,D}(m, d) - \iint_{\mathcal{X}_i(0)} f(x \mid m, d) dx dF_{M,D}(m, d),$$

where $\mathcal{X}_i(s)$ is i ’s area of attraction under the counterfactual positions $(Y_{\text{CDU}} + s_{\text{CDU}}, Y_{\text{AfD}} + s_{\text{AfD}})$. Two features of this payoff should be kept in mind. First, it is computed over the 2025 electorate with participation held fixed: the exercise reallocates voters who turned out, and does not model the turnout margin, although a move could itself mobilise or demobilise voters. Second, it is a national payoff, the population-weighted sum of the East and West changes.

Parties are assumed to know the distribution of voters’ ideal points, which the researcher only knows up to the identified set. I therefore draw $K = 500$ distributions from the identified set and solve the game for each. For draw k the payoffs $u_i^{(k)}(s)$ define a normal-form game $G^{(k)}$, and I report how outcomes are distributed across draws.

Sequential moves. I study the sequential version of the game in which the CDU, L , moves first and the AfD, F , observes s_L before choosing s_F . The follower’s best-response correspondence is

$$B_F^{(k)}(s_L) = \arg \max_{s_F \in S_F} u_F^{(k)}(s_L, s_F),$$

and the leader solves

$$s_L^{*(k)} \in \arg \max_{s_L \in S_L} \min_{s_F \in B_F^{(k)}(s_L)} u_L^{(k)}(s_L, s_F),$$

the inner minimum resolving ties in the follower’s response pessimistically for the leader. A profile $s^{*(k)}$ with $s_L^{*(k)}$ solving the leader’s problem and $s_F^{*(k)} \in B_F^{(k)}(s_L^{*(k)})$ is a **Stackelberg equilibrium** of $G^{(k)}$. The sequential formulation matches the question being asked, namely whether the establishment

party should move towards its populist rival, and it sidesteps a well-known difficulty of simultaneous-move spatial competition: in more than one dimension a pure-strategy Nash equilibrium exists only under restrictive conditions on preferences (Osborne, 2025), and in the present game it does not exist.

When the AfD stands still. I first ask whether moving right pays for the CDU when the AfD does not respond. Panel (a) of Figure 14 reports the CDU’s median net change in vote share for each Left–Right move, its Anti-Elite position and the AfD held fixed. The answer is yes, and increasingly so: a 0.1 step gains 0.3 percentage points of the electorate, a 0.2 step 2.5 points (interquartile range 2.3 to 2.6), while steps to the left cost 0.8 and 2.5 points. Panel (a) of Figure 15 shows what lies behind the 0.2 step: the CDU captures the wedge of the AfD’s area that is closest to it on Left–Right (shaded) and loses a thin strip on its left flank. The wedge is large because, as Figure 6 showed, there are substantial masses of AfD voters who hold relatively pro-elite and radical-right positions. The CDU’s move brings it closer to those voters than the AfD is. If the AfD cannot respond, accommodation pays, and pays well.

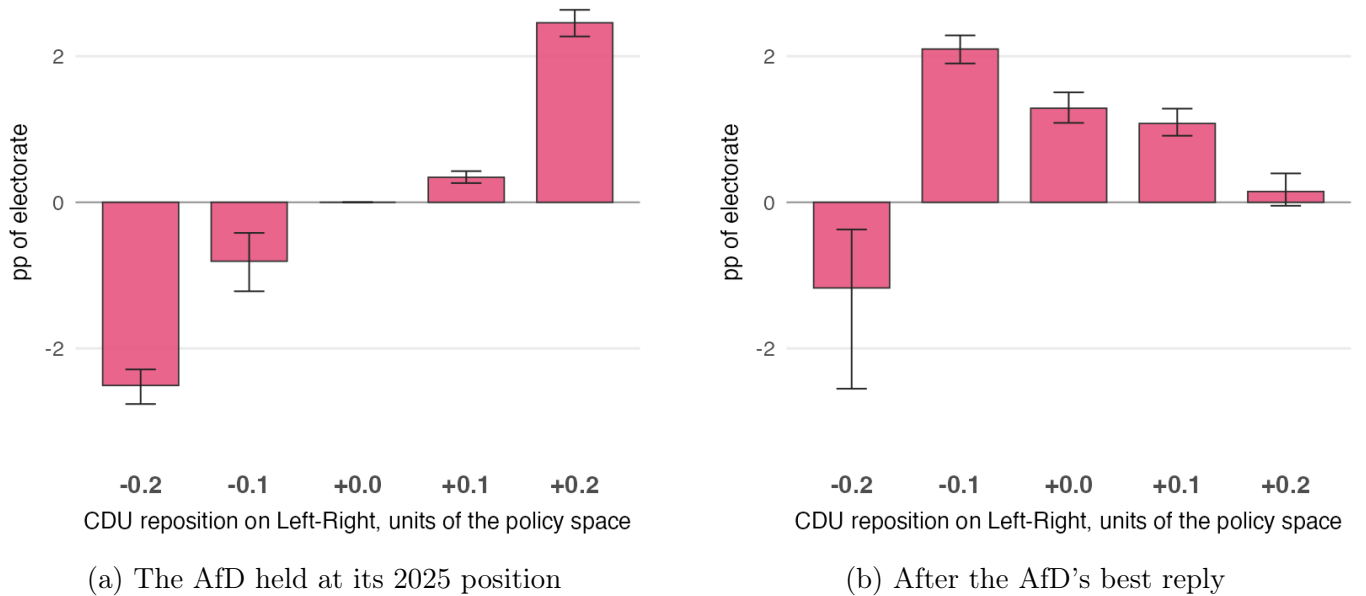
When the AfD responds. That conclusion rests on the AfD standing still while the CDU takes its voters. Panel (b) of Figure 15 shows the AfD’s best reply to the CDU’s 0.2 step: it moves one step further right. The far-right wedge the CDU had captured returns to the AfD (the purple area).

Panel (b) of Figure 14 reports the CDU’s net change after the AfD’s best reply, with the CDU pairing each Left–Right move with its best admissible Anti-Elite adjustment. The 0.2 step to the right now yields 0.15 points (interquartile range -0.05 to 0.4). The game is not zero-sum, since the other parties also gain and lose voters along the way, but the AfD’s reply is aimed mostly at the CDU. This is the populist party’s response to accommodation: when the establishment party moves towards it, it does not concede the contested voters but radicalises further, moving away from the challenger along the same dimension.

Once the AfD’s response is anticipated, the CDU’s best move is a small step to the left. The Stackelberg outcome is CDU $(-0.1, -0.2)$ followed by AfD $(-0.2, +0.1)$ in 56% of the draws; the CDU’s Left–Right component is -0.1 in 96% of draws and its Anti-Elite component is negative in 78%, while the AfD replies with $(-0.2, +0.1)$ in 89% of draws. The CDU’s equilibrium gain is 2.1 points (interquartile range 1.9 to 2.3), more than any move to the right yields once the AfD responds; the AfD gains 2.8 points. Under the equilibrium, the AfD follows the CDU left into the space it has vacated and at the same time sharpens its anti-elite profile, which costs it its least anti-elite voters, who go to the CDU (0.7 points). The larger part of the CDU’s gain (1.4 points) comes from the parties to its left, which do not move in this exercise. The leftward equilibrium is therefore partly an artefact of holding the other parties fixed: competition on the CDU’s left flank is one-sided here,

whereas competition on its right flank is a genuine two-player game. What is not an artefact is the sign of the answer to the question this section asks: with the AfD able to respond, moving right is not the CDU's best move.

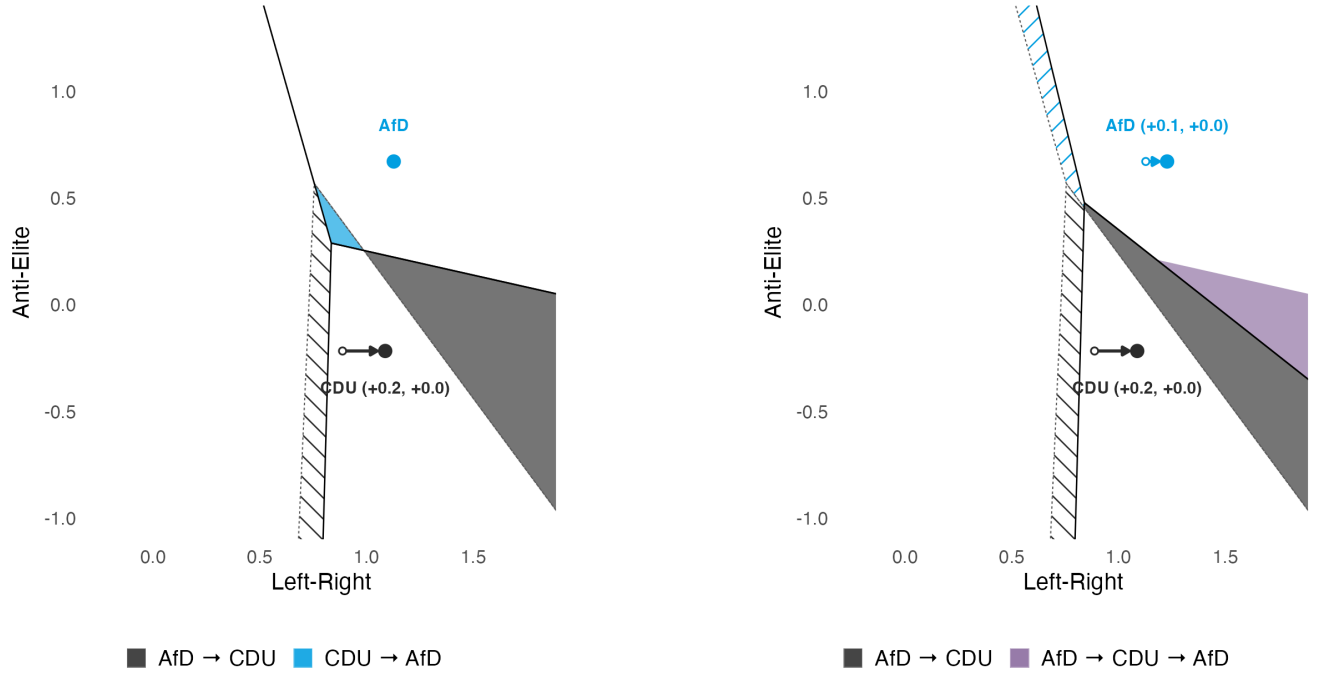
The Anti-Elite dimension is essential to these results. The CDU and the AfD compete for right-wing voters who differ in how much the AfD's anti-elite message appeals to them, and both parties use the anti-elite lever: the AfD escapes the CDU's rightward push by moving further right and more anti-elite, and the CDU's best move combines a small step left with a step towards the establishment pole. In a version of the game in which both parties can move only on Left–Right, the AfD still answers a 0.2 step by the CDU with a step right in every draw, but it lets a 0.1 step go unanswered in 87% of draws, and that 0.1 step becomes the CDU's best move (82% of draws), worth a net gain of only 0.3 points. The stronger prediction of the two-dimensional game, that the CDU should step left and away from the anti-elite pole, follows from the AfD being able to use the second dimension in reply.



Notes. Bars show the median CDU net change across 500 draws from the identified set, in percentage points of the electorate; whiskers show interquartile ranges. Payoffs are population-weighted sums of the East and West changes. In panel (a) both parties keep their 2025 Anti-Elite positions and the AfD does not move. In panel (b) the CDU pairs each Left–Right move with its best admissible Anti-Elite adjustment and the AfD responds on both dimensions; the AfD's ties are resolved pessimistically for the CDU.

Figure 14. The CDU's net change in vote share by Left–Right move, 2025

The main lesson is that profitable rightward deviations for the CDU do exist under the AfD's status-quo position, but the AfD can respond, and it is precisely this response that keeps the CDU from moving right in equilibrium. Accommodation can work when the challenger cannot respond;



(a) The CDU moves $(+0.2, 0)$, the AfD stands still

(b) The AfD replies $(+0.1, 0)$

Notes. Each panel shows the (Left–Right, Anti–Elite) policy space with the CDU (dark grey) and the AfD (blue) at their 2025 national medians and the other parties fixed; open circles and arrows mark the moves. Dashed grey lines are the boundaries of the two parties’ areas of attraction at the status quo, solid black lines the boundaries after the moves. Shaded areas are voters who change between the two players: dark grey, from the AfD to the CDU; blue, from the CDU to the AfD; purple, won back by the AfD after having been captured by the CDU in panel (a). Hatching marks areas lost to the other parties. Partitions are drawn with the estimated West weight $w_{AE} = 0.2$; the payoffs in the text are at a national level.

Figure 15. Areas of attraction under the counterfactual moves, 2025

the AfD can, and its response is to radicalise further, which leaves the establishment party no better off and pulls the policy debate to the right. Considering the populists’ reaction is thus essential when assessing the establishment’s options, a channel long emphasised in the theoretical literature on party competition and entry (Palfrey, 1984) but rarely quantified.

9 Coalition-building and the firewall against cooperation with the far right

The previous section asked how mainstream parties could foreclose the rise of populism by repositioning in a two-dimensional policy space. This section studies the welfare consequences of a different measure the establishment parties could use in their electoral competition with populists: the firewall against cooperation with them. Since the AfD entered the Bundestag in 2017 there has

been an unwritten rule of no cooperation with it³⁹; comparable rules were enacted in France, Sweden and Belgium, and in most cases abandoned once the radical-right party became pivotal for some governing majority.

Until 2025 the German rule was largely non-binding at the national level: the AfD’s vote share was modest and its parliamentary group small. In the 2025 elections it outpolled the SPD (historically one of the two largest establishment parties) in the party-list vote, and, absent the firewall, would have been a plausible coalition partner for the CDU (i.e. CDU/CSU).

Among other things, critics of the firewall policy argue that it leaves a large share of the electorate unrepresented, and this section tests that argument via coalition formation. Given the vote shares, only two coalitions were feasible: the centrist CDU–SPD and the right-wing CDU–AfD. Taking the estimated 2025 voter preferences as given, I ask (i) which coalition maximizes social welfare and (ii) which coalition is preferred by a majority of voters.

Figure 16a illustrates the logic behind the coalition analysis. The policy implemented by the coalition is assumed to be the midpoint of its members’ positions – blue and red diamonds for the CDU+AfD and CDU+SPD coalitions, respectively. As before, the analysis uses cells as the unit of analysis; some cells are unambiguously closer to one coalition policy than to the other, while some cells are divided (as illustrated in the figure). I evaluate the coalitions using two different metrics – social welfare and majoritarian choice.

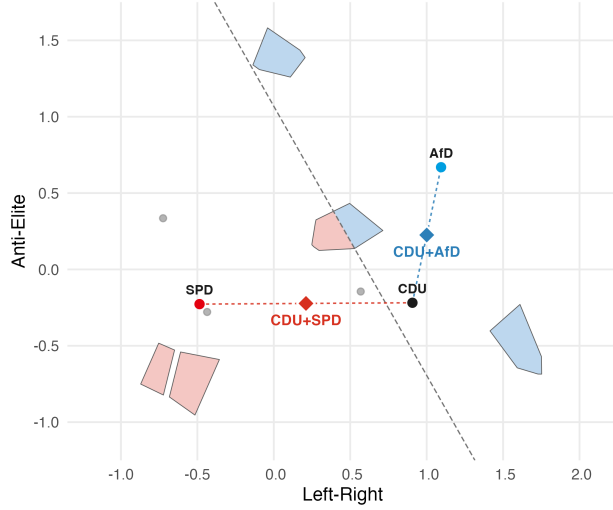
Social welfare from coalitions. Let $Y_{\text{CDU+AfD}}$ and $Y_{\text{CDU+SPD}}$ denote the two coalitions’ implemented policies; and as before let $f(x \mid m, d)$ be the density of ideal points among eligible voters in state m and demographic group d , and $F_{M,D}$ their joint distribution. The target parameter is the difference of weighted Euclidean distances $d_W(\cdot, \cdot)$ from voters’ bliss points to the two coalitions’ policies:⁴⁰

$$\theta_{\text{Welfare}} = \iint_{\mathcal{R}} \left[d_W(x, Y_{\text{CDU+AfD}}) - d_W(x, Y_{\text{CDU+SPD}}) \right] f(x \mid m, d) dx dF_{M,D}(m, d).$$

The bounds for θ_{Welfare} are in the upper part of Figure 16(b), where negative values mean that the CDU+AfD policy is closer to voters on average than the CDU+SPD policy. The estimated

³⁹The rule is informal at the federal level, but the CDU codified it in its 2018 incompatibility resolution (*Unvereinbarkeitsbeschluss*). At the municipal level compliance is high but not universal: [Schroeder, Ziblatt and Bochert \(2025\)](#) find no cooperation in about 81% of district-council votes in which cooperation was feasible between 2019 and 2024.

⁴⁰As before, I report sharp bounds: for each cell c of the partition I compute the range $[D_{\min}(c), D_{\max}(c)]$ of the difference $D(x) = d_W(x, Y_{\text{CDU+AfD}}) - d_W(x, Y_{\text{CDU+SPD}})$ over $x \in c$, and the lower (upper) bound on θ_{Welfare} is the minimum (maximum) of the cell-mass-weighted sum of $D_{\min}(c)$ ($D_{\max}(c)$) over the identified set of the main linear program. These bounds are sharp: they impose a common preference distribution for both coalitions and, within each cell, a common bliss-point location for both distances.



<i>Welfare difference, $CDU+AfD - CDU+SPD$</i>	
National	$[-0.18, 0.03]$
East	$[-0.29, -0.10]$
West	$[-0.13, 0.08]$
<i>Share of the electorate closer to $CDU+AfD$</i>	
National	$[61.9\%, 72.8\%]$
East	$[69.9\%, 83.8\%]$
West	$[58.5\%, 68.1\%]$

(b) Bounds on the target parameter values, θ_{Welfare} and θ_{Majority}

(a) Illustration: Coalition policy positions and random cells of voters' locations

Notes: Illustration of the coalition-welfare exercise (Panel (a)). Dots mark the 2025 national median positions of the parties. Each coalition is assumed to implement the midpoint of its members' positions (diamonds): CDU+AfD in blue, CDU+SPD in red. The dashed grey line is the set of voters indifferent between the two coalition policies. The polygons are illustrative cells of the preference partition: a cell is shaded blue (red) when every point in it is closer to the CDU+AfD (CDU+SPD) policy; the cell crossed by the indifference line is split, with each part shaded by the closer policy — within-cell locations are not identified, so such cells can only be bounded. Panel (b) reports bounds over the identified set of preference distributions. Distances are in the region-specific w_2 -weighted metric; no assumption is made on the location of mass within a cell. The welfare-difference rows bound $\mathbb{E}[d(\theta, p_{\text{CDU+AfD}}) - d(\theta, p_{\text{CDU+SPD}})]$ directly in one program per bound (sharp); negative values mean the CDU+AfD policy is closer on average. The majority rows bound the mass of cells lying entirely closer to the CDU+AfD policy (lower) and with any point closer (upper); they are invariant to any increasing transformation of the distance. National bounds combine East and West by their electorate weights (0.30/0.70); the regional programs are independent, so the combination is sharp.

Figure 16. Welfare difference and majority preference between the two 2025 coalitions

bounds at the national level straddle zero, so under this welfare rule neither coalition can be ranked unambiguously, although the bounds lean negative. This negative lean is driven by Eastern Germany, where the difference is bounded strictly below zero; in the West the bounds are roughly centered around zero, favoring neither coalition.

Note that this welfare estimation is not invariant to the distance metric; Table J1 provides the same welfare analysis for the squared Euclidean distance.

Majoritarian choice of the coalition. Another way to estimate which coalition is preferred by society is via the share of the electorate strictly closer to one coalition policy than to the other. To

estimate this, I formulate the target parameter as the share of voters preferring the CDU+AfD policy,

$$\theta_{\text{Majority}} = \iint_{\mathcal{R}} \mathbb{1}\left\{d_W(x, Y_{\text{CDU+AfD}}) < d_W(x, Y_{\text{CDU+SPD}})\right\} f(x \mid m, d) dx dF_{M,D}(m, d).$$

For θ_{Majority} , the lower bound counts the mass of voters lying in the cells c with $D_{\max}(c) < 0$ (entirely closer to CDU+AfD, as in the colour coding in Figure 16a) and the upper bound the mass of cells with $D_{\min}(c) \leq 0$.

The results are in the lower part of Figure 16(b). For every preference distribution in the identified set, a majority of voters in both parts of the country is closer to the CDU+AfD policy: at least 69.9% in the East and at least 58.5% in the West. Hence the CDU+AfD coalition would win a pairwise vote against the CDU+SPD coalition. The underrepresentation argument is therefore nuanced and depends on the decision rule one has in mind when making such claims. This analysis nevertheless highlights once again the regional differences: in the worst scenario only 16.2% of voters in Eastern Germany actually prefer the coalition with the SPD over the one with the AfD; this highlights that coalition building under the firewall policy could create an additional feeling of exclusion and underrepresentation among Eastern Germans in particular.

The contrast between the two measures of coalition preferences highlights an essential fact about the present-day political landscape in Germany: many voters on the left of the political spectrum hold very strong preferences against a coalition of the center with the AfD — strong enough to keep the welfare comparison ambiguous even though a clear majority sides with CDU+AfD.⁴¹

Figure J15 provides the same analyses as above, but under the assumption that the implemented coalition policy is the vote-share-weighted average of the coalition parties' own positions.

10 Discussion

The framework developed in this paper rests on a number of choices about the model, the ideological dimensions, and the identifying assumptions. I now discuss possible extensions of the framework, as well as how it compares to alternative modelling choices.

Probabilistic Voting Model. The model considered in the main text is Downsian: voters care only about the policy platforms of the candidates. Probabilistic voting models augment this framework

⁴¹The results broadly align with survey data, both in terms of regional heterogeneity and in terms of the strong dislike of a CDU–AfD coalition among some groups of voters. Figure J16 plots the survivor functions of the coalition desirability ratings in the West and the East. In both regions many respondents place the CDU–AfD coalition at the very bottom of the scale (47% in the West, 36% in the East, against 21% and 26% for CDU–SPD). But in the West the CDU–SPD curve dominates over most of the scale, whereas in the East the CDU–AfD curve lies above it at every positive threshold.

with another component, interpreted as an individual-specific taste for a candidate (Persson and Tabellini, 2002). I now show that the estimated model can be viewed as a probabilistic voting model with particular restrictions on the distribution of the error term.

The estimated model is a model with *random parameters*: the object of estimation is the distribution of a parameter — here, the bliss point. Let X_i be the average bliss point of type i , ϵ_i an additive shock defining the deviation from it, and Y_j the position of candidate j :

$$U_{ij}^{RPM} = -\text{dist}(X_i + \epsilon_i, Y_j).$$

The probabilistic voting model is instead a *random utility* model,

$$U_{ij}^{RUM} = -\text{dist}(X_i, Y_j) + \bar{\epsilon}_{ij},$$

with $\bar{\epsilon}_{ij}$ usually assumed to be i.i.d. across candidates and independent of X_i . For ease of exposition here and below in this section I assume the quadratic distance function and omit subscripts. Expanding the utility,

$$U_j^{RPM} = -(X + \epsilon - Y_j)^2 = -(X - Y_j)^2 + \underbrace{2\epsilon(Y_j - X) - \epsilon^2}_{\nu_j}.$$

The random-parameter model is therefore itself a random utility model, and a probabilistic voting model, $U_j^{RPM} = -(X - Y_j)^2 + \nu_j$, in which the distribution of the error is induced by the distribution of the bliss-point shock rather than assumed directly. The two models differ only in the restrictions placed on the joint distribution of the errors.

I assume the bliss-point shock is independent of both voter type and candidate positions, $\epsilon_i \perp (X_i, Y)$. Independence from X_i requires that the distribution of deviations does not vary systematically with the type's position; independence from Y rules out candidates positioning in response to the realisation of the shock. The former just states that given the type, the deviations are pure noise. The latter is the stronger identification restriction, discussed above: it rules out that, within a type, deviations are shifted towards the candidates' positions.⁴²

Note that this treatment of the framework also clarifies the consequences of potential mis-

⁴²Two properties of ν_j follow from ϵ_i being a single voter-level shock rather than a voter-candidate-level shock. First, the term ϵ^2 is common to all candidates and cancels from every utility comparison, so effectively $\nu_j = 2\epsilon(Y_j - X)$ up to a candidate-invariant constant. Second, because all ν_j are driven by the same draw of ϵ , the errors are correlated across candidates, the more strongly the closer the candidates' positions in the policy space. Consequently, whenever the number of candidates exceeds the dimension of the policy space plus one, no distribution of the bliss-point shock reproduces an i.i.d. error vector: the induced errors are confined to a lower-dimensional subspace. Exact equivalence obtains, e.g., in the case of two candidates.

specifications, such as dropping the probabilistic voting term, omitting important dimensions, or measurement error in the candidate positions. I briefly discuss the effect of the first below, as it belongs to the probabilistic voting model, and turn to the others in the following paragraphs.

If there is a non-spatial *taste component* Z_{ij} orthogonal to both X_i and Y_j , dropping it does not bias the recovered bliss points, as orthogonality rules out any systematic tilt towards particular positions. It could nonetheless inflate the dispersion of the estimated distributions for the bliss points, as choices driven by *taste* are rationalised *spatially*. It could also widen the identified sets for weights, as part of the choice variation becomes uninformative about weights, flattening the criterion around its minimum; the relative weights remain unbiased.

More than two dimensions. The framework restricts attention to two dimensions, and one could argue that others matter—nationalism, anti-globalism, or attitudes towards the European Union. In principle the framework accommodates more dimensions; in practice this is infeasible with the data at hand. Moreover, the candidate surveys contain no items suitable for estimating positions on nationalism or anti-globalism, though several allow me to construct an EU dimension.

I hence estimate the model with the alternative plausible pair of dimensions – Left-Right and European Union. The results suggest that the Anti-EU motivations are clearly important in 2017 and 2025, and less so in other years. Section G discusses this exercise and its results in more details.

An important caveat is that additional dimensions are likely to be correlated with the existing ones, which could complicate estimation. Pooling candidates across parties and years, the correlation between the Left-Right and Anti-Elite dimensions is indistinguishable from zero, whereas it is 0.46 between Left-Right and EU, and 0.60 between Anti-Elite and EU. Figure K4 plots median party positions for each pair of dimensions across years, together with the corresponding correlation coefficients. Figure K3 documents the same patterns on a broader sample, using expert estimates of European parties’ positions across relevant policy dimensions.

How would my results be affected if there is another dimension which my estimates miss? Consider the cleanest case: one studied dimension with weight w and positions Y_j , and one omitted dimension with weight w^o , positions $Y_j^o = \gamma Y_j + e_j$, $\gamma > 0$ (dimensions bundled at the candidate level), and voter component X^o . Set $e_j = 0$ first.⁴³ True utility is

$$U_j = -w(X - Y_j)^2 - w^o(X^o - Y_j^o)^2.$$

⁴³With $e_j \neq 0$, further terms arise: a voter-level factor error $2w^o X^o e_j$, absorbed into the bliss-point shock when X^o is independent of (X, Y) within type; and candidate-specific intercepts $-2w^o \gamma e_j Y_j - w^o e_j^2$, which a pure characteristics model rationalizes spatially. All are proportional to w^o and vanish in the same limit.

Collecting the terms that depend on j and dropping alternative-invariant ones,

$$U_j \sim -(w + w^o \gamma^2) Y_j^2 + 2(wX + w^o \gamma X^o) Y_j = -(w + w^o \gamma^2) (Y_j - \tilde{X})^2 + \text{const},$$

where

$$\tilde{X} = \frac{wX + w^o \gamma X^o}{w + w^o \gamma^2}, \quad \tilde{w} = w + w^o \gamma^2.$$

The misspecified one-dimensional model thus retains the correct form but has distorted primitives: the recovered bliss point is $\tilde{X}$, a weighted average of the true position on the studied dimension and the omitted-dimension component, and the recovered weight is $\tilde{w} = w + w^o \gamma^2$, inflated by the omitted dimension's importance and its correlation with the studied one. Both distortions vanish as (1) $\gamma \rightarrow 0$: dimensions along which candidates' positions are unrelated to the studied ones can be omitted without bias; or (2) $w^o \rightarrow 0$: unimportant dimensions can be omitted without bias, regardless of bundling.

Hence, my approach is not robust to omitting relevant dimensions correlated with the included ones. The correlation patterns in Figure K3 suggest that the Left–Right dimension, being highly correlated with plausibly relevant dimensions such as Nationalism and Immigration, may capture those dimensions. Immigration enters the Left–Right dimension by construction, as questions concerning immigration are used to build it. Nationalism, by contrast, could not be accounted for, as no survey questions capture it. Under this reading, the Left–Right results could be interpreted as those for a weighted bundle of economic, cultural, nationalist, and immigration policies. Similarly, the Anti-Elite dimension is highly correlated with the Anti-EU dimension, so its results can likewise be read as referring to a bundle.

Candidate positions estimated with an error. Assume the candidate positions are measured with error: the estimation uses $\bar{Y}_j = Y_j + \xi_j$ in place of the true Y_j . The effective utility is then

$$U_j^{RPM} = -(X + \epsilon - Y_j - \xi_j)^2,$$

so the error enters as a perturbation of the candidate's location rather than as an additional disturbance: unlike ϵ_i , which varies at the voter level, ξ_j is common across voters, and the recovered bliss points are located relative to the perturbed positions. The relevant question is therefore whether the objects of interest are stable under small perturbations of the positions. Figure J17 assesses this for the West 2025 results: I add an i.i.d. $N(0, 0.01)$ shock to every estimated district-level party position (and to the national anchor of the counterfactual parties) and re-estimate the target parameter

of Section 6.2.2 for each of 10 draws.⁴⁴ The shocks move the bounds without widening them, and on average pull them towards the uninformative composition,⁴⁵ but preserving the overall results. Moderate error in the measured positions therefore weakens but does not systematically overturn the conclusions.

Left–Right as a combination of Economics and Culture. So far the paper has treated the Left–Right dimension as a bundle of economic and cultural policies without unpacking the two. Estimating the model in three dimensions is infeasible given the data, so I instead re-estimate it in two: Economics and Culture.

To do so, I first split the Left–Right items into those concerning culture and those concerning economics; see Table K2 for the classification.⁴⁶ I then follow exactly the estimation procedure of Section 6, applied to these two dimensions.

Figure J19 reports the estimated weights by year and part of the country, and Figure J20 reports the counterfactual motivation exercise, which splits the AfD into two hypothetical parties: one identical to the AfD on culture but economically liberal, the other identical on economics but culturally liberal. The decomposition shows a gradual realignment of voting motives, from economic ones, which clearly dominate in 2013, to cultural ones, which clearly dominate in 2025. In the West, voters attach a much larger weight to the cultural than to the economic dimension from 2017 onward; in the East the two weights are closer, although imprecisely estimated. These results are partly in line with Danieli et al. (2023), who attribute the rise of radical-right parties largely to an increase in the weight voters attach to cultural relative to economic issues.

The Role of the Supply Side. To what extent the results presented above are driven by the supply side repositioning versus the demand side changes (genuine evolution of preferences, or compositional changes)? To answer this question I decompose the evolution of the values of the target parameter from the Section 6.2.2 into the preference evolution and supply-side repositioning, using Shapley values approach (Section H provides more details on the exercise).⁴⁷

The results of this exercise suggest that the evolution over time in the share of Right-motivated voters documented in this paper is not driven by parties changing their positions, but by genuine

⁴⁴A standard deviation of 0.01 is not small relative to the cell partition: about 35% of the cells are thinner than 0.02 (two s.d.).

⁴⁵This resembles OLS attenuation bias: the shocks break the cross-election structure of the cell labels that the "correct" distributions exploit, and the bounds are biased towards the raw share of AfD cells in the "Right" versus "Anti-Elite" areas (39% Right; see Figure 7a).

⁴⁶Two coding decisions are worth noting: immigration items are classified as cultural (Alesina and Tabellini, 2024), and the Left–Right self-placement item is dropped.

⁴⁷See Shapley (1953) for the original reference, or Guriev et al. (Forthcoming); Longuet-Marx (2025) for recent applications in political economy.

evolution of preferences of AfD electorate.

In this setup it is hard to pin down demand-side changes and attribute them to either the composition or the preference channel (see Table 18 for the wide bounds of Shapley values on both). I now provide two pieces of evidence that preference evolution within demographic groups is consistent with increasing Right motivation. First, within-person evolution towards Right motivations is consistent with survey-panel evidence showing that repeat AfD voters (i) become both more conservative and more anti-elite over time, and (ii) move faster along the Left–Right dimension.⁴⁸ Second, as shown earlier, there are consistent within-group patterns of preference evolution—for instance, voters without higher education and young voters becoming increasingly Right-motivated.

Approach of Merlo and De Paula (2017). Section E re-estimates the results of Section 6.2.2 with the method of Merlo and De Paula (2017), who obtain point estimates of the same spatial model by restricting the distribution of bliss points to a semi-nonparametric class and estimating it by sieve minimum distance. Applied to the same data, demographic groups and counterfactual masks, their point estimates lie inside or close to the sharp bounds in 2017–2025 and reproduce the overall reversal from an anti-elite to a right-motivated majority of AfD voters. The two approaches have their own benefits: the parametric restriction delivers a single estimate, at the cost of fixing the shape of each group’s density, which affects the estimation of both the weights and the bliss points; the bounds impose no such shape but identify the parameters only up to a set.

11 Conclusion

Is populist support rooted in Left–Right ideology or in protest against the elites? This paper develops an approach to recovering voters’ preferences from aggregate electoral data in a multidimensional spatial voting model, and uses it to address exactly this kind of question. Applied to the radical-right Alternative for Germany (AfD), the method reveals that the party’s support became increasingly grounded in its conservative left–right positions rather than its anti-elite stance. Counterfactual exercises suggest that establishment parties could recapture a substantial share of AfD voters by modestly hardening their Left–Right positions—particularly on immigration.

These findings speak to a large political economy literature on the rise of populism, which has focused primarily on identifying the causal forces driving voters towards populist parties and candidates. This paper takes a different route: it recovers *the preferences of populist voters themselves*. Taking this route helps inform the design of responses to populism.

⁴⁸The results are based on a 2016–2021 panel of all voters, taking their self-reported issue positions as their ideal points; see Figure J18 and Table J2.

That question of response is important. Populism appears to carry real economic and institutional costs (Funke, Schularick and Trebesch, 2023), so how mainstream parties react is consequential. Responses fall broadly into two classes: those addressing voters’ anti-elite grievances, and those addressing their ideological left–right demands. The former may take the form of commitment politics (Bellodi et al., 2025), among other approaches; the latter calls for more economically or culturally conservative platforms. Knowing which margin moves a given electorate therefore determines which response would be most effective. In the German case, the evidence points towards the left-right margin, especially in later years.

More broadly, the paper offers a framework for unpacking the bundle of characteristics embodied in any political option. Any voting option—a candidate, a party, a referendum answer—can be seen as a product with multiple attributes, and recovering preferences over these attributes is far from straightforward. Voting departs from most settings in empirical industrial organization: in a spatial model, individuals hold unobserved ideal points on each dimension, so one cannot impose the linear characteristic–utility relationship standard in that literature. The question “*Do you prefer sweeter or less sweet cereal?*” does not translate into “*Do you prefer a more or less right-wing candidate?*”—voters differ not in how much of an attribute they want, but in where their ideal point lies.

The method developed in this paper is portable to any electoral setting in which variation in candidate choice sets provides identification—across districts, in simultaneous elections, or under ranked-choice voting. Such settings together cover a substantial share of real-world elections. The method can further be extended to incorporate turnout, candidate valence, or additional dimensions of conflict.

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A Identification of dimension weights

This section establishes that the identified set for the relative weight in the two-dimensional weighted spatial voting model of Section 2 is contained in a closed interval of $\mathbb{R}_+$ that contains the true weight.

The intuition behind the result is as follows. The weight w_2 determines which cells of the MRP are non-empty, and weights are therefore identified only up to the differences in the cells they make available. The argument has two steps. First, a single cell is non-empty only for weights in an interval. Second, any weight consistent with the data must make every cell the data require non-empty; the set of such weights is therefore an intersection of availability intervals, which is again an interval.

A.1 Setting and notation

I use the weighted spatial voting model of Section 2 with the normalization $w_1 = 1$, so the single parameter is the relative weight $w_2 = w > 0$ on the second dimension. The policy space is the box $\mathcal{R} = [a_1, b_1] \times [a_2, b_2]$. Recall that the cell chosen by candidate j under a position vector y is $\mathcal{X}_{j;w}(y) = \{x \in \mathcal{R} : d_W(x, Y_{ij}) \leq d_W(x, Y_{ik}) \ \forall k \neq j\}$, with $d_W(x, Y_{ij})^2 = (x_1 - Y_{ij,1})^2 + w(x_2 - Y_{ij,2})^2$.

Superimposing the partitions induced by every $y \in \mathcal{Y}$ at weight w yields the minimal relevant partition $\mathbb{X}_w$.⁴⁹ A cell of $\mathbb{X}_w$ is indexed by a *choice profile*

$$\mathbf{c} = (j(y))_{y \in S}, \quad j(y) \in \mathcal{J}, \quad S \subseteq \mathcal{Y},$$

recording the candidate chosen under each environment in S , and equals the intersection

$$\mathcal{X}_w(\mathbf{c}) = \bigcap_{y \in S} \mathcal{X}_{j(y);w}(y). \tag{A1}$$

Taking $S = \mathcal{Y}$ gives the cells of $\mathbb{X}_w$; a proper subset S gives a *partial* profile, which indexes a union of such cells. The Fréchet constraints of Section 2.3 concern pairs of environments, $|S| = 2$; everything below applies to full and partial profiles alike. The profile $\mathbf{c}$ is *available* at weight w when $\mathcal{X}_w(\mathbf{c}) \neq \emptyset$; only available profiles enter the prediction of observed vote choices (cf. Equation 5). A partition $\mathbb{X}_w$ is thus summarized by its set of available profiles,

$$\mathcal{C}(\mathbb{X}_w) = \{\mathbf{c} : \mathcal{X}_w(\mathbf{c}) \neq \emptyset\},$$

and two weights are observationally equivalent whenever they induce the same $\mathcal{C}(\mathbb{X}_w)$.

The cell as a system of linear constraints. By (A1), $\mathcal{X}_w(\mathbf{c})$ is an intersection of half-planes with $\mathcal{R}$: one half-plane per environment $y \in S$ and rival $k \neq j(y)$, for a total of $M = \sum_{y \in S} (|y| - 1)$ constraints. The set of bliss points at least as close to an own position s as to a rival s' is the half-plane $\{x : d_W(x, s)^2 \leq d_W(x, s')^2\}$, which after expansion reads $2(s'_1 - s_1)x_1 + 2w(s'_2 - s_2)x_2 \leq$

⁴⁹This coincides with the coarsest partition of Section 2: superimposing the per-environment partitions merges exactly those bliss points that choose identically under every $y \in \mathcal{Y}$.

$(s_1'^2 - s_1^2) + w(s_2'^2 - s_2^2)$.⁵⁰ Indexing the constraints by $m = 1, \dots, M$, each takes the form

$$\alpha_m x_1 + w \beta_m x_2 \leq \gamma_m + w \delta_m, \quad m = 1, \dots, M, \quad (\text{A2})$$

with fixed constants $\alpha_m = 2(s_1' - s_1)$, $\beta_m = 2(s_2' - s_2)$, $\gamma_m = s_1'^2 - s_1^2$, $\delta_m = s_2'^2 - s_2^2$, where for constraint m arising from environment y and rival k the own position is $s = Y_{i,j(y)}$ and the rival is $s' = Y_{ik}$. The support adds $a_1 \leq x_1 \leq b_1$ and $a_2 \leq x_2 \leq b_2$. The key structural feature is that w enters (A2) only linearly, both in the coefficient of x_2 and in the threshold, and the same scalar w is shared across all M constraints.

The *availability set* of a profile $\mathbf{c}$ is the set of weights at which its cell is non-empty:

$$\mathcal{W}(\mathbf{c}) = \{w > 0 : \mathcal{X}_w(\mathbf{c}) \neq \emptyset\} = \{w > 0 : \exists x \in \mathcal{R} \text{ satisfying (A2)} \forall m\}. \quad (\text{A3})$$

A.2 Set-identification of weights

Lemma 1. *For every choice profile $\mathbf{c}$, the availability set $\mathcal{W}(\mathbf{c})$ is an interval, closed in $(0, \infty)$.*

Proof. Fix $w > 0$ and apply the change of variables $(z_1, z_2) = (x_1, wx_2)$, with inverse $x_2 = z_2/w$. For each fixed $w > 0$ this is a bijection on $\mathbb{R}^2$, so $\mathbf{c}$ is available at w if and only if the transformed system

$$\alpha_m z_1 + \beta_m z_2 \leq \gamma_m + w \delta_m, \quad m = 1, \dots, M, \quad a_1 \leq z_1 \leq b_1, \quad wa_2 \leq z_2 \leq wb_2, \quad (\text{A4})$$

admits a solution $z \in \mathbb{R}^2$. The substitution linearizes the bilinear term $w\beta_m x_2$ into $\beta_m z_2$, so w now appears only in the thresholds, including those of the support. Define the augmented set in (z_1, z_2, w) -space

$$\Omega = \left\{ (z_1, z_2, w) \in \mathbb{R}^3 : \alpha_m z_1 + \beta_m z_2 - w \delta_m \leq \gamma_m, \quad m = 1, \dots, M; \quad a_1 \leq z_1 \leq b_1; \quad wa_2 \leq z_2 \leq wb_2 \right\}. \quad (\text{A5})$$

Each defining inequality is linear in the joint variable (z_1, z_2, w) , so Ω is a polyhedron in $\mathbb{R}^3$.

Let $\pi : \mathbb{R}^3 \rightarrow \mathbb{R}$ be the projection $\pi(z_1, z_2, w) = w$. By construction,

$$\mathcal{W}(\mathbf{c}) = \pi(\Omega) \cap (0, \infty).$$

The projection of a polyhedron is a polyhedron (Fourier–Motzkin elimination of z_1, z_2), so $\pi(\Omega)$ is a closed interval of $\mathbb{R}$, and $\mathcal{W}(\mathbf{c})$ is an interval, closed in $(0, \infty)$. The elimination writes $\mathcal{W}(\mathbf{c}) = \{w > 0 : \underline{w}(\mathbf{c}) \leq w \leq \overline{w}(\mathbf{c})\}$ explicitly, with endpoints that are ratios of nonnegative linear combinations of the constants in (A2) and of a_ℓ, b_ℓ . $\square$

Proposition 3. *Let $\mathcal{C}^+$ be any finite collection of choice profiles whose cells are required to carry positive mass. Then the set of weights at which every profile in $\mathcal{C}^+$ is available,*

$$\mathcal{W}(\mathcal{C}^+) \equiv \bigcap_{\mathbf{c} \in \mathcal{C}^+} \mathcal{W}(\mathbf{c}),$$

⁵⁰Availability is defined by non-emptiness with weak inequalities. If the cell has positive Lebesgue measure for some w , the set of such weights is the same system with strict inequalities, an open interval whose closure is $\mathcal{W}(\mathbf{c})$ below; the two differ only at the endpoints, which is immaterial for the containment in Proposition 3.

is an interval, closed in $(0, \infty)$. Moreover, if the data are generated at the true weight w_0 , then $w_0 \in \mathcal{W}(\mathcal{C}^+)$, and the identified set of weights satisfies $W^* \subseteq \mathcal{W}(\mathcal{C}^+)$.

Proof. By Lemma 1, each availability set $\mathcal{W}(\mathbf{c})$ is a closed interval, and a finite intersection of closed intervals is again a closed interval; since $\mathcal{Y}$ is finite, $\mathcal{C}^+$ is finite. If the data are generated at w_0 , they are generated by a distribution supported on the cells of $\mathbb{X}_{w_0}$, so any cell that the data require to carry positive mass is available at w_0 ; hence $w_0 \in \mathcal{W}(\mathbf{c})$ for every $\mathbf{c} \in \mathcal{C}^+$, and $w_0 \in \mathcal{W}(\mathcal{C}^+)$. Finally, any weight $w \in W^*$ must rationalize the data, which requires every cell in $\mathcal{C}^+$ to be non-empty at w ; therefore $W^* \subseteq \mathcal{W}(\mathcal{C}^+)$. $\square$

Outer versus sharp set. $\mathcal{W}(\mathcal{C}^+)$ is an outer set: availability of the required cells does not guarantee that the mass the data leave unassigned can be placed on the cells available at w . Let S^* be the cells of $\mathbb{X}_{w_0}$ carrying positive mass under f and $I^* = \bigcap_{\mathbf{c} \in S^*} \mathcal{W}(\mathbf{c})$. Because the predicted shares in (9) depend only on the masses of profiles and not on the geometry of the cells at w , the true mass function reproduces the data at every w in the interior of I^* , so $I^* \subseteq W^* \subseteq \mathcal{W}(\mathcal{C}^+)$. The outer set is therefore sharp whenever $\mathcal{W}(\mathcal{C}^+) \subseteq I^*$, that is, when the cells that determine the endpoints of I^* are among those the data force. This becomes more plausible as the number of elections grows: each additional election adds constraints of the form $s_i(y) + s_j(y') > 1$ and forces new cells, so $\mathcal{C}^+$ expands and $\mathcal{W}(\mathcal{C}^+)$ shrinks towards W^* (see the simulations in Appendix B). Since S^* is unobserved the condition cannot be verified; the grid search approach in the paper approximates W^* directly.

B Weights estimation: simulation

This section provides a simulation comparing the weight estimator based on 1) the method used in this paper, and 2) an approach that allows for a simultaneous recovery of bliss points and weights from Merlo and De Paula (2017).

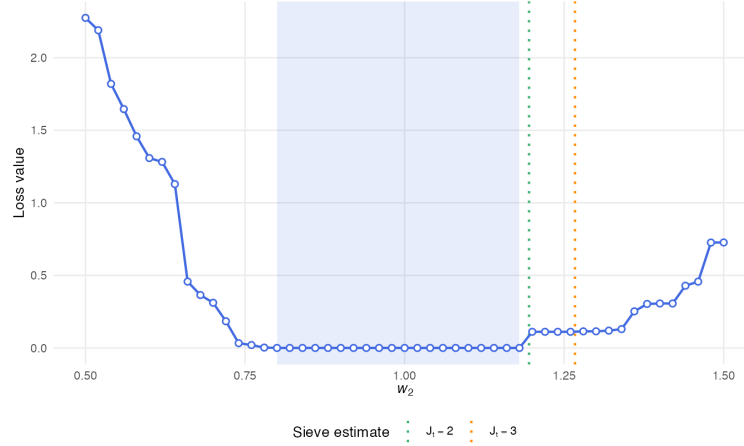
I simulate a population of 10,000 voters voting in twenty-four elections with six candidates drawn at random. Bliss points are drawn from a mixture of two dominant clusters and a diffuse background. Voters choose the candidate closest to their bliss point under the weighted Euclidean metric with true weight $w_2 = 1$; the researcher observes only the resulting candidate vote shares in each election.

For each candidate value of w_2 on a grid, the implied superimposed Voronoi tessellation determines the set of feasible cells (joint vote patterns). The question then is whether voter mass can be distributed across these cells so as to reproduce the observed shares exactly, subject to masses being non-negative and summing to one. The set of weights for which an exact fit exists is the identified set: since every cell is feasible only on an interval of weights, and concentrated vote shares force particular cells to carry mass, the weights consistent with the data are contained in an interval around the truth; in this simulation, the exact-fit set is itself an interval.

Figure B1 plots the residual fit against w_2 . The loss is exactly zero on the interval $[0.80, 1.18]$, which contains the true value $w_2 = 1$, and rises outside it. The cell geometry alone thus set-identifies the weight without any assumption on the distribution of bliss points.

The dotted lines mark the estimates obtained from the sieve estimator at polynomial orders $J_t = 2, 3$, which impose a flexible parametric form on the bliss-point density. Because a low-order polynomial correction of a Gaussian base cannot reproduce the distribution of voters' bliss points, the

resulting weight estimates are far from the truth as well, at 1.20 and 1.27 for $J_t = 2, 3$ respectively.

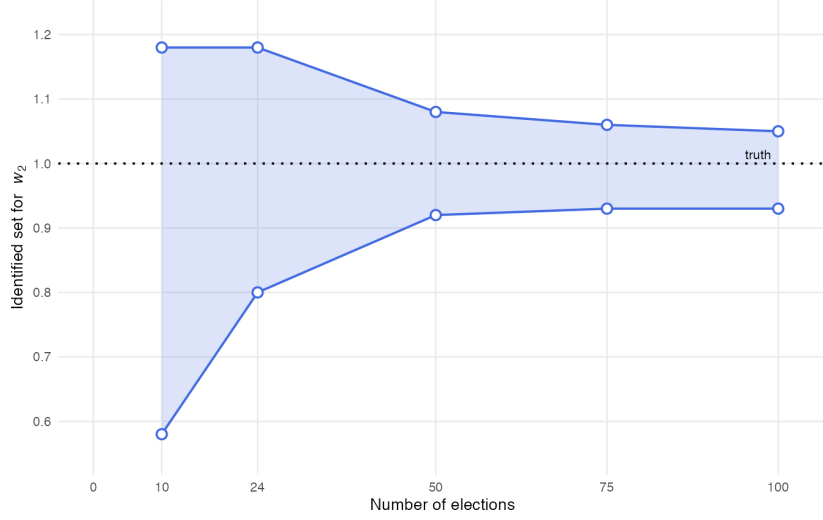


Notes: Monte Carlo with 10,000 voters, true $w_2 = 1$, twenty-four six-candidate elections; bliss points follow a two-cluster mixture with a diffuse background, and only vote shares are observed. The solid curve is the minimal L^1 distance between observed shares and shares generated by any distribution of voter mass over the cells of the superimposed tessellation at each candidate weight. The shaded region is the identified set $[0.80, 1.18]$, where the fit is exact; it contains the truth. The dotted lines mark the sieve estimates of Merlo and De Paula (2017) at $J_t = 2, 3$ ($\hat{w}_2 = 1.20, 1.27$), both of which lie outside the identified set.

Figure B1. Set identification of the weight via the method used in this paper, and sieve estimates from Merlo and De Paula (2017)

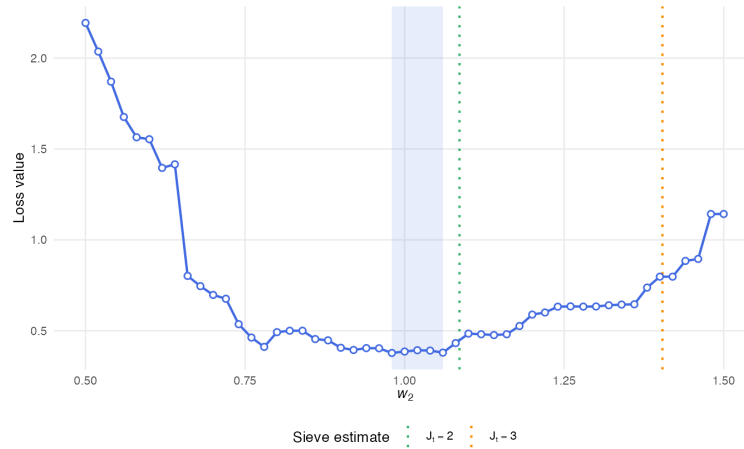
Number of Elections. The identified set shrinks as the number of elections grows. Because each additional election contributes a further set of cells that the observed shares may or may not be able to populate, more elections impose more restrictions on which weights admit an exact fit. Figure B2 traces the identified set for w_2 against the number of elections, holding the data-generating process fixed: the set narrows monotonically around the true value, from $[0.58, 1.18]$ with ten elections to $[0.93, 1.05]$ with one hundred. Note that while this simulation illustrates that the size of the identified set shrinks with the number of elections, its size at any given number of elections depends on the positions of the candidates and the distribution of preferences.

Noisy perceptions of candidate positions. Realistically, however, there is always noise in the data. I next simulate the behaviour of both estimators under the assumption that each voter observes every candidate's position with an idiosyncratic mislocation error, distributed $\mathcal{N}(0, 0.1^2 I)$, while the estimation procedures remain unchanged. Under noise an exact fit is no longer attainable at any weight. The question, however, is where the minimum is located and how large the basin delivering a near-identical fit is. The region of near-minimal losses becomes *smaller*: the loss now develops a strict interior minimum, and the near-feasible set (weights whose fit is within one percentage point of share mass of the best attainable) shrinks from the noise-free interval $[0.80, 1.18]$ to $[0.98, 1.06]$, still containing the truth (Figure B3). Intuitively, the smoothed shares are best approximated by the cell structure of the true weight, so mild noise converts the flat exact-fit region into a well-defined basin centred near $w_2 = 1$. The sieve estimator yields $\hat{w}_2 = 1.09$ and 1.40 at $J_t = 2, 3$, both falling outside the near-feasible set. Note that in this simulation the results obtained with $J_t = 2$ are closer to the truth; this suggests that additional flexibility in the estimator does not necessarily translate into more precise estimates.



Notes: Each point pair gives the lower and upper bounds of the identified set for w_2 (the weights admitting an exact fit), computed as in Figure B1 but varying the number of elections. The dotted line marks the true value $w_2 = 1$.

Figure B2. Identified set for the weight at different numbers of elections



Notes: Same setup as Figure B1, except that each voter perceives every candidate's position with an idiosyncratic $\mathcal{N}(0, 0.1^2 I)$ mislocation error. The solid curve is the minimal L^1 distance between observed shares and shares generated by any distribution of voter mass over the cells of the superimposed tessellation at each candidate weight; with noise the loss is strictly positive everywhere. The shaded region is the near-feasible set $[0.98, 1.06]$ (fit within 0.01 of the minimum), which contains the true value $w_2 = 1$. The dotted lines mark the sieve estimates of Merlo and De Paula (2017) at $J_t = 2, 3$ ($\hat{w}_2 = 1.09, 1.40$), both outside the near-feasible set.

Figure B3. Set identification of the weight via the method used in this paper, and sieve estimates from Merlo and De Paula (2017) with voter perception noise

C Exploiting congressional district borders for identification

Longuet-Marx (2025) exploit differences in candidate positions across congressional district borders to identify voters' preferences, but rely on a different model of voters' choice. In this section I illustrate how this type of identifying variation could in principle be used to estimate the primitives of vote choice using the model considered in this paper.

For simplicity, I assume a one-dimensional model of choice, with the policy space normalized to the interval $[1, 5]$, and a population that is homogeneous in its observed characteristics across space (i.e. the entire population belongs to a single group d , for example educated females aged 30–35). This assumption simplifies the exposition without affecting the substantive argument.

Suppose that precincts with positions y and y' are paired. Then $f(x | y, m, d) = f(x | y', m, d)$ by the assumption that congressional district borders are exogenous. Likewise, for another pair y'' and y''' we have $f(x | y'', m, d) = f(x | y''', m, d)$. Crucially, this approach does not assume $f(x | y, m, d) = f(x | y'', m, d)$; that is, preferences may differ across distinct pairs of precincts.

Consider a two-party system, which is the most demanding case, as it provides the least variation. Each choice situation is then characterized by the positions of two candidates on a one-dimensional space. From the perspective of a spatial voting model, the object that uniquely defines this situation is the cut-point between the two candidates. Hence choice situation y yields a cut-point y_c , while y' yields a cut-point y'_c . The researcher observes vote shares for both candidates in both precincts, as illustrated in Figure C1.



Figure C1. Two paired precincts. Candidate positions (A, B), the implied cut-point (thin vertical line), and observed vote shares differ across the pair, while the underlying distribution of bliss points is common by the exogenous-border assumption.

Since the distribution of bliss points is identical within a pair of contiguous precincts, one can overlay the two sets of results to obtain a single, more finely partitioned distribution, as shown in Figure C2.

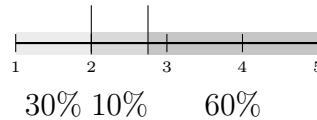


Figure C2. Overlaid distribution for the paired precincts of Figure C1. The two cut-points partition the line into three intervals, yielding refined shares of the common bliss-point distribution.

Repeating this exercise across many pairs of contiguous precincts yields a collection of refined distributions of bliss points. Note, however, that these distributions are assumed to differ across pairs and therefore cannot be overlaid with one another, even when, as in the case considered here, the demographics are identical. What the researcher can do instead is draw inferences from this collection of distributions, using the known population size of each pair of precincts.

For example, suppose the precinct pair above has a population of one thousand. The numbers of individuals lying in each of the three intervals are then 300, 100, and 600.

Now consider a second precinct pair of 1,500 individuals, with cut-points and shares as displayed in Figure C3. The corresponding numbers of individuals in each interval are 450, 300, and 750.

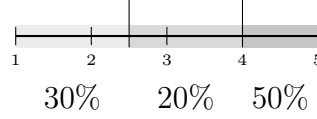


Figure C3. Overlaid distribution for a second precinct pair, with distinct cut-points and shares. This distribution is specific to the pair and cannot be combined with that of Figure C2.

What can be learned from the two distributions jointly? Although they differ and their electoral results cannot be pooled, one can still pose partial-identification questions in the spirit of this paper. For instance, what share of individuals of the given demographic hold left-wing views, defined as a bliss point below 2? The first pair’s cut-points lie at 2 and 3.5, and the second pair’s at 2.5 and 4. For the first pair, the threshold 2 coincides with the first cut-point, so the 300 individuals lie below the threshold. For the second pair, the threshold falls strictly inside the first interval $[1, 3)$; since the within-interval distribution is unrestricted, the 450 individuals in that interval may lie anywhere in it, so the pair contributes between 0 and 450. Hence, out of 2,500 individuals in total, the share holding left-wing views is set-identified within $[0.12, 0.30]$.

Note that identifying the dimension weights in this setup would not be possible without further assumptions. As discussed in the main text, the weights are identified through variation in the structure of the partition that candidate positions induce on the policy space, that is, in the composition of the resulting cells. Overlaying only two elections within a pair of contiguous electoral districts generates little to no variation in this cell composition, and is therefore insufficient to pin down the weights.

D Stated preferences

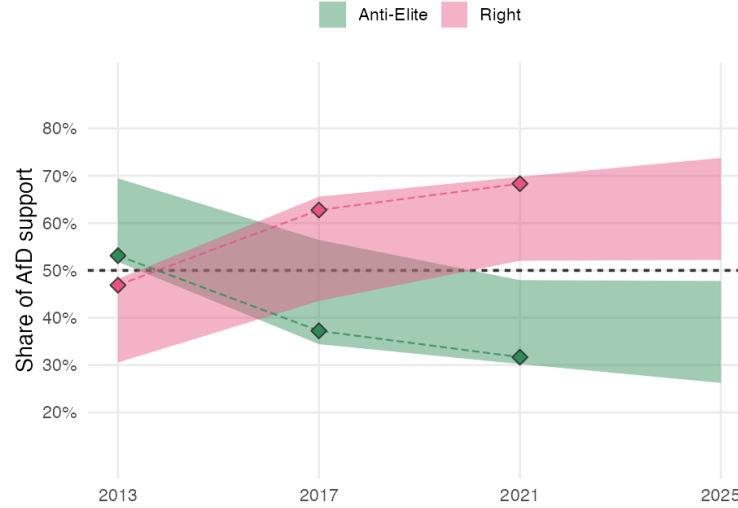
This section runs an exercise similar to the one in Section 6.2.2 using stated-preference survey data.

Before turning to the exercise and its results, I want to stress that recovering the weights from these data remains infeasible, so I take the estimated weights from the analysis in this paper. This is so for at least two reasons. First, respondents are surveyed throughout the year, and one cannot expect their perceptions of party positions to remain constant over time; restricting attention to those surveyed no more than three months before the election leaves at most 300 AfD voters in any given year. Second, the problem of voters adjusting their preferences so as to be closer to the party is especially acute for weight estimation, since such closeness would be read as evidence that a dimension is important—which may be precisely the opposite of the truth if individuals are adjusting their stated positions on the issues that matter least to them.

For this exercise I use data from the Politbarometer survey for 2013–2021 (2025 is not yet available). I approximate voters’ Left–Right and Anti–Elite ideal points by their answers to the questions on Left–Right self-placement and satisfaction with the government, as these are the only items available across all years and across all waves within each year. To construct the nationwide

AfD position, I take the mean of these answers across AfD voters in a given year. While this need not approximate the nationwide AfD position as accurately as in the main exercise, it is a reasonable approximation and avoids having to place the estimated AfD median values on the same scale as the survey items. I then proceed exactly as in Section 6.2.2.

Figure D1 plots the stated-preference estimates over the results from Figure 7b. The survey results replicate the trend well and remain within the estimated bounds from the revealed-preference estimates, while they tend to lie close to the upper (lower) bound of the right (anti-elite) shares. This might reflect the specific questions used for the construction of the measure, or the fact that the within-sample means are a poor approximation of the positions used in the revealed preference exercise. Nevertheless, the main takeaway of this exercise is that the identified patterns are consistent with the survey results.



Notes: Survey benchmark constructed from the GESIS Politbarometer surveys (2013, 2017, 2021). Each respondent's bliss point is built from two items: the left–right self-placement (11-point scale), and satisfaction with the government, rescaled to the same 11-point range. For each year I compute the national survey-weighted mean bliss point among all voters and proceed with the analysis as in Section 6.2.2. The resulting shares of AfD voters are point estimates, not partial-identification bounds. The 2025 survey data are not yet available. The estimates are weighted throughout using the survey weights provided by GESIS. The bands are the main estimates without second-choice moments.

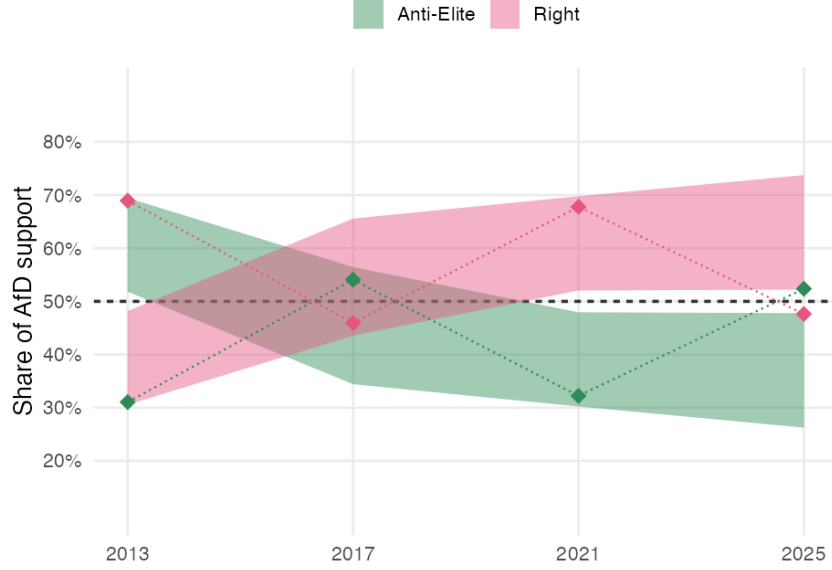
Figure D1. Comparison to the survey results

E Comparison with the Merlo–de Paula point estimates

The bounds of Section 6.2.2 are derived under no assumptions on the functional form of the distributions of bliss points. Merlo and De Paula (2017) instead obtain point estimates of the same spatial model by restricting the distribution to the semi-nonparametric (SNP) class of Gallant and Nychka (1987) and estimating it by sieve minimum distance. This section applies their estimator to the same data and target parameters. The bliss-point density of each of the 24 demographic groups is an SNP density, $f_g(t) \propto P_g(z)^2 \phi(z)$ with $z = R^{-1}(t - b_g)$, where P_g is a polynomial of degree at most two, b_g is the group mean and R a common scale matrix; the municipal vote shares are the mixture of the group densities integrated over the parties' areas of attraction, with the turnout-adjusted demographic composition as mixture weights, exactly as in the linear program. The weight matrix W is estimated jointly. Following Merlo and De Paula (2017), group means are anchored at the party centroids with a bounded deviation. The AfD composition shares are then computed from

the estimated densities under the same counterfactual masks as in the main text.

Figure E1 overlays the resulting point estimates on the sharp bounds. The point estimates lie inside or close to the bounds in every year except for 2013 and reproduce an overall reversal from an anti-elite to a right-motivated majority. They should be read as an illustration of what the parametric restriction adds rather than as an alternative estimate: the SNP class fixes the shape of each group’s density, so the point estimates depend on the anchoring and on the polynomial order, whereas the bounds do not.



Notes. Bands are the sharp bounds of Figure 7b. Diamonds are the sieve minimum-distance point estimates of the same shares under the SNP restriction of Merlo and De Paula (2017) (anchored group means, polynomial order two), aggregated over East and West with AfD-vote weights. Estimation uses the same municipality sample, demographic groups and counterfactual masks as the bounds; the weight matrix is estimated jointly and agrees with the grid-searched w_{AE} of the main text.

Figure E1. Merlo–de Paula point estimates vs. sharp bounds

F Model Estimation

This section describes how the empirical restrictions implied by the spatial voting model are translated into a finite-dimensional optimization problem, and how weights and bounds for target parameters are computed.

F.1 Weights estimation

This appendix details how the relative weight w_{AE} on the anti-elite dimension is computed. Throughout, the left–right weight is normalised to $w_{LR} = 1$, so the only free parameter is $w_{AE} \in \mathbb{R}_+$.

The exercise is run separately for each region (East, West) and year. As discussed in Appendix B, weight estimation requires a great deal of variation in candidate positions; pooling electoral districts within each region helps achieve this. Within a region, the market m is the *state*, the submarket s is the *municipality* (with $s \in m$), and d indexes the 24 age×gender×education demographic groups, with observed demographic shares $q_{s,d}$, $\sum_d q_{s,d} = 1$. An *election* y is the candidate configuration of an electoral district; $\mathcal{Y}_R$ collects the elections of region R , and each municipality s belongs to one

election $y(s)$ and one state $m(s)$. The choice set is the six parties.

For a candidate value w_{AE} , the weighting matrix is $W = \text{diag}(1, w_{AE})$. Since w_{AE} enters the distance $d_W(x, Y_{ij}) = \sqrt{(x - Y_{ij})^\top W (x - Y_{ij})}$, it rescales the anti-elite axis relative to left-right and thus tilts the bisecting hyperplanes separating any two parties, reshaping each attraction region $\mathcal{X}_{j;w}(y)$ and hence which cells (joint choice sets) appear in the MRP.

Overlaying all elections in $\mathcal{Y}_R$ yields the minimal relevant partition (MRP) $\mathbb{X}_w$, and $\mathbb{X}_{j;w}(y)$ denotes the cells of $\mathbb{X}_w$ on which j wins under y . Because the number of feasible cells governs how well any given partition can fit the data, and this count varies with w_{AE} , I hold it fixed across weights: I restrict attention to the AfD attraction region, retain the $N = 500$ largest cells, and merge each smaller fragment into the neighbouring cell with which it shares the longest boundary. Let $\mathbb{X}_w$ denote the resulting 500-cell mesh.

The unknowns are the state-and-group mass functions

$$\phi(\mathcal{X} \mid m, d) \geq 0, \quad \sum_{\mathcal{X} \in \mathbb{X}_w} \phi(\mathcal{X} \mid m, d) = 1,$$

one per state m in region R and demographic group d . The weight w_{AE} is held common across all states and groups of the region-year. Given ϕ and w_{AE} , the predicted AfD vote share in municipality s is the finite-dimensional analogue of (9),

$$s_{\text{AfD}}(y(s), s; \phi, w_{AE}) = \sum_d q_{s,d} \sum_{\mathcal{X} \in \mathbb{X}_{\text{AfD};w}(y(s))} \phi(\mathcal{X} \mid m(s), d).$$

In finite samples the exact restriction (9) generically has no solution. For each w_{AE} I choose the mass functions that minimise the mean absolute municipal AfD-share error,

$$\mathcal{E}(w_{AE}) = \min_{\{\phi(\cdot \mid m, d)\} \in \Phi} \frac{1}{|\mathcal{S}_R|} \sum_{s \in \mathcal{S}_R} \left| s_{\text{AfD}}(y(s), s) - s_{\text{AfD}}(y(s), s; \phi, w_{AE}) \right|,$$

where $\mathcal{S}_R$ is the set of municipalities in region R and $s_{\text{AfD}}(y(s), s)$ the observed share. Introducing slacks $u_s \geq 0$, this is the linear program

$$\min_{\{\phi(\cdot \mid m, d)\}, u \geq 0} \sum_{s \in \mathcal{S}_R} u_s \quad \text{s.t.} \quad -u_s \leq s_{\text{AfD}}(y(s), s; \phi, w_{AE}) - s_{\text{AfD}}(y(s), s) \leq u_s,$$

solved over $\phi \in \Phi$.

The program is solved for every candidate weight on the grid

$$\mathcal{W} = \{0.2, 0.3, \dots, 0.9, 1.0\} \cup \{1/0.9, \dots, 1/0.2\} \quad (17 \text{ points}),$$

symmetric around 1 in log scale, and the estimate is the fit-minimising weight

$$\hat{w}_{AE} = \arg \min_{w_{AE} \in \mathcal{W}} \mathcal{E}(w_{AE}).$$

Values $\hat{w}_{AE} < 1$ mean voters tolerate larger distances on the anti-elite dimension (the left-right

dimension weighs more); $\hat{w}_{AE} > 1$ the reverse. Because the data are noisy, the minimizer is generically a single point. Estimation is carried out independently by region and year. The number of elections per state is capped at 20. The 500-cell mesh is rebuilt at every w_{AE} .

F.2 Target parameters estimation

This appendix describes the bounds used for the main Anti-Elite versus Right exercise. Estimation of bounds for the remaining exercises follows the same logic. It follows the generic target-parameter LP and adds the two features specific to that exercise: the demographic-group masses are made *state-specific* but are pooled onto a single region-wide partition, and they are tied together across states of a region by a *cross-state coupling* restriction. The exercise is run separately for each region (West, East) and year, with the relative weight fixed at the estimate $\hat{w}_{AE}$ from Appendix F.1.

Under the weight $\hat{w}_{AE}$ I overlay the elections of *all* states of a region to form one minimal relevant partition, and restrict attention to the AfD attraction region, fixing it at $n_c = 500$ cells, as in the weight-estimation procedure. Cells are indexed $c \in \{1, \dots, n_c\}$ and demographic groups (age $\times$ gender $\times$ education bins) by $d \in \{1, \dots, D\}$. The masses are indexed by state m : for each state m in the region and each group d ,

$$\boldsymbol{\pi}_d^{(m)} = (\pi_{d1}^{(m)}, \dots, \pi_{dn_c}^{(m)})^\top \in \mathbb{R}_+^{n_c}, \quad \mathbf{1}^\top \boldsymbol{\pi}_d^{(m)} = 1,$$

so each state has its own bliss-point distribution over the shared region mesh. Municipalities are indexed $r \in \{1, \dots, R\}$, each belonging to one state $m(r)$; y_r is the observed AfD share and $w_{r,d}$ the turnout-adjusted share of group d among eligible voters in r . With $a_{r,c}^{\text{AfD}} \in \{0, 1\}$ the model-implied AfD choice indicator in r and $\mathbf{a}_r^{\text{AfD}}$ their vector, the predicted AfD share is

$$\hat{y}_r(\boldsymbol{\pi}) = \sum_{d=1}^D w_{r,d} (\mathbf{a}_r^{\text{AfD}})^\top \boldsymbol{\pi}_d^{(m(r))}. \quad (\text{F1})$$

Let $(\bar{X}^{RR}, \bar{X}^{AE})$ be the AfD position (national median value in a year) on the left–right and anti-elite dimensions. I split the AfD attraction region into two target areas by which of two reference points is nearer under the weighted metric $W = \text{diag}(1, \hat{w}_{AE})$:

$$C^{AE} = (\bar{X}^{RR} - 1, \bar{X}^{AE}), \quad C^R = (\bar{X}^{RR}, \bar{X}^{AE} - 1).$$

The *Anti-Elite* region collects the cells nearer C^{AE} ; the *Right* region collects the cells nearer C^R , which pulls on the anti-elite axis and retains the right-wing voters. Writing $b_c^{AE}, b_c^R \in \{0, 1\}$ for the resulting cell indicators, the target parameters are the shares of the region electorate who vote AfD and hold ideal points in each area,

$$\bar{\theta}^\bullet(\boldsymbol{\pi}) = \sum_m \sum_{d=1}^D (\mathbf{b}_d^\bullet)^\top \boldsymbol{\pi}_d^{(m)}, \quad \bullet \in \{AE, R\},$$

where $\mathbf{b}_d^\bullet$ weights the in-area AfD cells by the population mass of group d in state m .⁵¹

The masses satisfy, besides non-negativity, normalisation, and the fit slacks $-u_r \leq \hat{y}_r(\boldsymbol{\pi}) - y_r \leq$

⁵¹For simplicity, I discuss the case of the target parameter with respect to eligible voters. The parameter with respect to the AfD vote share is analogous, but requires dividing by that share.

u_r , the second-choice moment restrictions, namely the share of AfD voters for whom the Left is the second-best choice. Writing $s_{r,c}^{\text{Left}} \in \{0, 1\}$ for the indicator that in r 's election the Left is the second-nearest party (after the AfD) on cell c , the vector $\mathbf{v}_{m,d}^{\text{Left}}$ has entries $v_{m,d,c}^{\text{Left}} = \sum_{r: m(r)=m} e_r w_{r,d} a_{r,c}^{\text{AfD}} s_{r,c}^{\text{Left}}$, with e_r the number of eligible voters in r , so that $(\mathbf{v}_{m,d}^{\text{Left}})^\top \boldsymbol{\pi}_d^{(m)}$ counts the state- m , group- d AfD voters with the Left as second choice:

$$(\ell_m + \tau_{\text{SC}}) A_m \geq \sum_d (\mathbf{v}_{m,d}^{\text{Left}})^\top \boldsymbol{\pi}_d^{(m)} \geq (\ell_m - \tau_{\text{SC}}) A_m \quad \forall m,$$

with $A_m = \sum_d X_{m,d} T_{m,d}$ the state AfD mass ($T_{m,d}$: eligible voters of group d in state m ; $X_{m,d}$: the group's observed AfD support rate — both data, so the restriction is linear in $\boldsymbol{\pi}$) and ℓ_m the observed AfD to Left second-choice rate; $\tau_{\text{SC}} = 0.1$.

Another restriction ties the states of a region together so that pooling is informative without forcing them to be identical. I coarsen the policy space into blocks B on a 0.5 grid and, for each coarse marginal $g \in \{\text{With Higher Education, Without Higher Education, Young, Old, Male, Female}\}$ and block B , require every state's within-marginal mass share on B to stay within a factor $(1 + \tau_{\text{state}})$ of the population-weighted regional mean. With $e_{m,g,B} = \sum_{d \in g} T_{m,d} \sum_{c \in B} \pi_{dc}^{(m)}$ the state's block mass, $T_{m,g} = \sum_{d \in g} T_{m,d}$, and an auxiliary regional mean share $M_{g,B}$,

$$\sum_m e_{m,g,B} = \left(\sum_m T_{m,g} \right) M_{g,B}, \quad \frac{M_{g,B}}{1 + \tau_{\text{state}}} \leq \frac{e_{m,g,B}}{T_{m,g}} \leq (1 + \tau_{\text{state}}) M_{g,B} \quad \forall m.$$

The benchmark configuration uses $\tau_{\text{state}} = 2$ (each state's block share within a factor of three of the regional mean). Note that this is a loose coupling, as it is done only at the level of coarse demographic groups, for large "blocks", and allows for large differences across states within those group-block cells.

I first solve the best-fit program $\hat{Q} = \min_{\boldsymbol{\pi}, \mathbf{u}} \mathbf{1}^\top \mathbf{u}$ subject to all of the above. The approximately feasible set caps the total fit slack additively, $\mathbf{1}^\top \mathbf{u} \leq \hat{Q} + \epsilon_{\text{abs}}$, with $\epsilon_{\text{abs}} = 10^{-5}$, and the bounds are the two LPs

$$\underline{\theta}^\bullet = \min_{\boldsymbol{\pi}, \mathbf{u}} \bar{\theta}^\bullet(\boldsymbol{\pi}), \quad \bar{\theta}^\bullet = \max_{\boldsymbol{\pi}, \mathbf{u}} \bar{\theta}^\bullet(\boldsymbol{\pi}),$$

subject to the same constraints and the fit cap. This procedure is similar to the one in [Tebaldi, Torgovitsky and Yang \(2023\)](#).

The reference run pools each region under $\hat{w}_{AE}$, samples at most 100 municipalities per election with population below 10 thousand inhabitants, caps elections per state at 20, and fixes the mesh at 500 cells.

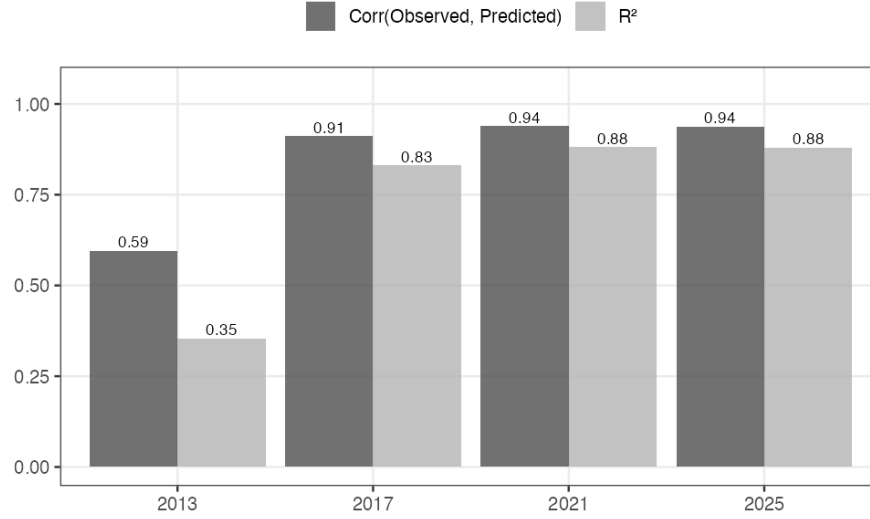
F.3 Model Fit

I now discuss model fit. Table [I4](#) presents summary statistics on model fit at the municipality level. In every election year, the median residual is exactly 0 by construction. The standard deviation ranges from 0.016 in 2013 to just above 0.03 in 2017 and 2021, and 0.048 in 2025. This increase mirrors an increase in AfD vote shares. The correlation between predicted and observed vote shares is above 0.9 in the later years, and about 0.6 in 2013. Overall, these results suggest a reasonable model fit, especially in the later years.

The correlations of predicted and observed municipal AfD shares with municipalities' demo-

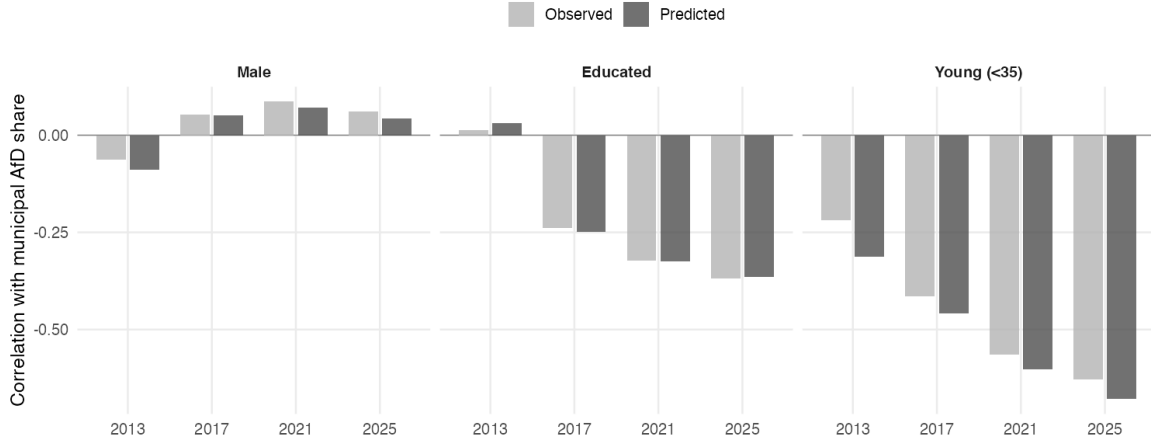
graphic characteristics are almost identical (Figure F2).

To sum up, both the model fit and the observed heterogeneity in the data suggest that the model performs well in explaining observed AfD voting patterns.



Notes. Bars plot the correlation between predicted and observed vote shares, and the R^2 from a regression of one on the other. Predicted AfD vote shares are chosen at a random seed.

Figure F1. Correspondence between predicted and observed AfD vote shares across municipalities.

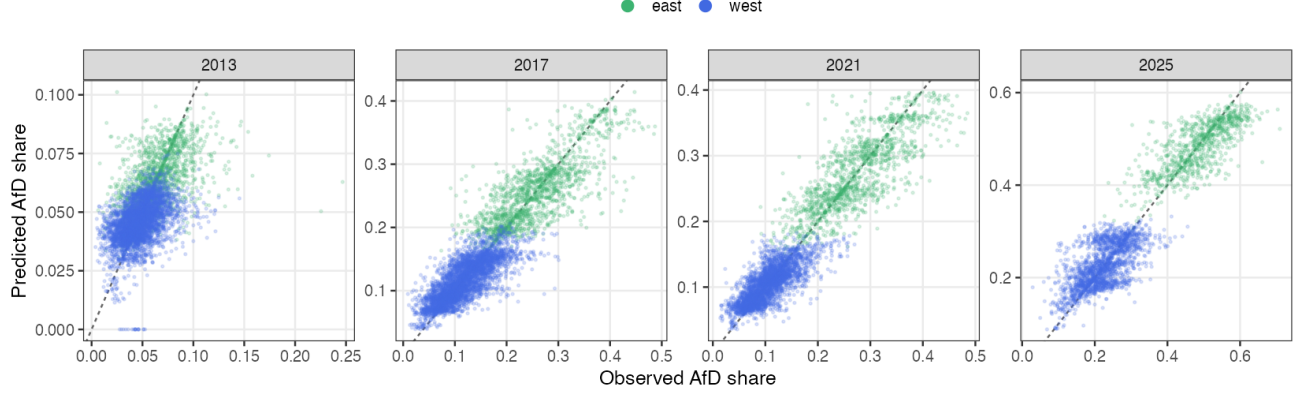


Notes. Each bar is an estimate of the correlation between the predicted (observed) AfD vote share and a given municipality demographic characteristic.

Figure F2. Correlation between predicted and observed AfD vote shares with demographic characteristics of municipalities.

F.4 Statistical inference

The target is the *Right share* of AfD support in the benchmark specification (Section 6.2.2). In this section I account for the sampling uncertainty in the sample of municipalities. The analysis was run only for 2021 to ease computation.



Notes. Each dot is a municipality, with the respective values for predicted and observed vote shares. Predicted AfD vote shares are chosen at a random seed.

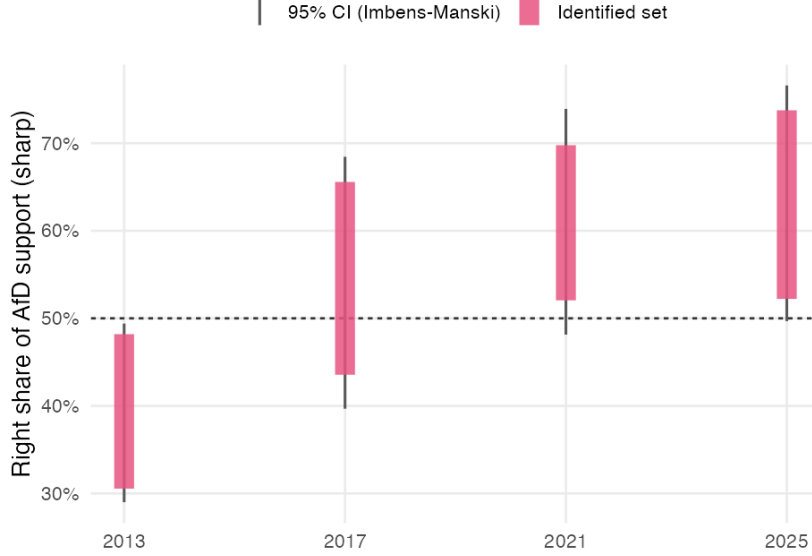
Figure F3. Correspondence between predicted and observed AfD vote shares in Eastern and Western Germany.

Sampling uncertainty is quantified with a nonparametric bootstrap. Each replication $b = 1, \dots, B$ ($B = 100$):

1. resamples municipality observations with replacement within each (state $\times$ election) group, preserving the rest of the design;
2. re-solves the full estimator on the resampled data, yielding endpoints $(\hat{\theta}_{(b)}^{lb}, \hat{\theta}_{(b)}^{ub})$.

From the bootstrap distribution of the endpoints I report an interval for the parameter following [Imbens and Manski \(2004\)](#), $[\hat{\theta}^{lb} - c \widehat{se}_{lb}, \hat{\theta}^{ub} + c \widehat{se}_{ub}]$ with c solving $\Phi(c + \hat{\Delta} / \max(\widehat{se})) - \Phi(-c) = 0.95$ and $\hat{\Delta}$ the estimated width of the identified set.

Such 95% CIs are presented in [Figure F4](#); the results remain quantitatively similar to the main results in the paper.



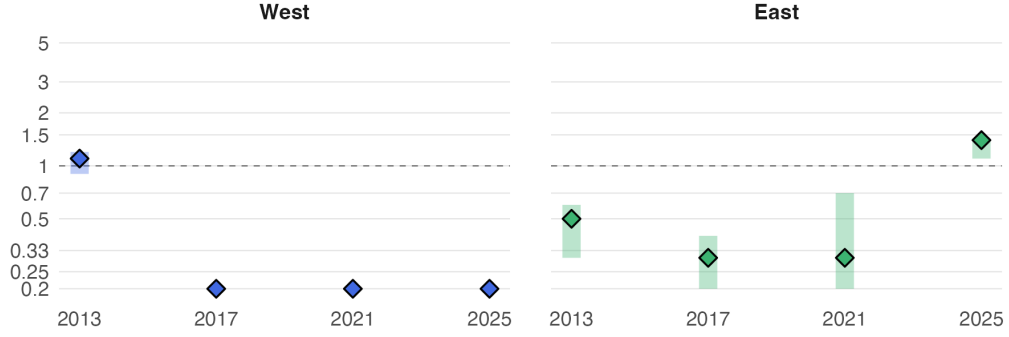
Notes. Bars show the sharp identified set for the Right share of AfD support, θ_{Right} , in the main specification without second-choice moments, aggregated over East and West with AfD-vote weights within each bootstrap replication. Whiskers are 95% confidence intervals for the parameter following [Imbens and Manski \(2004\)](#). Endpoint standard errors are obtained from a nonparametric bootstrap ($B = 100$) that resamples municipalities with replacement within state–election strata and re-solves the full linear program on each draw.

Figure F4. Bootstrapped 95% confidence intervals (Imbens–Manski), national

G Alternative dimension: Anti-EU

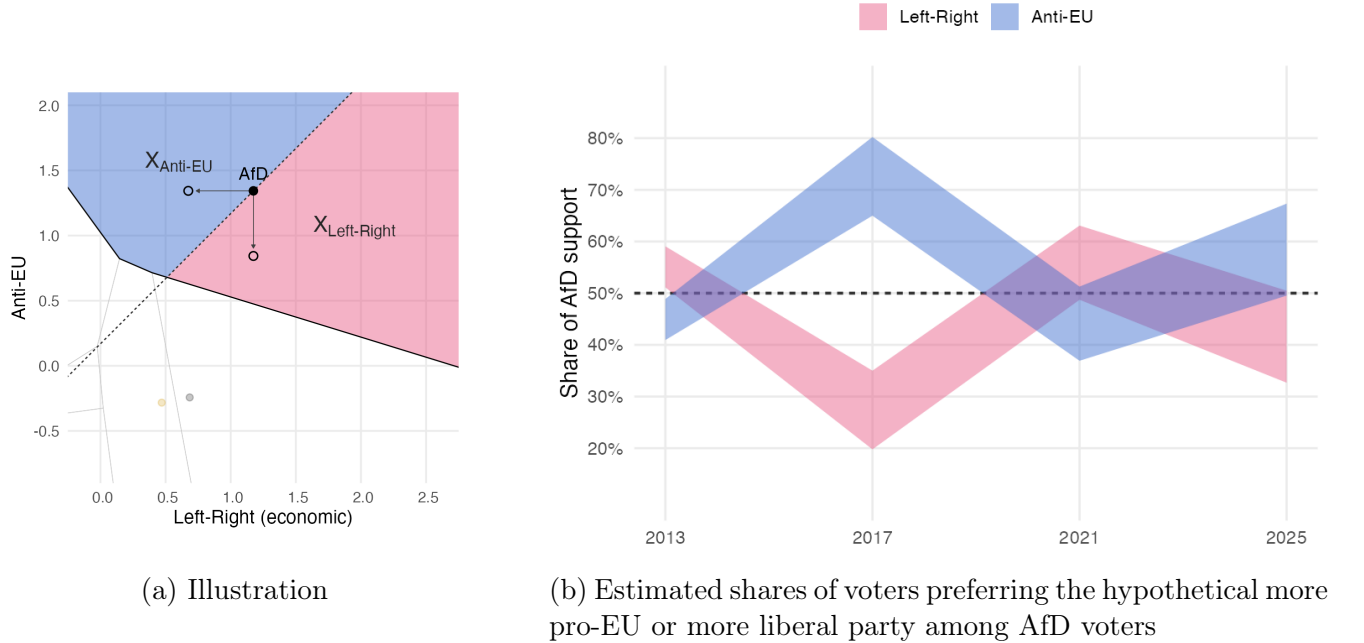
I construct the European Union dimension using a procedure similar to the one described in the main text. I use three questions from the GLES Candidate Survey: “EU Membership” (Good/Bad/Neither); “Should European unification be pushed further, or has it already gone too far?” (1, Pushed Further – 11, Too Far); and “EU Democratic Satisfaction” (1, Very Satisfied – 5, Very Dissatisfied). I then proceed with elastic net imputation to bridge the *Wen Wählen?* data with this data source, as is done for the main pair of dimensions (Left–Right and Anti-Elite). I then proceed with the estimation of the model in a vein similar to Section 6. Figure G1 reports the estimated weights by year and part of the country, and Figure G2 reports the counterfactual motivation exercise similar to the exercise in Section 6.2.2, which splits the AfD into two hypothetical parties: one identical to the AfD on the Left–Right dimension but more pro-EU, and another identical to the AfD on the Anti-EU dimension but more liberal on Left–Right.

The results suggest that the Left–Right dimension is more important for voters in Western Germany in all years except 2013 (when the weights are approximately equal); in the East the Left–Right dimension is somewhat more important in all years except 2025 (when the weights are approximately equal). The motivation exercise highlights the motivation behind the AfD vote: Anti-EU motivations are clearly dominant in 2017 (after the immigration crisis); in other years the motivations are either dominated by the Left–Right dimension (2013, 2021) or are unclear (2025).



Notes. $w_{LR} = 1$ is fixed throughout; w_{EU} should be interpreted as the relative weight placed on the anti-EU dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (state) has its own bliss-point distribution $f(x | m, d)$, while imposing the same weight w_{EU} for all groups within a given year and region. To run this exercise, the grid of possible weights is constructed. The grid is symmetric around 1 and implies relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios.

Figure G1. Weights for the anti-EU dimension w_{EU} relative to Left–Right $w_{LR} = 1$ across years and parts of the country



Notes: Panel (a) illustrates party positions at their national median, and hollow circles illustrate the counterfactual AfD shifts. The areas of AfD voters motivated more by the Left–Right (pink) or Anti-EU (blue) ideological stance of the AfD are plotted. The illustrative weight used is $w_{EU} = 1$. Panel (b) plots bounds on the shares of voters whose bliss points lie in the yellow and purple regions across years. The results are aggregated across the two parts of the country.

Figure G2. Illustration of a target parameter (a) and results (b)

H The Role of the Supply Side: Shapley Values

Let $\theta_{Right}(Pref_t, Comp_t, Post_t)$ denote the share of AfD voters whose ideal points fall in the *Right* region (Figure 7a), given voter preferences $Pref_t$ across demographic groups, the demographic

composition of AfD voters $Comp_t$, and national median party positions Pos_t . I decompose the change in this share between t and t' as follows:

$$\begin{aligned} & \theta_{Right}(Pref_{t'}, Comp_{t'}, Pos_{t'}) - \theta_{Right}(Pref_t, Comp_t, Pos_t) \\ &= \phi(Pref) + \phi(Comp) + \phi(Pos) \end{aligned} \tag{H1}$$

where each Shapley value, e.g. $\phi(Pos)$, is the weighted average marginal contribution of this factor across all 2^2 coalitions of the other two factors:

$$\begin{aligned} \phi(Pos) = & \frac{1}{3} \left[\theta_{Right}(Pref_{t'}, Comp_{t'}, Pos_{t'}) - \theta_{Right}(Pref_{t'}, Comp_{t'}, Pos_t) \right] \\ & + \frac{1}{6} \left[\theta_{Right}(Pref_t, Comp_{t'}, Pos_{t'}) - \theta_{Right}(Pref_t, Comp_{t'}, Pos_t) \right] \\ & + \frac{1}{6} \left[\theta_{Right}(Pref_{t'}, Comp_t, Pos_{t'}) - \theta_{Right}(Pref_{t'}, Comp_t, Pos_t) \right] \\ & + \frac{1}{3} \left[\theta_{Right}(Pref_t, Comp_t, Pos_{t'}) - \theta_{Right}(Pref_t, Comp_t, Pos_t) \right], \end{aligned}$$

and $\phi(Comp)$, $\phi(Pref)$ are defined analogously. The weights $\frac{1}{3}, \frac{1}{6}$ follow from $\frac{|S|!(n-|S|-1)!}{n!}$ with $n = 3$ ($|S| = 0, 2$ give $\frac{1}{3}$; $|S| = 1$ gives $\frac{1}{6}$). By construction $\phi(Pref) + \phi(Comp) + \phi(Pos)$ equals the left-hand side of (H1) exactly.

To evaluate each counterfactual share, I combine three ingredients. Party positions (Pos_t) are the estimated national median positions in year t ; voter preferences ($Pref_t$) enter as group-specific ideal-point distributions, estimated as a primitive of the model, $f(x | m, d)$.

Composition ($Comp_t$) comes from the Representative Electoral Statistics. These are unique data on voting choices for 12 age–gender groups per federal state.⁵² These data do not permit disaggregation by education, which I therefore omit.

Table H1 presents the results for $\phi(Pos)$ across every pair of consecutive years and regions of the country. The last row provides conservative bounds aggregated over the two parts of the country. Table I8 also presents the Shapley values for the other two channels (composition and preferences), but bounds for those channels are wide and uninformative.

The bounds on the Shapley values for the Positions channel suggest that the supply-side channel contributes positively to the increase in the share of Right-motivated voters over 2021–2025, while its contribution for the other pairs of years leans negative. What these results make clear is that the evolution over time in the share of Right-motivated voters documented in this paper is not driven by parties changing their positions.

⁵²They are based on a sample of polling districts in which marked ballots allow votes and turnout to be tabulated by gender and age group. Hence, these data are based on actual ballots, not surveys.

	2013→2017	2017→2021	2021→2025
<i>West</i>			
Positions	[−0.202, −0.026]	[−0.047, +0.000]	[+0.023, +0.117]
<i>East</i>			
Positions	[−0.103, −0.018]	[+0.000, +0.018]	[−0.041, −0.005]
<i>National (AfD-electorate weighted, conservative)</i>			
Positions	[−0.188, −0.025]	[−0.038, +0.004]	[+0.008, +0.092]

Note: Bounds on the Shapley contribution of the positions channel, derived from a single linear program over the pinned feasible set with both position-year catchments held jointly. Positive values raise the Right share. The national row combines the East and West position bounds by each year’s AfD electorate-population weights, taking the worst case over the two year-weights of the pair; it is conservative relative to separate regional bounds.

Table H1. Shapley Decomposition of the Change in the Right Share of AfD Support

I Tables & Figures

	<i>Perceived Party Position, L-R</i>			<i>Own L-R Position</i>
	All Parties	Chosen Parties	Chosen AfD	Chosen Parties
	(1)	(2)	(3)	(4)
<i>Panel A: no individual controls</i>				
Candidate, L-R	−0.007 (0.080)	0.276** (0.135)	1.207*** (0.213)	0.106 (0.166)
Demographic controls	No	No	No	No
<i>Panel B: with demographic controls</i>				
Candidate, L-R	−0.015 (0.078)	0.359** (0.140)	1.243** (0.408)	0.174 (0.169)
Demographic controls	Yes	Yes	Yes	Yes
State F.E.	Yes	Yes	Yes	Yes
Party F.E.	Yes	Yes	No	Yes
Observations	19,992	2,269	210	2,219

Note: Coefficient on the estimated economic L–R position of the respondent’s district candidate of the party; GLES 2021 pre- and post-election cross-sections, survey weights. Columns (1)–(3): dependent variable is the respondent’s placement of the party on the 1–11 left–right scale, for all parties, for the party chosen with the second vote (party-list), and for AfD voters only; column (4): the respondent’s own left–right self-placement. Panel B adds gender, age, age² and education level controls. Sample: districts of the estimation sample (districts with at most three imputed candidates). Standard errors clustered by party × state in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table I1. Correlation of LR Positions between Candidates (estimated) and Parties (reported), GLES 2021, estimation sample

	<i>Candidate LR Position</i>		
	All Parties	Chosen Parties	Chosen AfD
	(1)	(2)	(3)
<i>Panel A: no individual controls</i>			
Self, LR	0.001 (0.001)	0.002 (0.004)	0.000 (0.017)
Demographic controls	No	No	No
<i>Panel B: with demographic controls</i>			
Self, LR	0.002 (0.001)	0.004 (0.004)	0.000 (0.017)
Demographic controls	Yes	Yes	Yes
State F.E.	Yes	Yes	Yes
Party F.E.	Yes	Yes	No
Observations	19,450	2,219	205

Note: Estimation is based on the GLES 2021 (pre- and post-election cross-sections), which records each individual's electoral district; survey weights. Dependent variable: estimated economic L–R position of the party's district candidate. Self-reported views take values from 1 (very left) to 11 (very right). Panel B adds gender, age, age² and school education (four groups). Sample: districts of the estimation sample (districts with at most three imputed candidates). Standard errors clustered by party $\times$ state in parentheses. *p<0.1; **p<0.05; ***p<0.01.

Table I2. Candidates' LR positions are uncorrelated with voters' self-reported views

	CDU/CSU	SPD	Grüne	FDP	Linke	AfD
	(1)	(2)	(3)	(4)	(5)	(6)
Raw difference	4.37 (0.24)	−3.16 (0.19)	−2.38 (0.16)	−0.60 (0.06)	−1.00 (0.17)	2.09 (0.18)
Within state	4.44 (0.18)	−1.77 (0.13)	−2.23 (0.12)	−0.34 (0.05)	−1.47 (0.06)	1.11 (0.11)
Within district	3.55 (0.15)	−1.59 (0.11)	−1.53 (0.09)	−0.07 (0.04)	−1.33 (0.05)	0.72 (0.09)

Each cell is the coefficient on $\mathbf{1}\{\text{eligible voters} < 10,000\}$, in percentage points, from a separate regression of the municipality's second-vote share on that indicator, pooling the 2013, 2017, 2021 and 2025 elections. *Raw difference* holds only the election year fixed; *within state* adds a state $\times$ year effect; *within district* adds a district $\times$ year effect. Standard errors clustered by municipality in parentheses.

Table I3. Small-minus-large difference in second-vote shares, pooled over 2013–2025, at three levels of comparison

	2013	2017	2021	2025
Mean	0.001	0.001	0.001	-0.001
SD	0.016	0.033	0.032	0.048
P10	-0.017	-0.036	-0.034	-0.062
Median	0.000	0.000	0.000	0.000
P90	0.020	0.041	0.040	0.058

Note: The unit of observation is the municipality. Residual = observed AfD vote share – model-predicted share.

Table I4. Distribution of Residuals Across Municipalities

Year	Region	Total SD		Between-party		Within-party		AfD only	
		econ	anti-el.	econ	anti-el.	econ	anti-el.	econ	anti-el.
2013	East	0.63	0.60	0.66	0.58	0.21	0.29	0.22	0.11
2013	West	0.64	0.64	0.65	0.61	0.24	0.31	0.25	0.38
2017	East	0.66	0.58	0.67	0.56	0.24	0.28	0.29	0.26
2017	West	0.72	0.64	0.75	0.62	0.22	0.30	0.26	0.28
2021	East	0.72	0.63	0.71	0.56	0.32	0.36	0.48	0.43
2021	West	0.71	0.61	0.74	0.58	0.24	0.30	0.29	0.29
2025	East	0.69	0.42	0.72	0.33	0.22	0.29	0.23	0.32
2025	West	0.75	0.48	0.80	0.41	0.19	0.30	0.19	0.35

Notes: Standard deviations of district-level (Wahlkreis) candidate positions; between-party = SD of party means, within-party = SD of positions around the own-party mean. The between-party dispersion on the anti-elite dimension compresses sharply in 2025, reflecting party convergence and the thinner 2025 item battery.

Table I5. Dispersion of estimated candidate positions by year and region

Survey item (question)	Theme	GESIS	Scale	σ (1 pt)
<i>Anti-elite dimension</i>				
Satisfaction with how democracy works in Germany	Current instit.	c20	1–4	0.175
Parties are citizens' central intermediaries	Current instit.	d6c	1–5	0.144
Legislation reflects citizens' interests	Current instit.	d6b	1–5	0.137
Citizens have lost trust in politics	Current instit.	d6h	1–5	0.137
Referendums lead to badly designed laws	Direct democracy	d6g	1–5	0.138
There are enough opportunities to participate	Direct democracy	d6a	1–5	0.117
There should be federal-level referendums	Direct democracy	d6f	1–5	0.098
<i>Left–right dimension</i>				
Own position on immigration (self-placement)	Immigration	c6	1–11	0.032
Immigrants are good for the German economy	Immigration	c2i	1–5	0.093
Immigrants should adapt to German culture	Immigration	c2a	1–5	0.073
Same-sex marriage should be banned	Culture	c2d	1–5	0.096
Women should decide themselves about abortion	Culture	c2j	1–5	0.085
Providing social security is a core government task	Econ. policy	c2g	1–5	0.081
Government spending on development aid	Econ. policy	c18h	1–5	0.080
The government should stay out of the economy	Econ. policy	c2b	1–5	0.071
The government should reduce income differences	Econ. policy	c2h	1–5	0.070
Government spending on defense	Econ. policy	c18c	1–5	0.070

Notes. σ is the change in the standardized dimension score (Left–right or anti-elite) induced by moving a single item one scale point; it is identical across parties because the score is linear in the answers. To move a dimension score by a 0.25/0.5/1 step using a single item requires about $0.25/\sigma$, $0.5/\sigma$, $1/\sigma$ scale points: e.g. $\approx 1.4/2.9/5.7$ points on the strongest anti-elite item (satisfaction with democracy) and far more than the scale allows for the weak eleven-point immigration self-placement, so sizeable steps must come from moving many items at once. Units and item set are held fixed across years; the 2017 survey items are provided as an example.

Table I6. Item-level potency (2017): shift in the dimension score from a one-point move on each survey item

Survey item (question)	Theme	GESIS	Scale	σ (1 pt)
<i>Anti-elite dimension</i>				
Satisfaction with how democracy works in Germany	Current instit.	c25	1–4	0.245
Parties are citizens’ central intermediaries	Current instit.	d4c	1–5	0.201
Legislation reflects citizens’ interests	Current instit.	d4b	1–5	0.192
Citizens have lost trust in politics	Current instit.	d4g	1–5	0.192
There should be federal-level referendums	Direct democracy	d4f	1–5	0.138
<i>Left–right (economic) dimension</i>				
Own position on immigration (self-placement)	Immigration	c6	1–11	0.035
Immigrants are good for the German economy	Immigration	c2i	1–5	0.102
Immigrants should adapt to German culture	Immigration	c2a	1–5	0.080
Marriage should be only for heterosexual couples	Culture	c2d	1–5	0.106
Women should decide themselves about abortion	Culture	c2j	1–5	0.093
Providing social security is a core government task	Econ. policy	c2g	1–5	0.089
The government should stay out of the economy	Econ. policy	c2b	1–5	0.078
The government should reduce income differences	Econ. policy	c2h	1–5	0.076
Own position on taxes vs. social spending	Econ. policy	c9	1–11	0.030

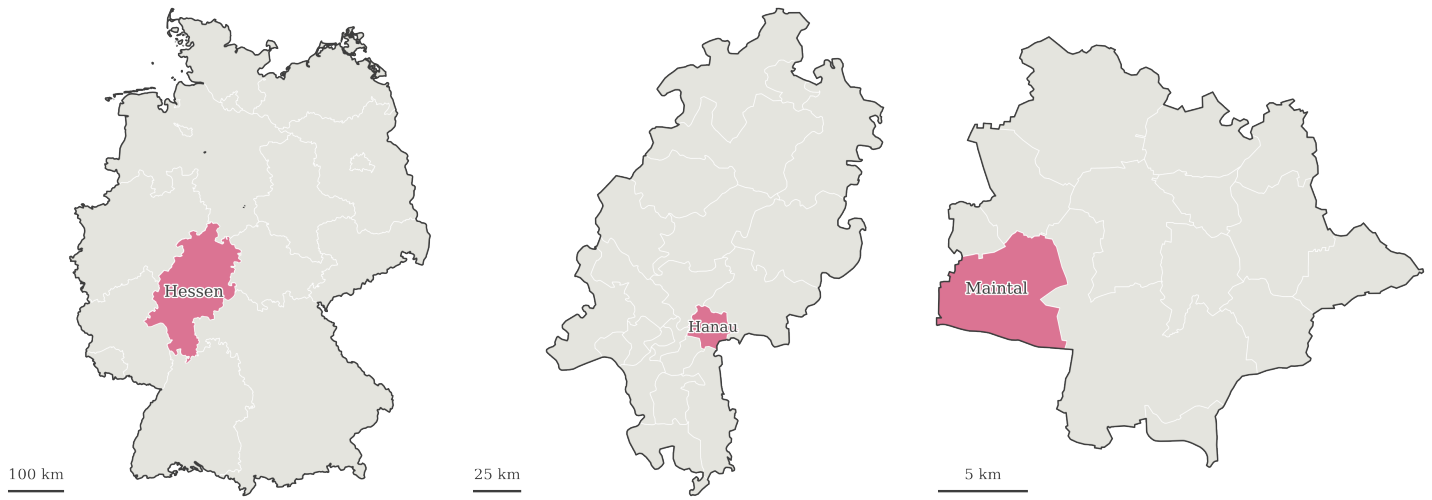
Notes. σ is the change in the standardized dimension score (Left–right or anti-elite) induced by moving a single item one scale point; it is identical across parties because the score is linear in the answers. To move a dimension score by a 0.25/0.5/1 step using a single item requires about $0.25/\sigma$, $0.5/\sigma$, $1/\sigma$ scale points: e.g. $\approx 1.4/2.9/5.7$ points on the strongest anti-elite item (satisfaction with democracy) and far more than the scale allows for the weak eleven-point immigration self-placement, so sizeable steps must come from moving many items at once. Units and item set are held fixed across years; the 2025 survey items are provided as an example.

Table I7. Item-level potency (2025): shift in the dimension score from a one-point move on each survey item

	2013→2017	2017→2021	2021→2025
<i>West</i>			
Composition	[−0.102, +0.141]	[−0.138, +0.119]	[−0.122, +0.106]
Positions	[−0.202, −0.026]	[−0.047, +0.000]	[+0.023, +0.117]
Preferences	[−0.287, +0.506]	[−0.432, +0.466]	[−0.345, +0.295]
<i>East</i>			
Composition	[−0.090, +0.048]	[−0.066, +0.042]	[−0.031, +0.045]
Positions	[−0.103, −0.018]	[+0.000, +0.018]	[−0.041, −0.005]
Preferences	[−0.384, +0.234]	[−0.233, +0.312]	[−0.302, +0.157]

Notes: Bounds on the Shapley contribution of Composition, Positions and Preferences channels, derived from a single linear program over the pinned feasible set with both position-year catchments held jointly. Positive values raise the Right share (θ_{Right} from Section 6.2.2).

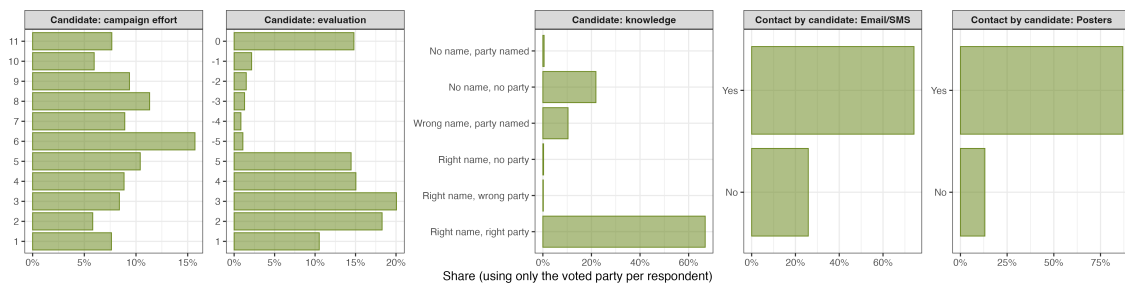
Table I8. Shapley Decomposition of the Change in the Right Share of AfD Support



(a) **Federal states.** Hessen high-
lighted in pink. (b) **Electoral districts** within
Hessen. Hanau highlighted in pink. (c) **Municipalities** within Hanau.
Maintal highlighted in pink.

Figure I1. Administrative and electoral hierarchy in Germany.

Municipalities are nested within electoral districts, which are in turn nested within federal states. The figure illustrates these levels for one example state and one example electoral district.



Notes. **Campaign effort (1–11):** 1 = mostly about the candidate; 11 = mostly about the party; **Candidate knowledge:** respondents' ability to name the constituency candidate and their party; **Contact:** whether the respondent was contacted via the listed means. Data are for the 2021 Bundestag election.

Figure I2. Voters' knowledge of and interaction with SMD candidates

Partei	Insgesamt		davon Wählerinnen und Wähler in der Geburtsjahresgruppe											
			1997 – 2003		1987 – 1996		1977 – 1986		1962 – 1976		1952 – 1961		1951 und früher	
			entspricht Altersgruppe etwa von ... bis ... Jahren											
			18 – 24		25 – 34		35 – 44		45 – 59		60 – 69		70 und mehr	
	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%	1.000	%

Mecklenburg-Vorpommern

Insgesamt

Insgesamt	935,0	100	52,1	100	91,9	100	135,3	100	241,1	100	205,3	100	209,4	100
ungültig	17,0	1,8	0,9	1,7	0,9	1,0	2,2	1,6	3,2	1,3	4,1	2,0	5,8	2,8
gültig	918,1	98,2	51,2	98,3	91,0	99,0	133,1	98,4	237,9	98,7	201,2	98,0	203,7	97,2
CDU	180,7	19,7	5,3	10,4	11,2	12,3	22,2	16,7	48,9	20,6	42,9	21,3	50,0	24,6
SPD	261,4	28,5	9,7	18,9	18,8	20,7	29,0	21,8	59,6	25,1	64,4	32,0	80,0	39,3
AfD	170,0	18,5	7,4	14,4	19,2	21,1	33,5	25,2	52,7	22,1	35,7	17,7	21,5	10,6
FDP	66,3	7,2	8,4	16,4	9,1	10,0	11,5	8,6	18,8	7,9	11,5	5,7	7,0	3,4
DIE LINKE	116,0	12,6	6,7	13,1	9,7	10,7	12,5	9,4	25,3	10,7	28,3	14,1	33,5	16,4
GRÜNE	62,7	6,8	8,4	16,5	12,7	14,0	11,4	8,5	15,0	6,3	8,9	4,4	6,3	3,1
CSU	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Sonstige	61,0	6,6	5,3	10,3	10,3	11,3	13,1	9,9	17,5	7,3	9,5	4,7	5,4	2,6
darunter:														
FREIE WÄHLER ¹	18,5	2,0	0,9	1,7	2,9	3,2	3,2	2,4	5,5	2,3	4,1	2,0	1,9	0,9
dieBasis ¹	15,4	1,7	0,8	1,5	1,9	2,1	3,8	2,8	5,3	2,2	2,2	1,1	1,4	0,7

Figure I3. Representative Electoral Statistics: Example

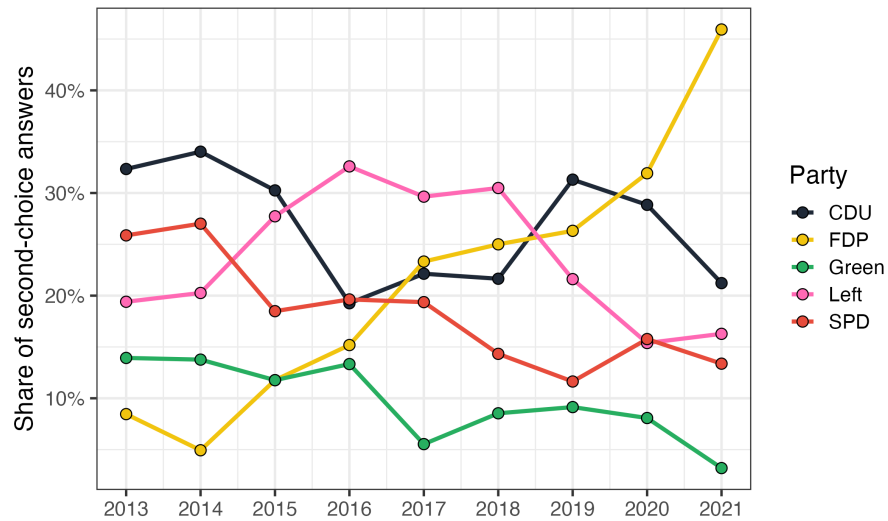
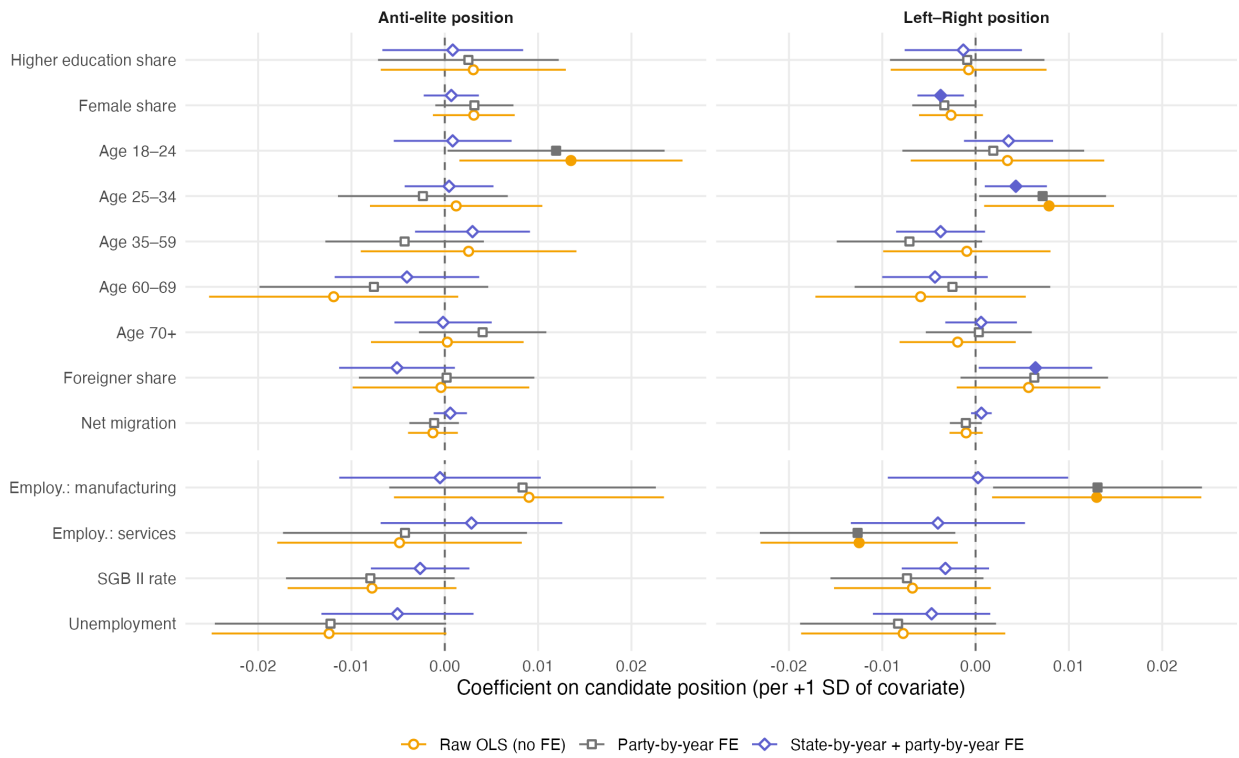
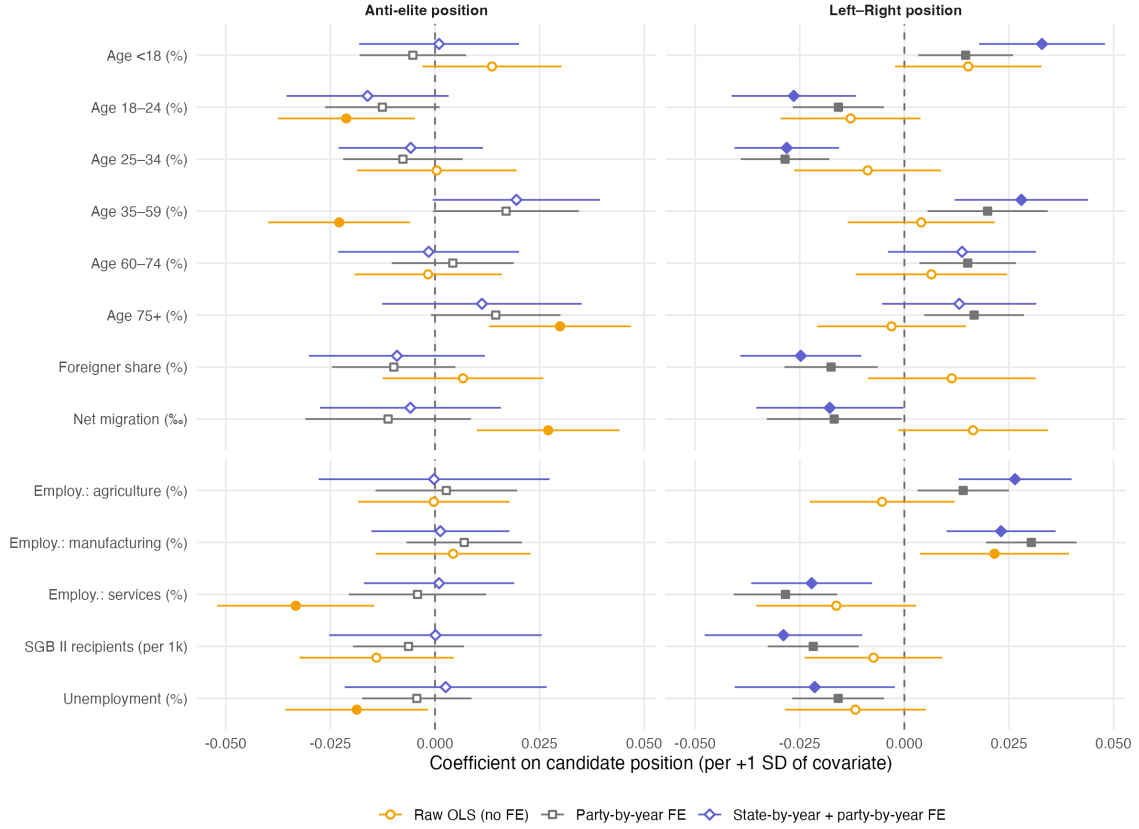


Figure I4. Second-best party choices of AfD voters, federal level



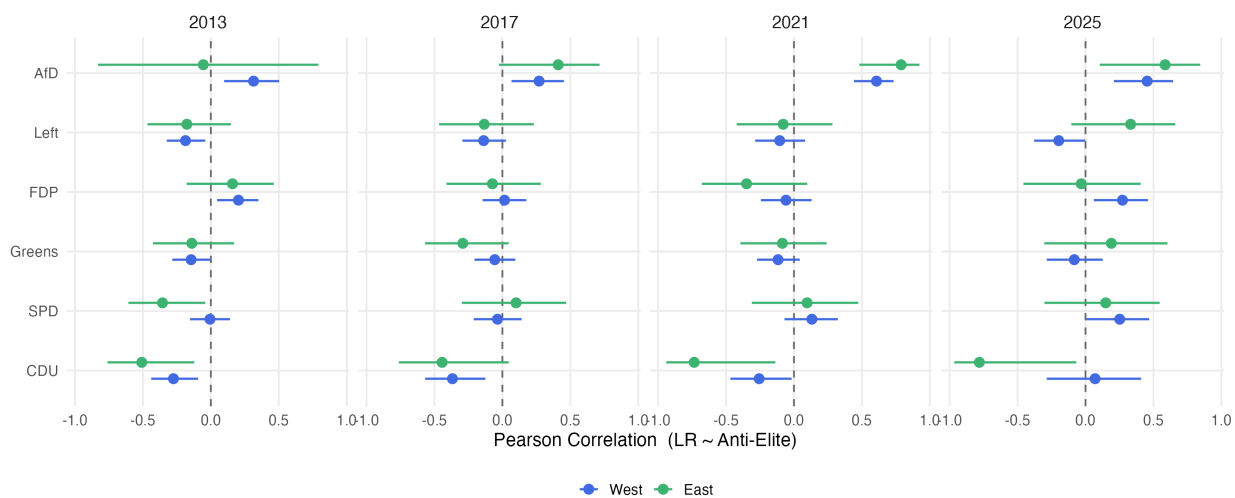
Notes. Each marker is the coefficient from a separate regression of a candidate position on one characteristic. The unit of observation is a municipality-by-party cell in the 2013–2025 elections. The sample consists of municipalities used in the estimation of the structural model (see Sections 4.2 and 5 for details). All covariates are standardized to mean zero and unit standard deviation. The specifications are (i) Raw OLS with no fixed effects, (ii) party-by-year fixed effects, and (iii) state-by-year plus party-by-year fixed effects. Standard errors are clustered by district-by-year. Horizontal bars are 95% confidence intervals. Filled markers denote coefficients significant at the 5% level ($|t| > 1.96$).

Figure I5. Association between candidates' positions on each dimension and municipalities' demographic and socio-economic characteristics.

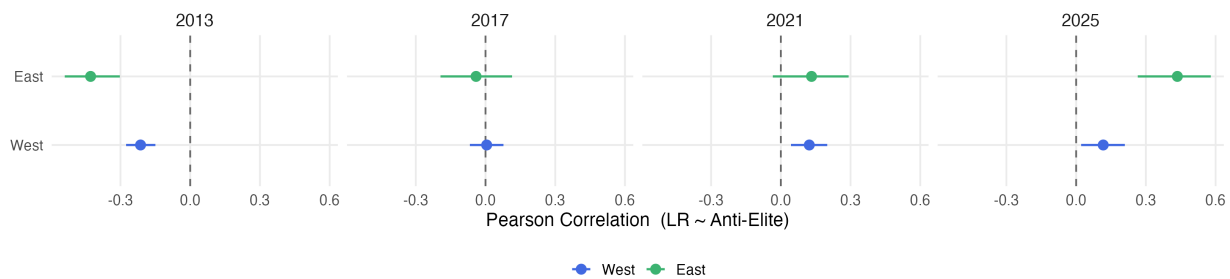


Notes. Each marker is the coefficient from a separate regression of a candidate position on one characteristic. The unit of observation is an electoral district-by-party cell in the 2013–2025 elections. The sample consists of electoral districts used in the estimation of the structural model (see Sections 4.2 and 5 for details). All covariates are standardized to mean zero and unit standard deviation. The specifications are (i) Raw OLS with no fixed effects, (ii) party-by-year fixed effects, and (iii) state-by-year plus party-by-year fixed effects. Standard errors are clustered by district-by-year. Horizontal bars are 95% confidence intervals. Filled markers denote coefficients significant at the 5% level ($|t| > 1.96$).

Figure I6. Association between candidates' positions on each dimension and their electoral districts' demographic and socio-economic characteristics.



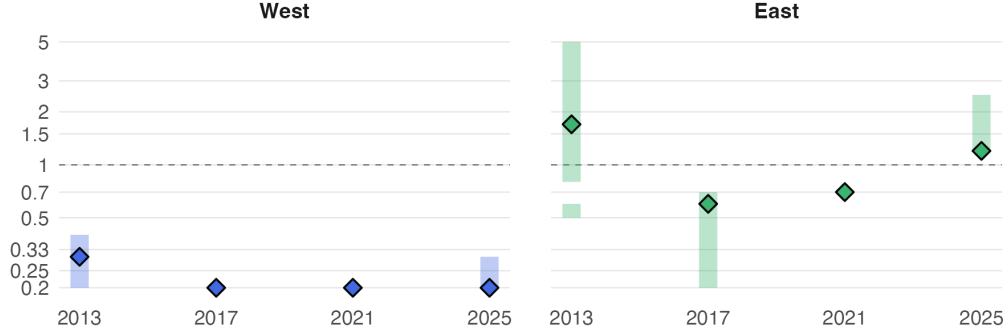
(a) Within parties



(b) Across parties

Figure I7. Correlation between candidates' Left-Right and Anti-Elite positions

J Additional Results



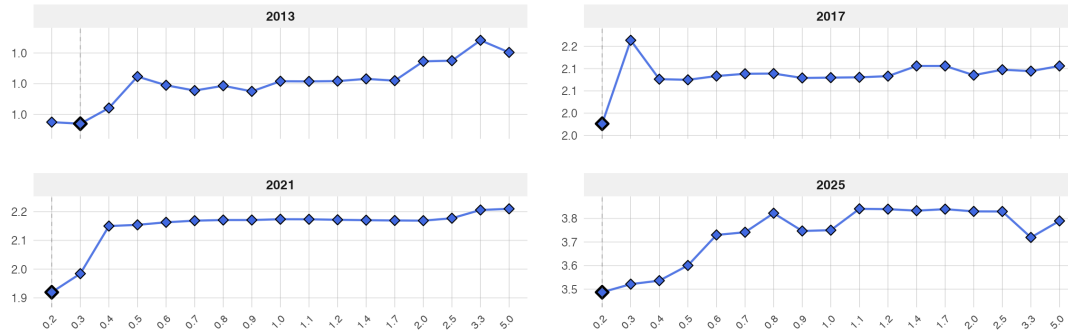
Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Points denote the point estimates corresponding to the best-fitting weight, shaded intervals denote weights within 1% of the minimum objective-function value. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss-point distribution $f(x | m, d)$, while imposing the same weight w_{AE} for all groups within a given year and region. To run this exercise, the grid of possible weights is constructed. The grid is symmetric around 1 and implies relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. See Figure J2 for full results over the grid.

Figure J1. Sets of best-fitting weights for the Anti-Elite dimension w_{AE} relative to Left–Right $w_{LR} = 1$ across years and parts of the country

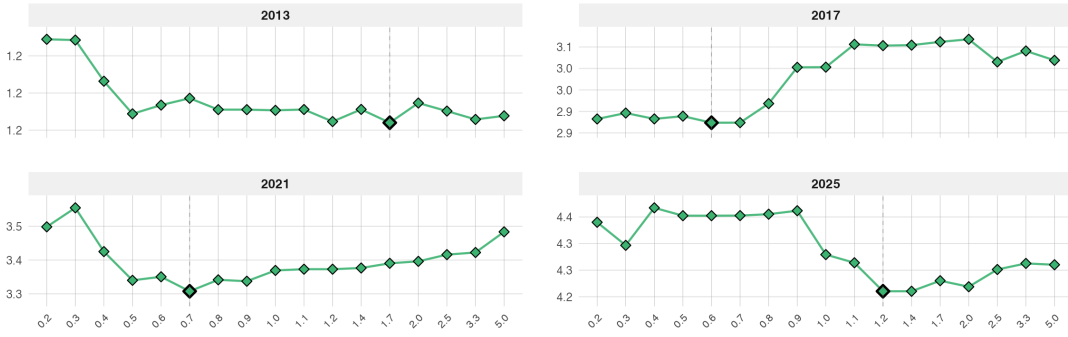
<i>Welfare difference (quadratic), CDU+AfD – CDU+SPD</i>	
National	[−1.43, 0.07]
East	[−3.33, −0.53]
West	[−0.62, 0.33]

Notes: Sharp bounds over the identified set of preference distributions on $\mathbb{E}[d_W^2(x, Y_{CDU+AfD}) - d_W^2(x, Y_{CDU+SPD})]$; negative values mean the CDU+AfD policy is closer to voters on average. Each coalition implements the simple average of its members' 2025 national median positions. National bounds combine East and West by their electorate weights; the regional programs are independent, so the combination is sharp.

Table J1. Welfare difference between the two 2025 coalitions under quadratic loss: equal-weighted policy



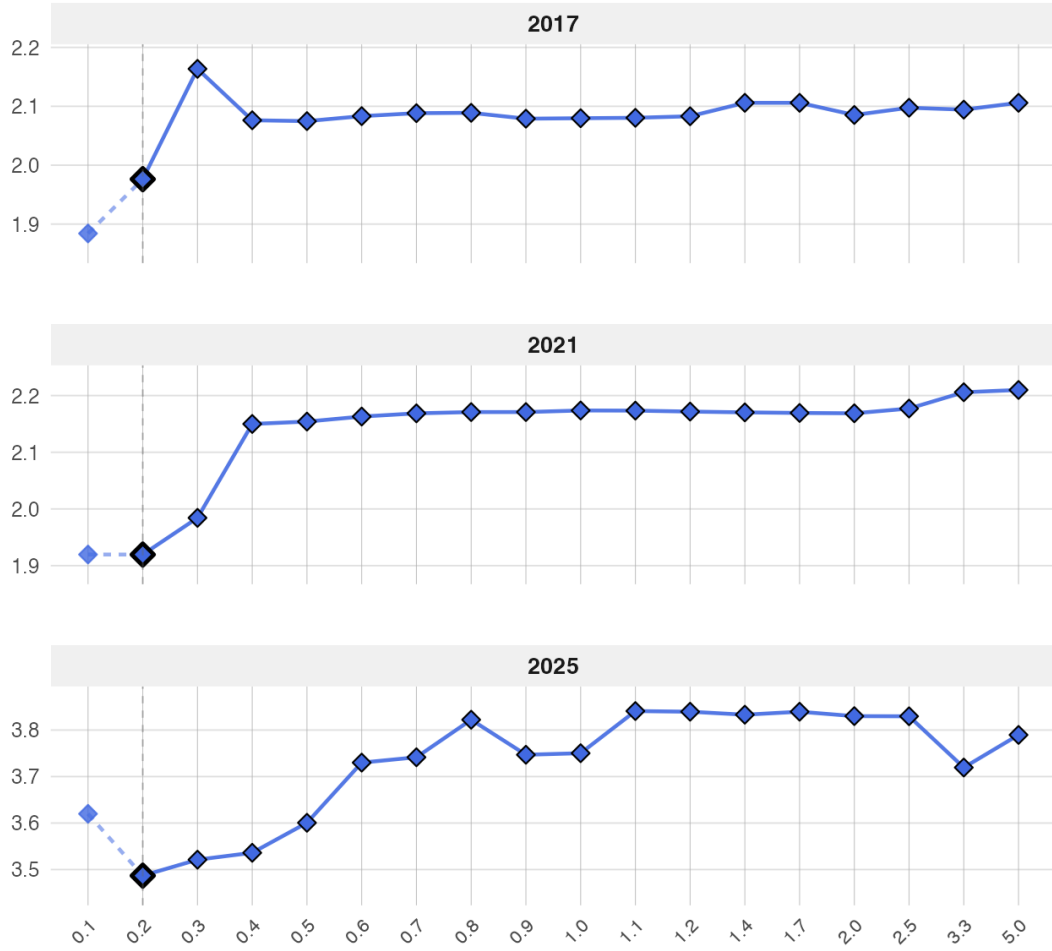
(a) West



(b) East

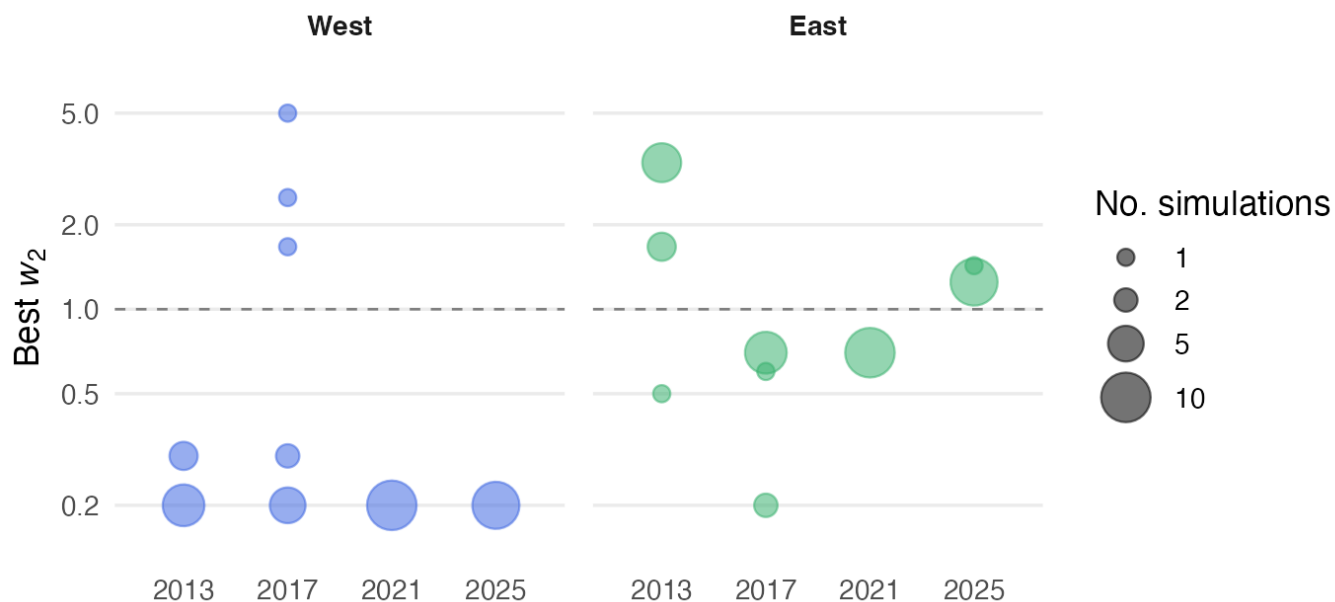
Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss-point distribution $f(x | m, d)$, while imposing the same weight w_{AE} for all groups within a given year and region. Note that the weights are symmetric around 1 and imply relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios.

Figure J2. Model Error Value by different relative weight values, years and parts of the country



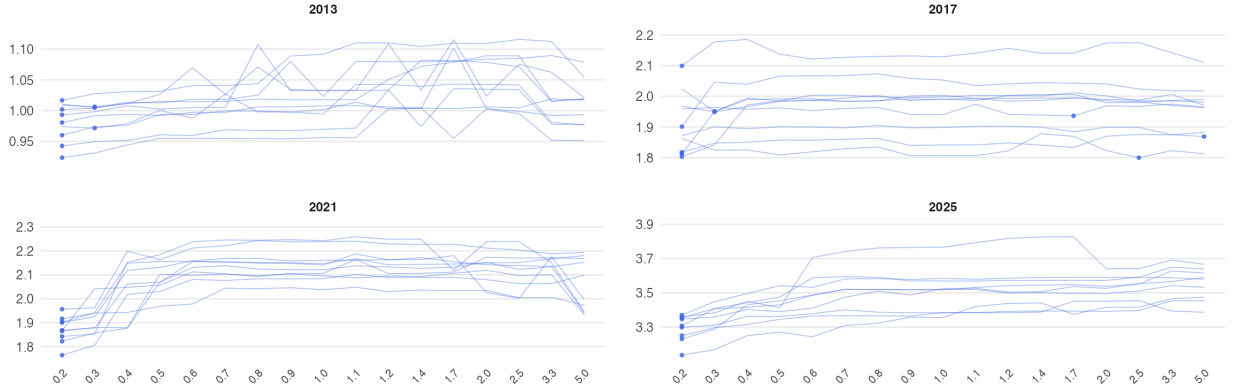
Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss-point distribution $f(x | m, d)$, while imposing the same weight w_{AE} for all groups within a given year and region.

Figure J3. Model Error Value by different relative weight values for the West 2017–2025 with a new point $w = 0.1$ (where the corner $w = 0.2$ is the best-fitting weight).

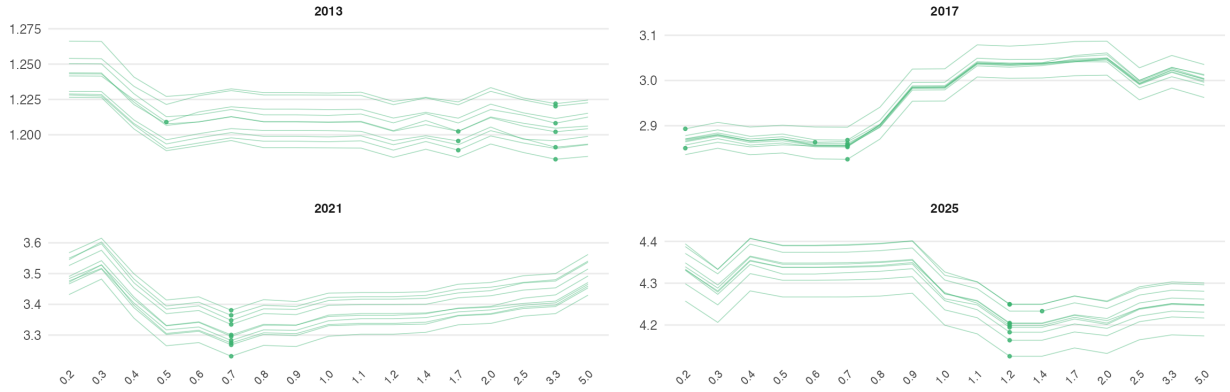


Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss-point distribution $f(x \mid m, d)$, while imposing the same weight w_{AE} for all groups within a given year and region. Note that the weights are symmetric around 1 and imply relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. The exercise was repeated 10 times for each part of the country and each year by subsampling at most 10 electoral districts from each state, with at most 50 municipalities in each district.

Figure J4. Optimal relative weight values by years and parts of the country



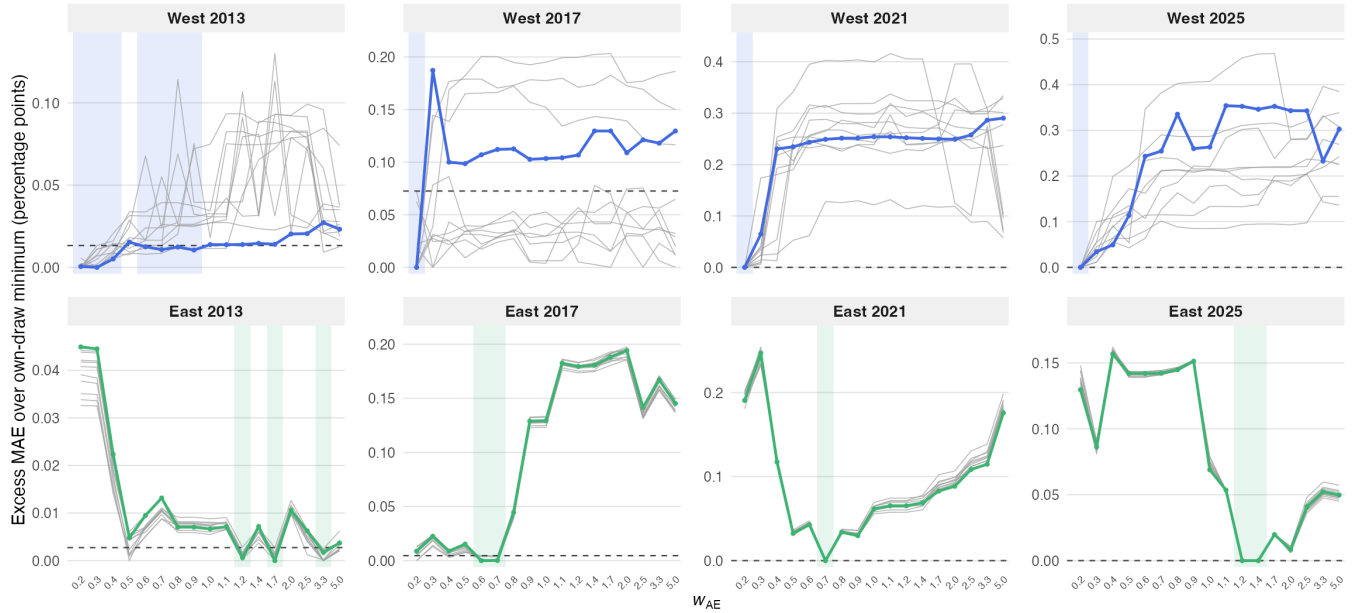
(a) West



(b) East

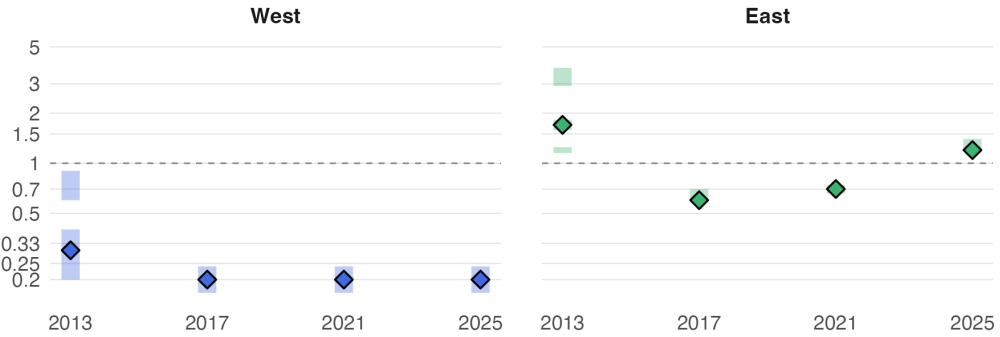
Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (federal state) has its own bliss-point distribution $f(x | m, d)$, while imposing the same weight w_{AE} for all groups within a given year and region. Note that the weights are symmetric around 1 and imply relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios. The exercise was repeated 10 times for each part of the country and each year by subsampling at most 10 electoral districts from each state, with at most 50 municipalities in each district.

Figure J5. Model Error Value by different relative weight values, years and parts of the country by subsamples



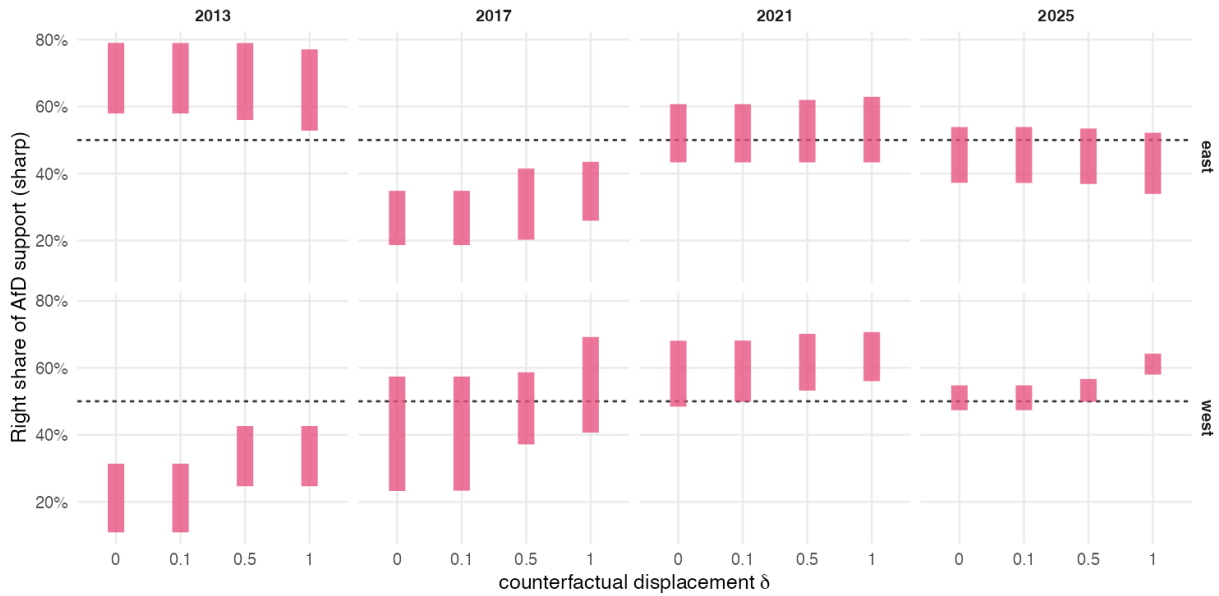
Notes. One panel per region-year. Grey lines are the $B = 10$ subsample MAE curves of Figure J5, each recentered on its own minimum ($D_{bj} = E_{bj} - \min_j E_{bj}$); recentering removes the vertical level shift between draws, which is what makes the raw curves look near-parallel, and leaves only the shape. The coloured line is the full-sample curve of Figure J2, recentered the same way ($\hat{d}_j$). The dashed horizontal line is the cutoff: the $(1 - \alpha)$ inverse-ECDF quantile of D_{bj^*} across draws at the full-sample minimiser j^* , with $\alpha = 0.05$.

Figure J6. Calibration of the weight confidence sets from the subsample re-runs



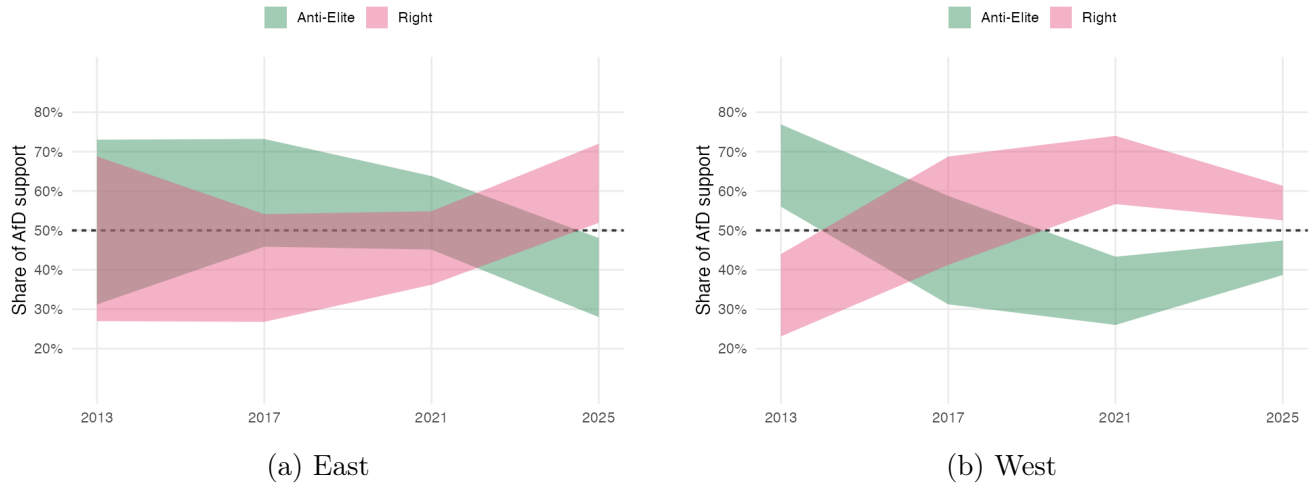
Notes. $w_{LR} = 1$ is fixed throughout; w_{AE} should be interpreted as the relative weight placed on the Anti-Elite dimension. Diamonds denote the point estimates corresponding to the best-fitting weight, identical to those in Figure J1. Shaded bars denote the confidence set CS , calibrated from the $B = 10$ subsample re-runs of Figures J4 and J5 rather than from the fixed 1% tolerance used in Figure J1. Each draw's MAE curve is recentered on its own minimum, $D_{bj} = E_{bj} - \min_j E_{bj}$, which removes the level shift between draws and retains only the curvature; the cutoff is the $(1 - \alpha)$ empirical quantile (inverse-ECDF, $\alpha = 0.05$) of D_{bj^*} . Separate exercises are run for East and West Germany. See Figure J6 for the calibration itself.

Figure J7. Confidence sets for the Anti-Elite weight w_{AE} , calibrated from the subsample re-runs



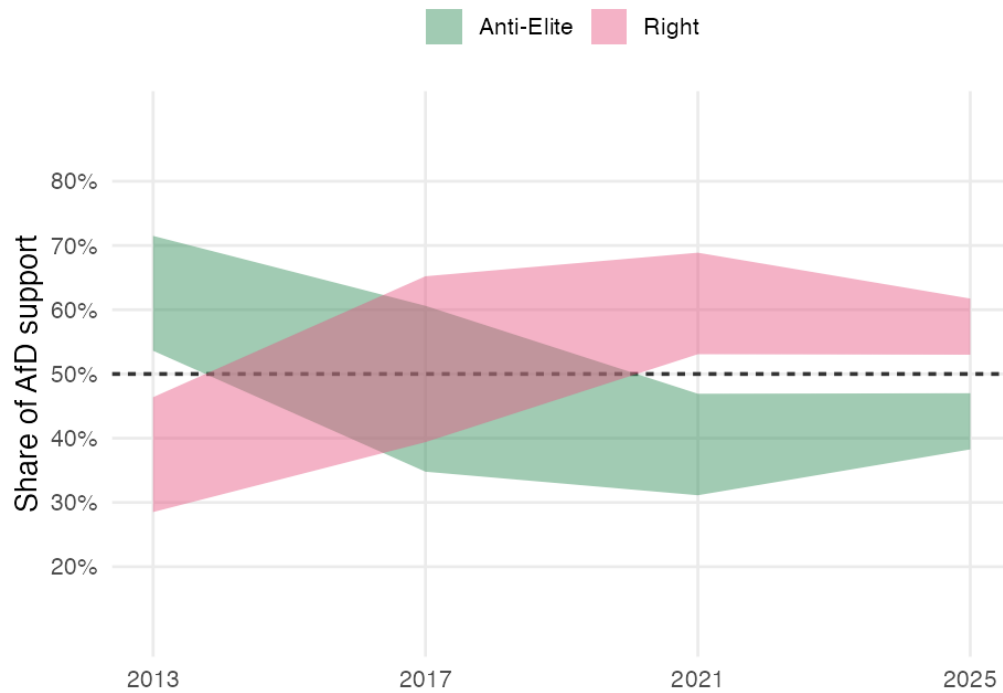
Notes: The smallest δ used is 10^{-6} (not exactly 0).

Figure J8. Bounds on the share of AfD voters preferring the hypothetical Right party, for different steps δ



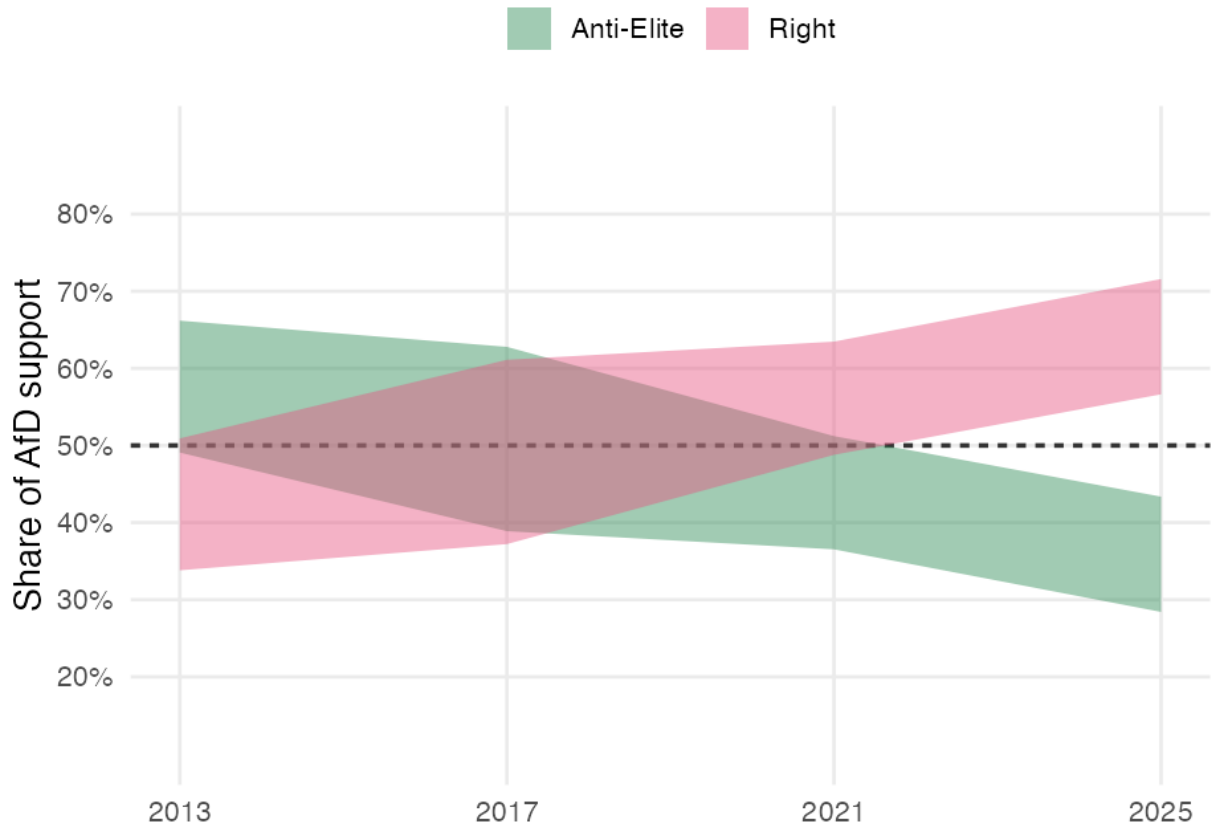
Notes: Panel plots bounds on the shares of voters whose bliss points lie in the attraction areas of two hypothetical parties that retain the AfD's median position within the state on one dimension while adopting a more moderate position on the other: the *Anti-Elite* party is more left-wing (green), whereas the *Right* party is more pro-elite (pink). Together, these areas recover the full AfD support. See Figure 7 for an illustration. The East panel is estimated on the three-survey candidate positions; the West panel allows conservative and moderate districts to differ by a factor of three, as in Figure J13.

Figure J9. Regional Heterogeneity: Results separately for West and East Germany.



Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions in Figure 7a across years are plotted. Second-choice moments are used in the estimation (benchmark specification); the main text reports the estimates without them.

Figure J10. Estimated shares of AfD voters in both subspaces, estimation with second-choice moments.



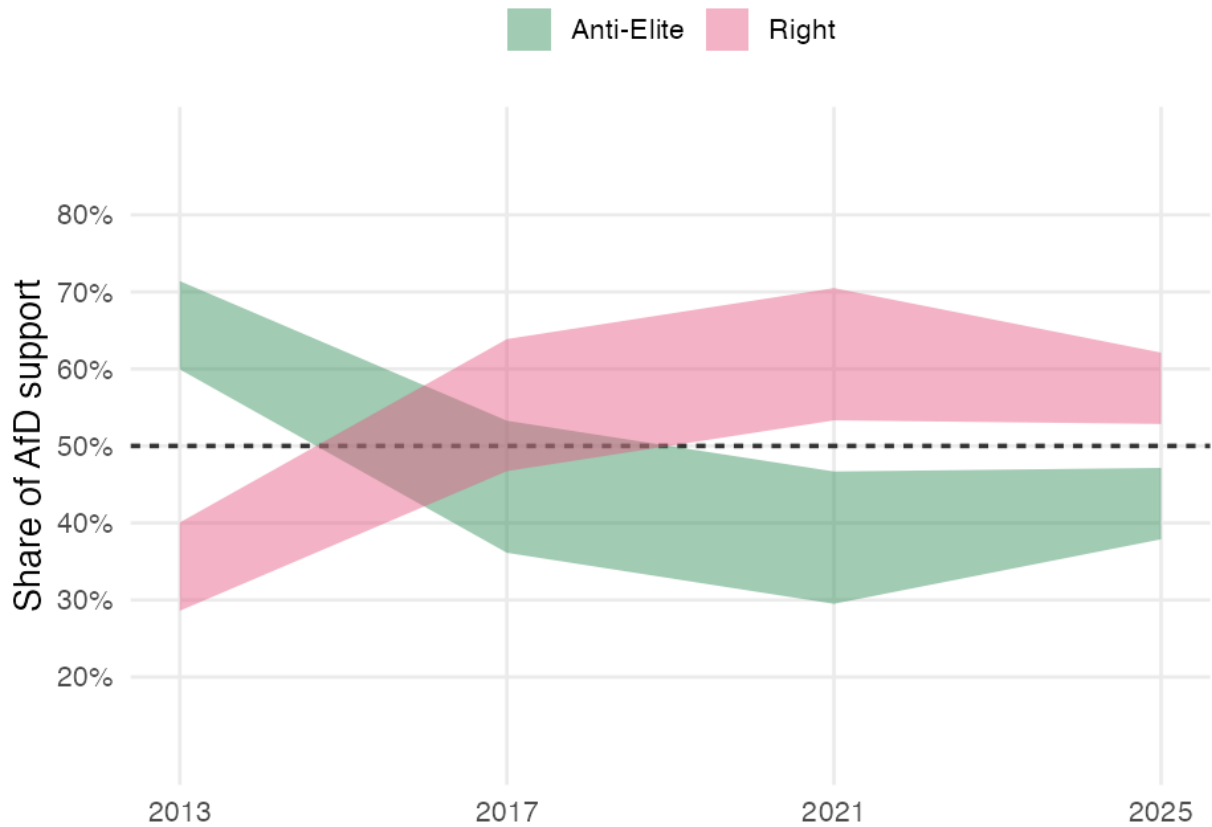
Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions in Figure 7a across years are plotted.

Figure J11. Estimated shares of AfD voters in both subspaces on the sample of municipalities with fewer than 5,000 inhabitants.



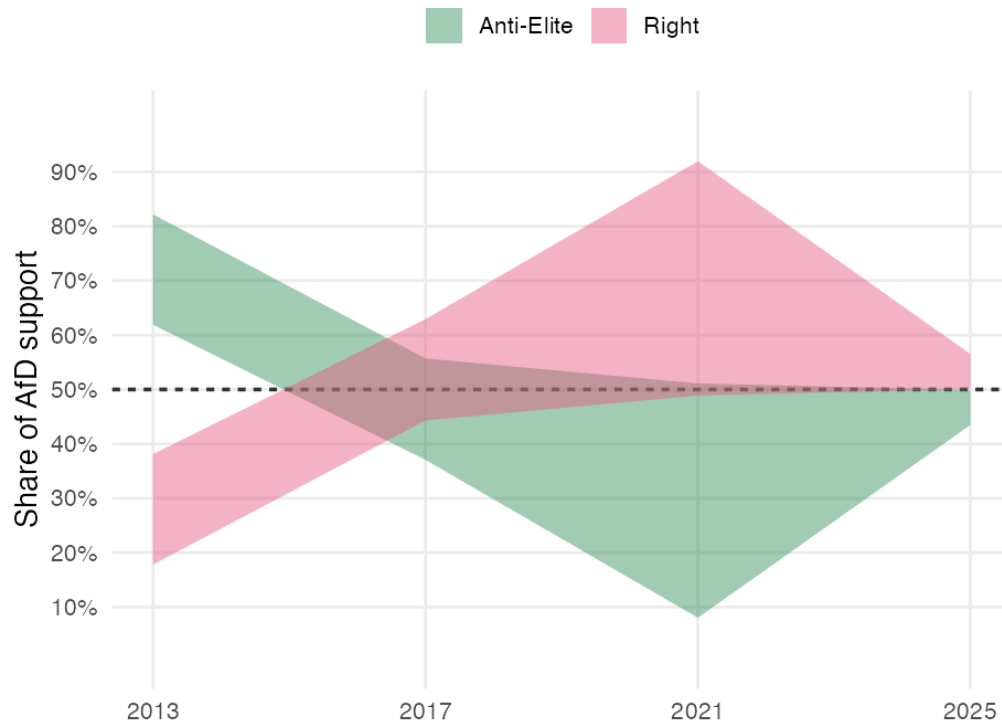
Notes: The figure plots bounds, across years, on the shares of voters whose bliss points lie in the green and pink regions of Figure 7a. Only districts in which the AfD candidate's vote share falls short of the winner's by more than 5% (10%) in the East (West) are retained, for the years 2017 and 2021 (2025). These restrictions drop 12 districts in the East in 2017 and 28 in 2021, and 22 districts in the West in 2025. The East in 2013 and the West in 2013–2021 are omitted because there were no serious contenders, so the results from the main text stand; the East in 2025 is omitted because the AfD won in almost every district, making the analysis infeasible.

Figure J12. Estimated shares of AfD voters in both subspaces on the sample of non-contender electoral districts.



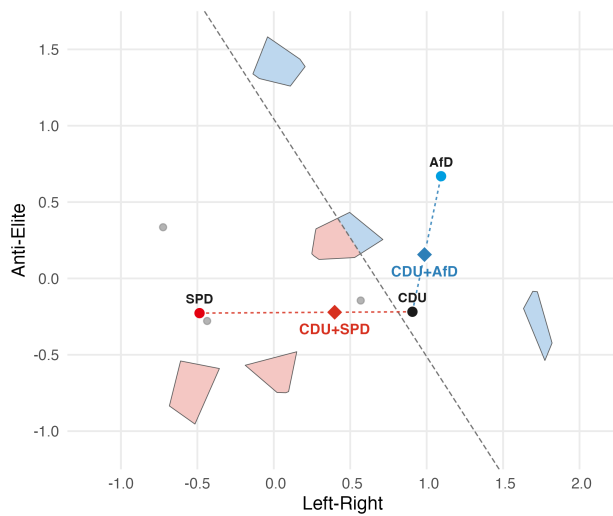
Notes: Bounds on the shares of voters whose bliss points lie in the green and pink regions in Figure 7a across years are plotted. The analysis allows the estimated ideal-point mass of each demographic group, in each partition cell, to differ by up to a factor of three between conservative and moderate electoral districts of the same state—i.e. the conservative-district mass may range from one-third to three times the moderate-district mass in every cell. A district is classified as conservative if its AfD candidate is positioned above the within-state median AfD position on both the Left–Right and the anti-elite dimension, and moderate otherwise. This analysis is similar to an instrument in [Tebaldi, Torgovitsky and Yang \(2023\)](#).

Figure J13. Estimated shares of AfD voters in both subspaces allowing for the differences in ideal points distributions between *conservative* and *moderate* electoral districts.



Notes. Bounds on the shares of voters whose bliss points lie in the green and pink regions in Figure 7a across years are plotted. The sample is restricted to municipalities adjacent to a border between two electoral districts within the same federal state. The preference distribution of each demographic group is imposed to be identical across the municipalities of the same border pair, and is allowed to differ across border pairs within a state by up to 20%; across states the coupling is the same as in the baseline specification.

Figure J14. Estimated shares of AfD voters in both subspaces, estimated on the sample of border municipalities.



Welfare difference, $CDU+AfD - CDU+SPD$

National	$[-0.13, 0.05]$
East	$[-0.24, -0.08]$
West	$[-0.08, 0.10]$

Share of the electorate closer to $CDU+AfD$

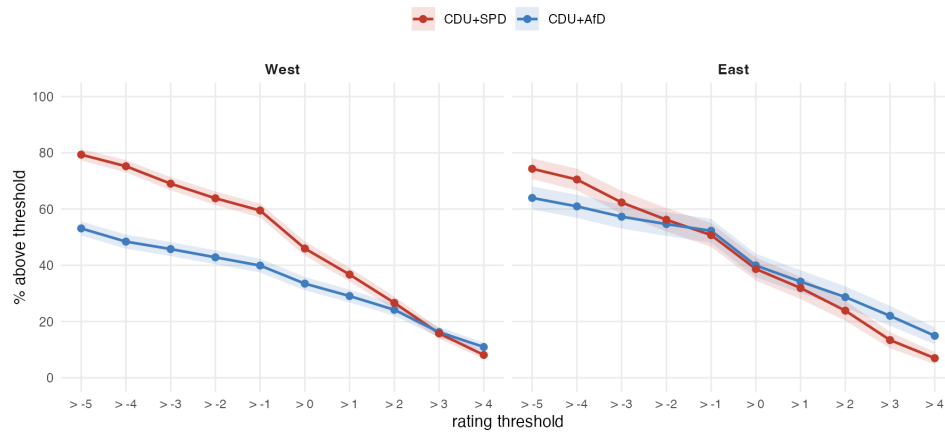
National	$[54.8\%, 70.8\%]$
East	$[69.9\%, 78.6\%]$
West	$[48.3\%, 67.5\%]$

(b) Bounds under the vote-share-weighted policy

(a) Illustration: Coalition policy positions and voters' locations

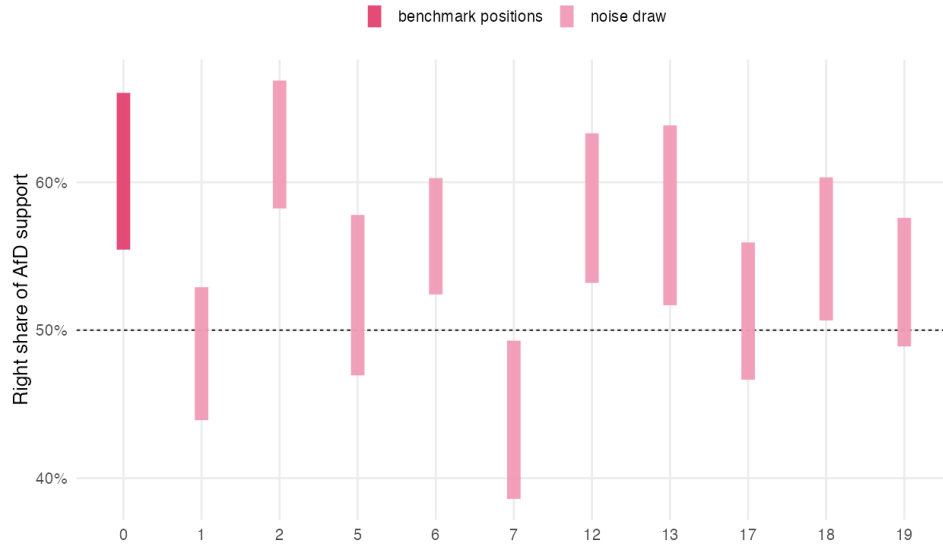
Notes: Same objects as Figure 16, but each coalition implements the vote-share-weighted average of its members' positions (diamonds in Panel (a)), with weights from the 2025 second-vote shares. Both policy points shift toward the CDU. Panel (b) reports bounds over the identified set of preference distributions, constructed exactly as in Figure 16.

Figure J15. Welfare difference and majority preference between the two 2025 coalitions: vote-share-weighted policy



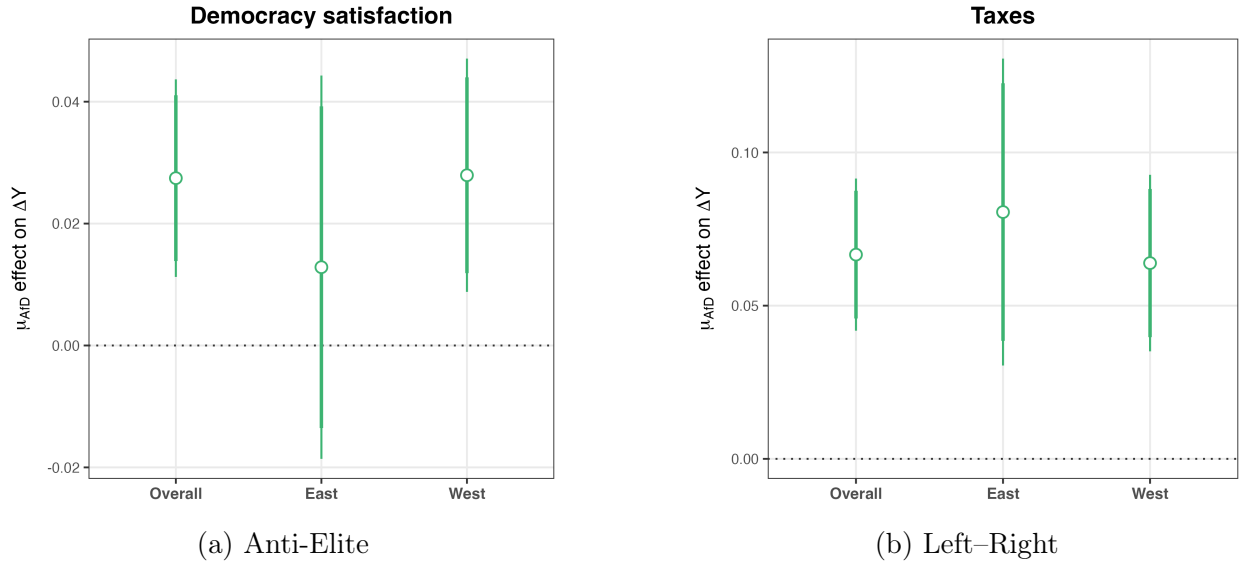
Notes. The figure plots the share of respondents rating each coalition strictly above a given threshold, for every threshold on the eleven-point desirability scale. Respondents are asked how desirable each coalition would be, on a scale from -5 ("not at all desirable") to $+5$ ("extremely desirable"). The same respondents rate both coalitions, so the two curves within a panel are directly comparable. Data are from the GLES Tracking, fielded 12–20 February 2025, one month before the federal election. Shaded bands are 95% confidence intervals.

Figure J16. Desirability of a CDU–SPD and a CDU–AfD coalition, West and East Germany



Notes. The figure shows the behaviour of the target parameter from Section 6.2.2, estimated on the West 2025 data, when candidate positions are perturbed by measurement error. In each of 10 replications, an i.i.d. $N(0, \delta^2)$ shock with $\delta = 0.01$ is added to every estimated candidate position on both dimensions and to the anchor of the hypothetical parties, and the parameter is re-estimated; the resulting values are plotted against the baseline obtained without perturbation. Relative to the spread of the West 2025 district positions, δ equals $0.013 \times \text{s.d.}(X_{LR})$ (s.d. 0.76) and $0.021 \times \text{s.d.}(X_{AE})$ (s.d. 0.48); relative to the AfD's own candidates, $0.05 \times \text{s.d.}$ and $0.03 \times \text{s.d.}$ respectively.

Figure J17. Robustness of the target parameter to measurement error in candidate positions



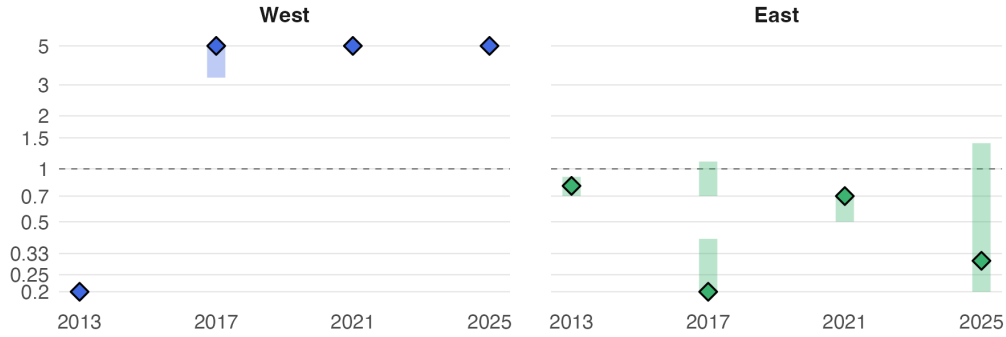
Notes: The panels plot the average yearly within-person evolution of preferences of repeat AfD voters relative to voters of other parties. Based on self-reported positions $Y_{i,t}$ I construct $\Delta Y_{i,t} = Y_{i,t} - Y_{i,t-1}$ for consecutive years $t-1, t$, and drop any respondent-year pair for which AfD status changes across the two waves. That is, I keep only pairs in which either $\text{AfD}_{i,t-1} = \text{AfD}_{i,t} = 1$ (consistent AfD voters) or $\text{AfD}_{i,t-1} = \text{AfD}_{i,t} = 0$ (consistent non-AfD voters). Plotted values are μ_{AfD} from: $\Delta Y_{i,t} = \mu + \mu_{\text{AfD}} \cdot \mathbf{1}\{\text{AfD}_{i,t-1} = \text{AfD}_{i,t} = 1\} + \delta_t + \varepsilon_{i,t}$. Higher values correspond to positions closer to the AfD. The point estimate of μ_{AfD} , together with 90% (thick) and 95% (thin) confidence intervals, is reported for the Overall, East, and West samples. Standard errors clustered at the respondent level.

Figure J18. Results for within-person preference evolution for repeat AfD voters relative to voters of other parties

	Overall		East		West	
	No FE	Year FE	No FE	Year FE	No FE	Year FE
μ	0.0229*** (0.0047)	0.0260*** (0.0048)	0.0165* (0.0099)	0.0183* (0.0102)	0.0249*** (0.0053)	0.0282*** (0.0054)
Year FE	No	Yes	No	Yes	No	Yes
Obs. (ΔY)	4806	4806	1155	1155	3651	3651
Individuals	2182	2182	506	506	1688	1688

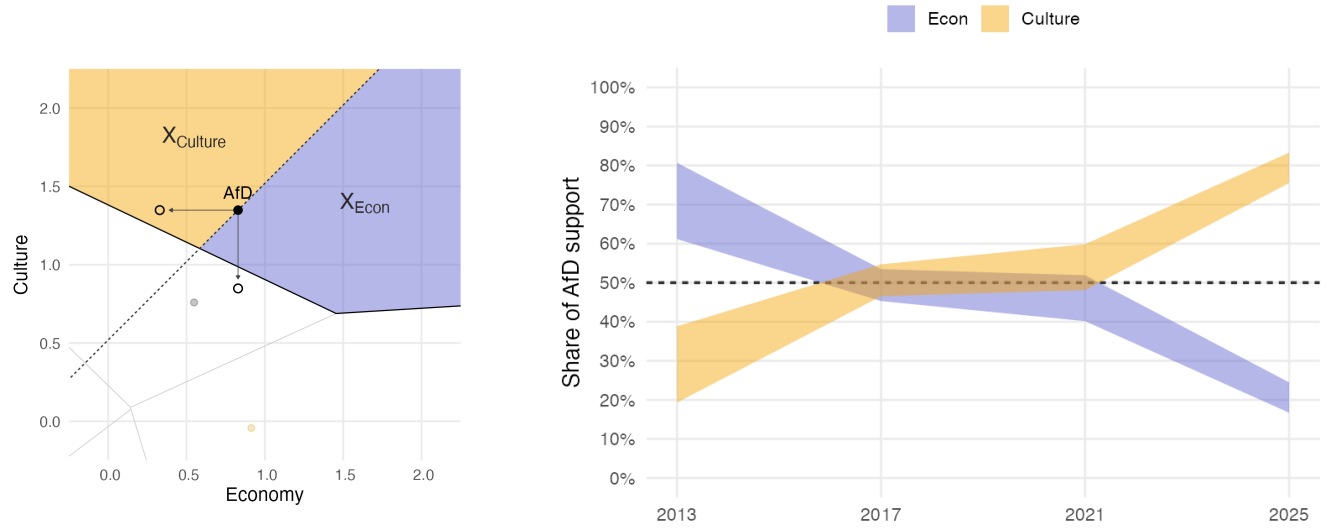
Notes: Sample: AfD voters only (from their first AfD vote onward). For each year t I compute the *median* AfD voter's position in the normalized (taxes, democracy-satisfaction) space, where both outcomes are oriented so that higher values are closer to the AfD, and draw a 45 degree line through the point, and define a dummy $\text{Below45}_{i,t}$ for being "below" the 45 degree line. I estimate, on consecutive-year pairs in which i is an AfD voter in t , $\text{Below45}_{i,t+1} - \text{Below45}_{i,t} = \mu + \delta_t + \varepsilon_{i,t}$. Standard errors are clustered at the respondent level. Columns are estimated separately on the Overall, East, and West subsamples.

Table J2. Annual drift in AfD voters' "below 45" dummy



Notes. $w_{econ} = 1$ is fixed throughout; w_{cult} should be interpreted as the relative weight placed on the cultural dimension. Separate exercises are run for East and West Germany. I allow ideal-point distributions to vary across federal states – that is, each demographic group d in each market m (state) has its own bliss-point distribution $f(x | m, d)$, while imposing the same weight w_{cult} for all groups within a given year and region. To run this exercise, the grid of possible weights is constructed. The grid is symmetric around 1 and implies relative weights across the two dimensions of $(0.2, 0.3, \dots, 0.9, 1)$ and the corresponding inverse ratios.

Figure J19. Weights for the Cultural dimension w_{cult} relative to Economic $w_{econ} = 1$ across years and parts of the country

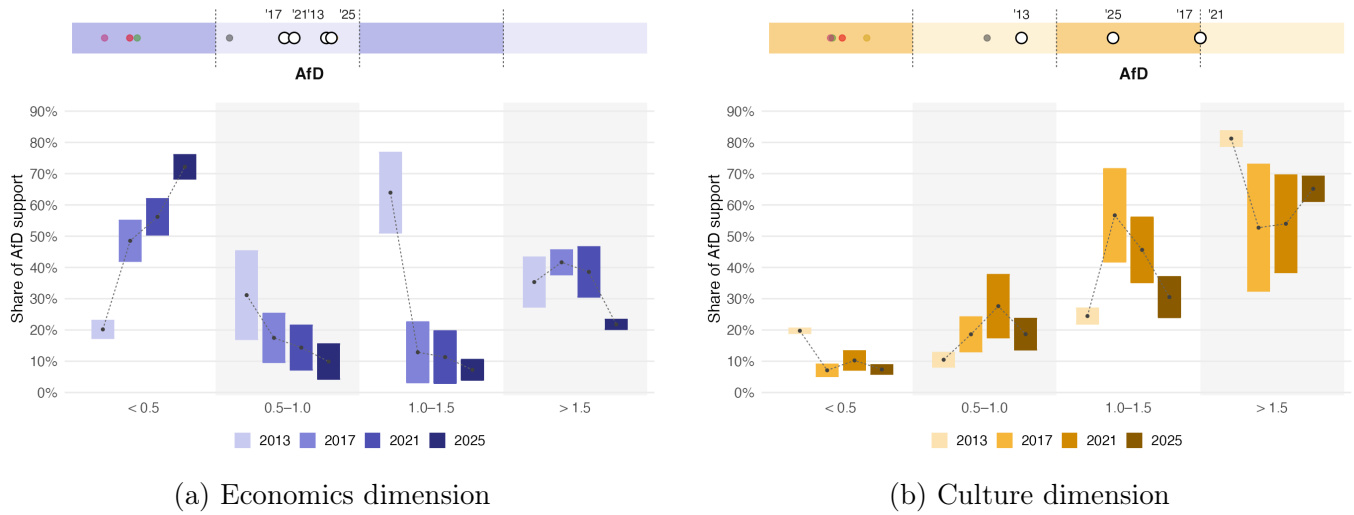


(a) Illustration

(b) Estimated shares of voters preferring the hypothetical Economically Conservative vs. Culturally Conservative party among AfD voters

Notes: Panel (a) illustrates party positions at their national median, and hollow circles illustrate the counterfactual AfD shifts. The areas of AfD voters motivated more by economic (purple) or cultural (yellow) ideological stance of AfD are plotted. The illustrative weight used is $w_{cult} = 1$. Panel (b) plots bounds on the shares of voters whose bliss points lie in the yellow and purple regions across years. The results are aggregated across the two parts of the country.

Figure J20. Illustration of a target parameter (a) and results (b)



Notes: The analogue of Figure 6 for the Economics $\times$ Culture split of the Left–Right dimension. Each panel plots sharp bounds on the share of AfD voters whose bliss points lie in each 0.5-wide interval of one dimension, the other dimension left unrestricted; within each interval, bars run from 2013 (lightest) to 2025 (darkest). The strip below each panel places the intervals on the dimension, with the AfD’s national median position by year (open circles) and the pooled medians of the other parties (faded dots). Bounds are computed separately for East and West Germany and aggregated; adjacent intervals share mesh cells, so the shares can sum to more than one within a year.

Figure J21. Distribution of AfD supporters’ bliss points along the Economics and Culture dimensions, 2013–2025

K Candidate data

K.1 Candidate nominations across state lists and single-member districts

This subsection provides descriptive information on nomination patterns across state lists and single-member districts (SMDs). The data on all nominated politicians, with their names and party affiliations, were obtained by scraping the non-partisan platform *Wen Wählen?* (*Whom to Vote For?*). I focus on candidates from the main parties considered in the paper (AfD, CDU/CSU, FDP, Left, Greens, SPD).

Under the mixed system, party nominations proceed through two channels: single-member district nomination and state-list nomination. The nomination of *single-member district* candidates is formally decentralized to constituency-level members' or delegates' assemblies, and state party actors do not nominate district candidates themselves, though they retain gatekeeping power. This is because the composition of state lists is fully controlled by state party actors, and the two channels are closely linked: about 64% of candidates who appear on state lists also run in an SMD, and about 74% of those who run in an SMD also appear on a state list (Table K1 reports the exact shares across parties and years). The state lists, chosen by the state leadership, therefore constrain the district-level choices made by local party units.

Candidates nominated in single-member districts also occupy higher list positions: across all years and lists their median position is 10, against 22 for list-only candidates. District strength predicts both access and rank. Candidates placed first or second in their district hold a list place in 84% of cases, against 68% of those placed third or lower; conditional on being listed, they stand at the 28th percentile of their list at the median, against the 44th for the remainder. The pattern also holds within lists: among candidates on the same state-party list, the top tercile of district vote share sits at a median position of 7, against 12 for the bottom tercile, and an additional percentage point of district vote share is worth 0.44 list places (s.e. 0.09).

I use these data to explore the patterns of candidate nomination and renomination to single-member districts. Across 2013, 2017, 2021, and 2025 there were 7,001 unique nominations at the candidate-election level.⁵³ Of these, 1,546 were repeat nominations from one electoral cycle to the next: the same candidate could be nominated in 2013 and 2017, in 2017 and 2021, or in 2021 and 2025. Repeat nominations account for 29.6% of all nominations in 2017, 2021, and 2025. Among them, only 91 (5.9%) are in a different electoral district, and only 9 (0.6%) in a different federal state.

Renomination patterns differ by electoral success. Among candidates who win their district, 56.3% are renominated, against 24.1% of those who lose; within each of the largest parties the rate rises with the first-vote share—from 17.1% in the bottom quartile of a party's candidates to 35.4% in the top, or 1.0 percentage points per percentage point of first-vote share (s.e. 0.08). Winning candidates who are renominated return to the same district in 99.6% of cases.⁵⁴ Among losing candidates who are renominated, 91.5% return to the same district.

⁵³The maximum possible number is 7,176. It is slightly lower because some parties—chiefly the AfD and the Greens—did not field candidates in all districts in some years.

⁵⁴There are two cases in which a candidate was nominated to a *different* district following a change in district boundaries. Albert Helmut Weiler won the district of Gera-Jena-Saale-Holzland-Kreis in 2013, which was dissolved before the 2017 elections; he was nominated in, and won, the district of Saalfeld-Rudolstadt-Saale-Holzland-Kreis-Saale-Orla-Kreis, which was partly composed of the district in which he had run in 2013. Kay-Uwe Ziegler won the district of Anhalt in 2021, which was merged into Anhalt-Dessau-Wittenberg before the 2025 elections; he was nominated in, and won, the neighbouring district of Mansfeld, whose boundaries were redrawn at the same time.

The strong attachment of candidates to their electoral districts (and federal states) is due in part to where they reside at the time of the elections. To examine this, I use the question in the GLES Candidate Survey that asks Bundestag candidates how long they have lived in their electoral district. Participation in the survey is voluntary, so not all candidates respond, and the information is self-reported. The share of candidates born in the district in which they run was 28%, 16%, and 31% in 2013, 2017, and 2021, respectively, and the mean number of years of residence in the district was 25, 25.3, and 25.9 across the same years. About 20% of candidates had lived in their district for less than ten years in every year. While most candidates thus have long-standing ties to their district, a non-negligible share moved not long before the elections, possibly for reasons related to their political careers.

	2013	2017	2021	2025
SPD	69%	62%	67%	65%
Left	76%	71%	71%	77%
Greens	76%	77%	64%	78%
FDP	80%	80%	77%	82%
CSU	41%	41%	25%	28%
CDU	52%	47%	46%	46%
AfD	28%	57%	76%	71%
All	62%	63%	63%	66%

	2013	2017	2021	2025
SPD	96%	99%	96%	97%
Left	48%	46%	46%	45%
Greens	72%	71%	85%	85%
FDP	86%	90%	88%	95%
CSU	64%	64%	50%	55%
CDU	81%	79%	81%	82%
AfD	49%	47%	45%	56%
All	73%	72%	73%	76%

(a) Of state-list candidates, share also nominated in a district

(b) Of district candidates, share also nominated on a state list

Notes. The data are drawn from the non-partisan candidate platform Wen Wählen?, which provides information on the candidate lists and state lists for federal elections. These data provide the universe of all party nominations. Panel (a) is computed over state-list candidacies, panel (b) over district candidacies, matched by candidate name and year. The Piraten are excluded from both panels, including the *All* rows.

Table K1. Overlap between state-list and single-member-district nominations

K.2 Estimation of candidates' positions

1. Estimation of candidates' positions using GLES Candidate Survey Data Candidate positions on two dimensions—left–right and anti–elite—are estimated from the GLES Candidate Studies for 2013, 2017, 2021 and 2025. For each dimension, a set of survey items is first harmonized (see Table K2 for the full list of questions). Each item is then z-scored using a pooled mean and standard deviation computed across all waves in which it appears. This strategy places all items on a common metric and preserves comparability across years.

Each candidate's score on a dimension is the unweighted mean of all available z-scored items within that dimension. The anti–elite dimension comprises seven questions on satisfaction with democracy and current institutions; the left–right dimension uses questions covering redistribution, state intervention, immigration, and social and cultural issues, such as abortion and same-sex marriage. It is also supplemented by a left–right self-placement scale.

Because of the low response rate to this survey (around 35% across all years and parties), the survey was complemented with a non-partisan candidate survey, which also asks candidates about their positions on different issues in each electoral year.

2. Complementing GLES Candidate Survey Data with non-partisan candidate survey *Wen Wählen*. For each election year, candidates from the GLES Candidate Survey are merged with candidate data from the non-partisan platform *Wen Wählen*. Because the two surveys cannot be linked at the level of the individual candidate, the merge is at the district–party level: each *Wen Wählen* candidate is matched to the mean GLES-derived score of their party’s candidates in their district. The GLES scores themselves come from the pooled cross-year normalisation, so the ideological scale is comparable across waves. The elastic net procedure is then used to impute ideological scores for those candidates missing in the GLES Survey but present in *Wen Wählen*. Since the *Wen Wählen* questionnaire changes from election to election, the models are estimated separately by wave (2013, 2017, 2021, 2025).

Two separate elastic net models are estimated per wave, one for each ideological dimension. The outcome y_i is the GLES-derived district–party score attached to candidate i . Only candidates whose district–party cell contains at least one GLES-surveyed candidate enter the training set.

The feature matrix $\mathbf{X}$ contains, for candidate i , the full battery of that wave’s policy questions (65 in 2013, 76 in 2017, 69 in 2021, 50 in 2025; the same set enters both dimension models). Each question is encoded as one-hot indicators for the five response categories: *Yes*, *Rather Yes*, *Undecided*, *Rather No*, *No*, plus an explicit indicator for a missing answer. No year indicators are needed since each wave is fit separately.

The elastic net minimises a penalised least-squares criterion that interpolates between Ridge (ℓ_2) and Lasso (ℓ_1) regularisation:

$$\hat{\beta}(\lambda, \alpha) = \arg \min_{\beta} \frac{1}{n} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \beta)^2 + \lambda \left[\alpha \sum_j |\beta_j| + \frac{1-\alpha}{2} \sum_j \beta_j^2 \right]$$

where n is the number of candidates in the training set, $\lambda \geq 0$ controls the overall amount of regularisation, and $\alpha \in [0, 1]$ determines the mix between the two penalty types. Setting $\alpha = 1$ recovers the Lasso, which drives many coefficients to exactly zero (implicit variable selection); $\alpha = 0$ recovers Ridge, which shrinks all coefficients towards zero while retaining all predictors. Intermediate values of α combine both effects.

Both λ and α are selected jointly by 10-fold cross-validation on the labelled candidates (those with a GLES-based score). The procedure is:

1. The labelled training set is randomly partitioned into 10 roughly equal folds.
2. A full Cartesian grid of 550 hyperparameter combinations is constructed:
 - $\lambda \in \{10^{-4}, \dots, 10^2\}$ (50 values, log-uniform spacing);
 - $\alpha \in \{0.0, 0.1, 0.2, \dots, 1.0\}$ (11 values).
3. For each of the $550 \times 10 = 5\,500$ fold–hyperparameter combinations the fit is run exclusively on the nine training folds and then applied to the held-out validation fold.
4. The root mean squared error (RMSE) between predicted and observed scores is recorded for each held-out fold.

5. The pair $(\hat{\lambda}, \hat{\alpha})$ that minimizes the mean RMSE across the 10 folds is selected.

The winning model is then refit on the wave’s full labelled training set using $(\hat{\lambda}, \hat{\alpha})$, and this final fit is used to predict scores for the wave’s unlabelled candidates. The selected mixtures are Ridge-leaning ($\hat{\alpha}$ between 0.1 and 0.5 across waves and dimensions), retaining 20–119 non-zero coefficients per model. The most potent questions for each year and dimension are presented in Table K3. The most potent questions and directions are as expected given the underlying dimensions.

The model performs well, explaining a large share of variation in observed GLES-based Left–Right position scores (cross-validated $R^2 = 0.76$ pooled across waves, ranging from 0.72 in 2021 to 0.82 in 2025), with a root mean squared error of 0.336 against a standard deviation of the target score of 0.689 (RMSE/s.d. = 0.49). This indicates that the elastic net predictions are able to capture meaningful within-party variation in candidate positions across districts.

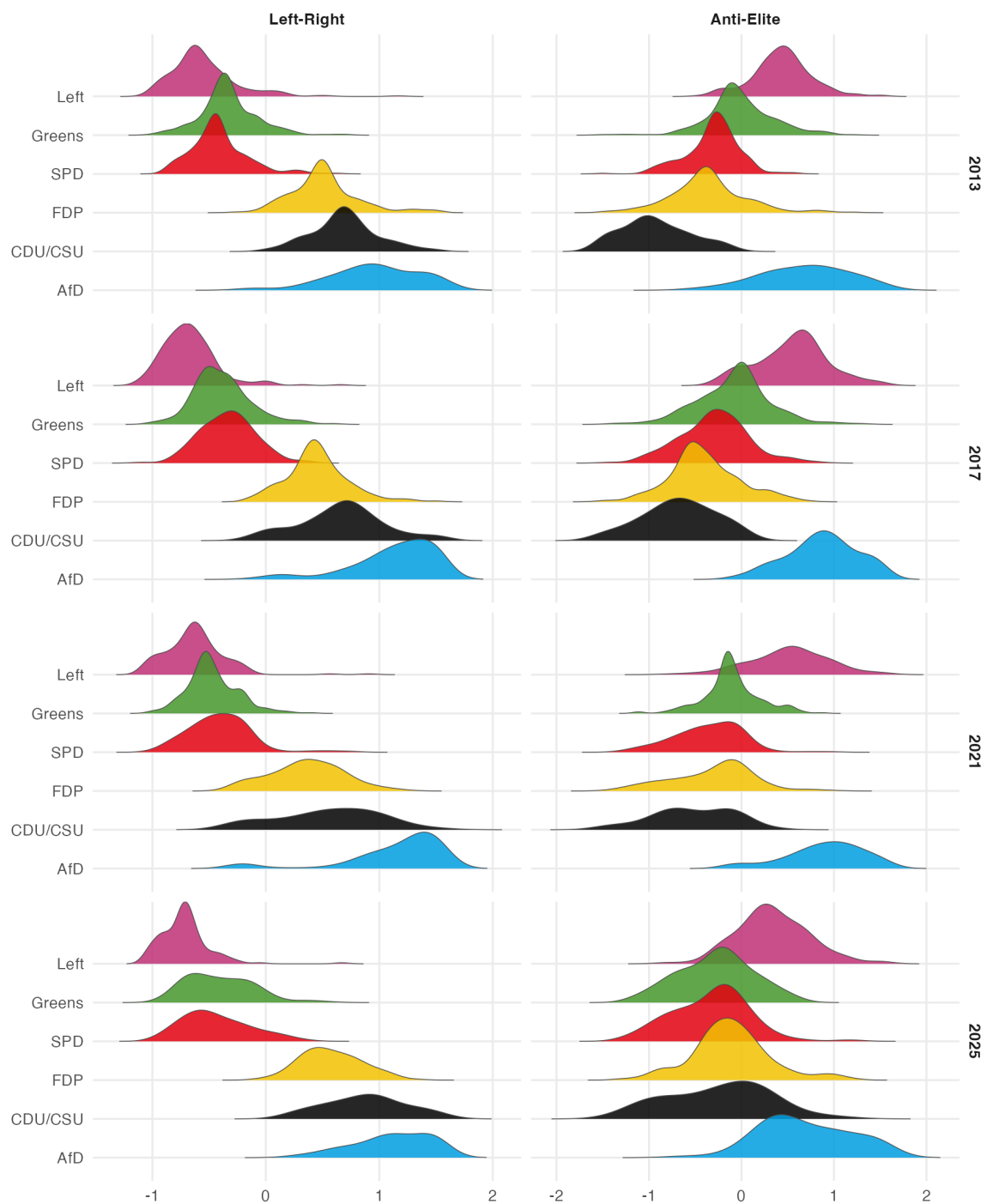
Performance is somewhat lower for the anti-elite score: an RMSE of 0.450 against a standard deviation of 0.648 (RMSE/s.d. = 0.69) and a cross-validated $R^2 = 0.53$ pooled, ranging from 0.38 in 2025 to 0.61 in 2017. The anti-elite score is harder to map from the policy-stance questions available in *Wen Wählen*. Nonetheless, the model explains about half of the variance in observed anti-elite scores, a clear improvement over mean imputation.

To mitigate overfitting, all performance metrics are computed from out-of-fold cross-validation predictions. Across the four waves, 1,147 *Wen Wählen* candidates carry a GLES-based training score and 1,348 receive an elastic-net imputed score; imputed scores are clamped to $[-1.5, 1.5]$ before entering the estimation data. Combining the two candidate surveys yields a locally estimated position for about 60% of all district×party cells. Missing positions are imputed with the within-state-wave median; if that median is also unavailable, the national within-wave median serves as a fallback.⁵⁵

⁵⁵The global median fallback is applied in 5% of the used electoral districts, and only in Eastern Germany in 2013 and 2017.

Dimension	Question	Values
Anti-Elite	Satisfaction with democracy in Germany	1–4
Anti-Elite	Democracy offers enough opportunities for citizens to participate	1–5
Anti-Elite	Laws reflect citizens’ interests	1–5
Anti-Elite	Political parties are the central intermediary between citizens and government	1–5
Anti-Elite	There should be nationwide referendums (direct votes) at the federal level	1–5
Anti-Elite	Referendums tend to lead to bad laws	1–5
Anti-Elite	Citizens have lost trust in politicians	1–5
<i>Economy:</i>		
Left–Right	In a crisis, the government should spend to stimulate economic growth	1–5
Left–Right	In a crisis, Germany should provide financial support to other countries/EU partners	1–5
Left–Right	Government should stay out of the economy	1–5
Left–Right	Providing social security should be a key goal of government	1–5
Left–Right	Government should reduce income differences	1–5
<i>Culture:</i>		
Left–Right	Immigrants should adapt to German culture	1–5
Left–Right	Same-sex marriage should be banned	1–5
Left–Right	Immigration is good for the German economy	1–5
Left–Right	Women should be free to decide about abortion	1–5
Left–Right	Immigration policy: make immigration easier vs. restrict it	1–11
<i>Overall:</i>		
Left–Right	Self-placement on the left–right scale	1–11

Table K2. Survey questions used to estimate candidate positions (GESIS Candidate Survey Data)



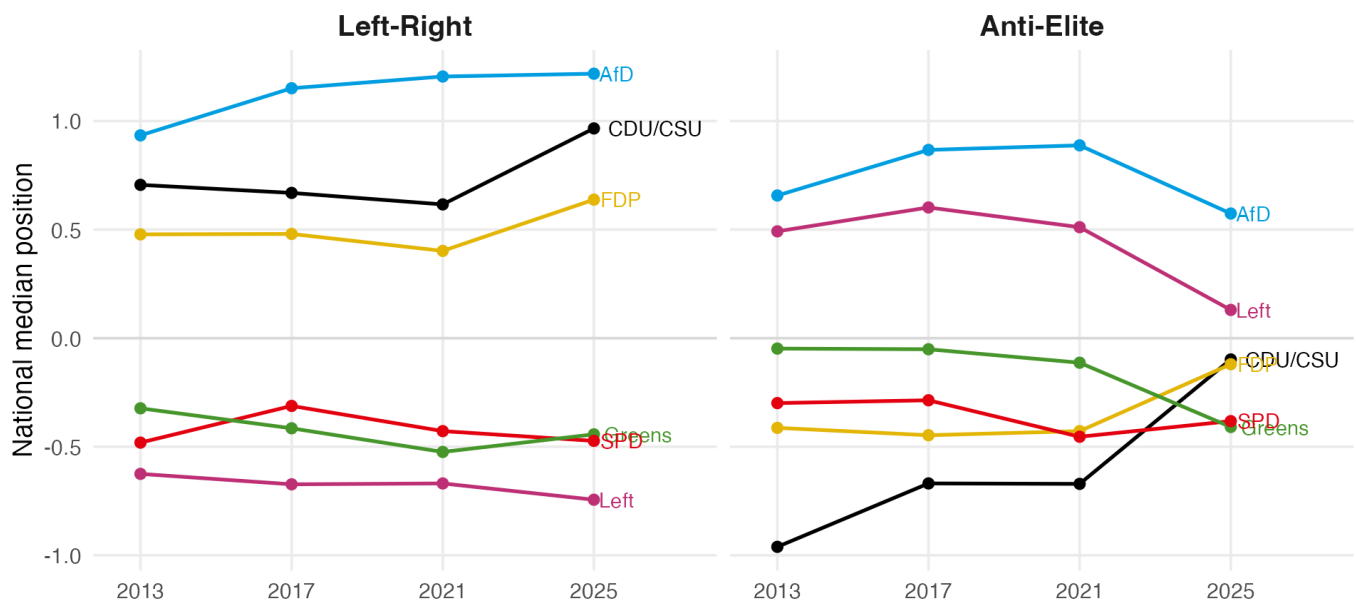
Notes. Kernel density estimates of the positions of single-member district candidates by party, separately for each election (rows) and dimension. Positions are estimated from the GLES Candidate Survey and the non-partisan platform *Wen Wählen*. Each density uses a Gaussian kernel with the bandwidth selected by Scott's rule computed separately for each party, year and dimension.

Figure K1. Distribution of candidate positions by party and election year

Year	Item (translated)	Answer	Coef.
<i>Left-Right (positive = further right)</i>			
2013	The welfare state is more important than reducing debt!	Yes!	−0.08
	More rights for homosexual couples!	Yes!	−0.08
2017	Repeal “marriage for all”!	Yes!	+0.12
	Reintroduce a wealth tax!	No!	+0.07
2021	Ban gender-inclusive language!	Yes!	+0.20
	Legalise soft drugs!	No!	+0.19
2025	Raise the minimum wage substantially!	Yes!	−0.39
	Statutory health insurance for everyone!	Yes!	−0.19
<i>Anti-Elite (positive = more anti-elite)</i>			
2013	Referendums at the federal level!	Yes!	+0.13
	Stop foreign deployments of the Bundeswehr!	Yes!	+0.16
2017	Referendums at the federal level!	Yes!	+0.25
	Stop the sanctions against Russia!	Yes!	+0.18
2021	Build a common European army!	No!	+0.14
	Reintroduce the Deutsche Mark!	Yes!	+0.13
2025	Invest more in defence!	Rather yes	−0.17
	Leave NATO!	No!	−0.12

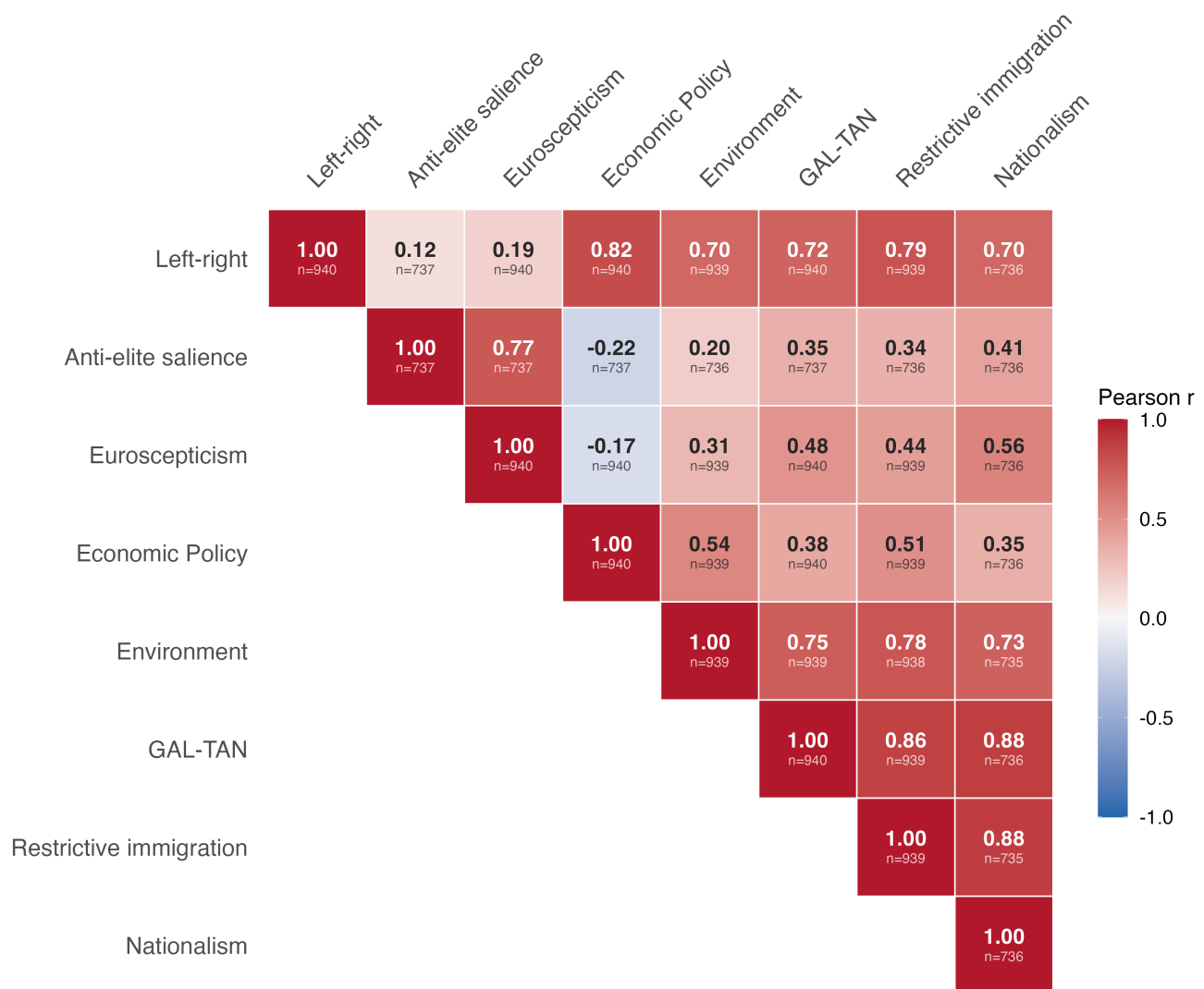
Notes. For each election year and dimension, the two *Wen Wählen* items with the largest contribution to the variance of the imputed score, measured as $\sum_a |\beta_a| \text{sd}(D_a)$ over the item’s answer dummies D_a ; the answer with the largest single contribution is shown. Coef. is the elastic-net coefficient β_a on that answer dummy: the change in the predicted GLES score (whose items are standardised, so roughly one standard deviation of the candidate score per unit) for a candidate giving this answer rather than any other answer to the item, holding all other answers fixed. The prediction targets are the district $\times$ party means of the GLES Left-Right and Anti-Elite scores; one elastic net per wave and dimension, λ and α chosen by 10-fold cross-validation. Answer categories “No answer”, “Undecided”, missing and pooled-rare levels are excluded from the ranking.

Table K3. Most Predictive *Wen Wählen* Items in the Elastic-Net Imputation



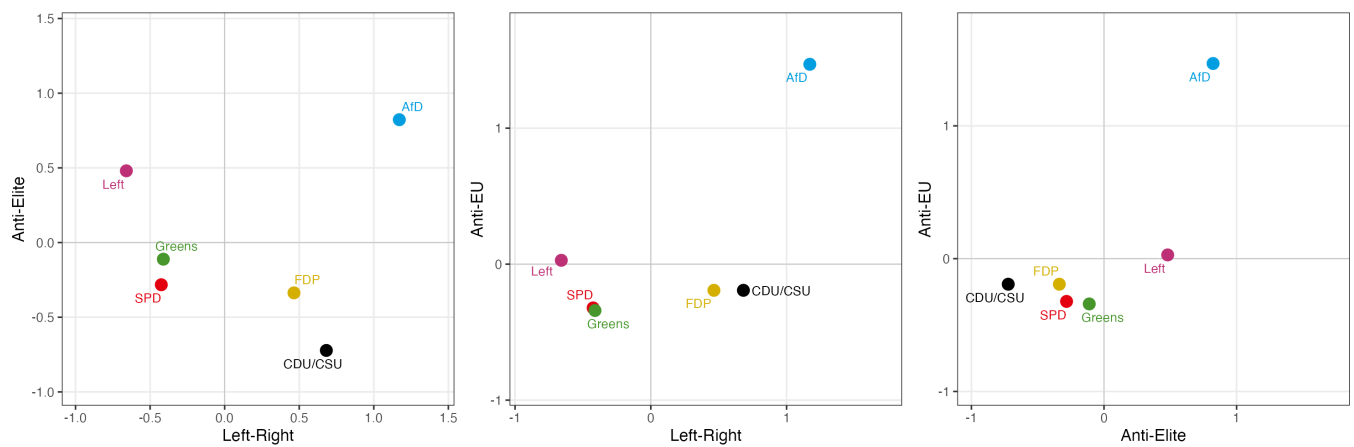
Notes. The data are based on the estimated single-member district candidate positions from the GLES Candidate Survey and the non-partisan platform *Wen Wählen*.

Figure K2. Medians of left-right and anti-elite positions across parties over years



Notes. Each cell reports the pairwise-complete Pearson correlation between two expert-estimated party-position dimensions, computed across party-year observations. Data are the Chapel Hill Expert Survey 1999–2024 (party-level expert means), restricted to European countries and to the waves from 2010 onwards ($N = 940$ party-year observations, 408 parties, 28 countries). All CHES policy scales run 0–10 and are coded so that higher values are more right, conservative or restrictive; *Economic Policy* is the economic left–right scale and *Euroscepticism* reverses the EU-position item. *Anti-elite salience* and *Nationalism* were not fielded in the 2010 wave. Selected correlations (95% Fisher- z CIs in brackets): Left–Right $\sim$ Anti-elite salience 0.12 [0.05, 0.19]; Left–Right $\sim$ Euroscepticism 0.19 [0.13, 0.25]; Anti-elite salience $\sim$ Euroscepticism 0.77 [0.74, 0.80].

Figure K3. Correlations between expert-estimated party positions across policy dimensions



Notes. Each dot is the median position of a party's single-member district candidates, pooled across all four elections (2013–2025) and all federal states. Positions are estimated from the GLES Candidate Survey and the non-partisan platform *Wen Wählen*. Raw Pearson correlations across all pooled candidates (95% Fisher- z CIs in brackets): Left–Right $\sim$ Anti-Elite -0.00 $[-0.04, 0.03]$; Left–Right $\sim$ Anti-EU 0.46 $[0.44, 0.49]$; Anti-Elite $\sim$ Anti-EU 0.60 $[0.58, 0.62]$. After removing party-by-year means, the latter two fall to 0.26 and 0.35 , while Left–Right $\sim$ Anti-Elite stays at 0.01 .

Figure K4. Party median positions on Left–Right, Anti-Elite and Anti-EU dimensions, pairwise